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Multi-Person Human Mesh Recovery with the Anny Body Model

Master's Thesis

by

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Eigenständigkeitserklärung

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Muhammad Saad Najib

Use of AI tools

An AI coding assistant (Claude, Anthropic) was used in this work under my direction: for writing and debugging code, for the analysis scripts that read the training logs and produce the tables and plots, and for drafting and revising the text of this thesis. Every number reported here was checked by me against the logs, scripts and checkpoints named in Appendix A, and the repository history records the assistant's contributions. I take full responsibility for the content of this thesis and can explain every component of it. **[TODO: Student: confirm the wording against RPTU's current rules on AI disclosure and sign off.]**

Abstract

Multi-person human mesh recovery finds every person in one image and builds a 3D body mesh for each. This thesis moves Multi-HMR, a network that does this in one forward pass, from the SMPL-X body model to Anny, an open body model whose shape is set by named values such as age, gender and height. I train it on the synthetic AnnyOne images and on the real 3DPW images, whose SMPL ground truth I converted to Anny with my own fitting script.

The model I started from showed three visible failures: thin, twisted necks, claw-like fingers, and adult bodies on children. Checks of the labels, the body model and the gradients ruled out a rig or decoding problem. Instead, I found a depth bug: the network places each body by its head, but my training targets were built at the pelvis, so it learned to put people too far away and to draw them too small. I also found that two neck bones and two hand bones were turned by half a turn; a constant rotation at the output, with no new parameters, turns them back. In the converted 3DPW labels I found and fixed three defects (reversed gender, a missing actor in two-person frames, missing labels), fitted all 87 people again and checked the new labels independently.

The final model combines both fixes with 30 % real images in every batch. A control run has the same training without the two fixes. On 100 held-out AnnyOne images the final model has a mean vertex error (PVE) of 73.7 mm, against 77.4 mm for the control and 70.5 mm for the starting model. These gaps lie within the 7.6 mm that separated two identical training runs. The errors the fixes target moved far more: against the control, the error of the body's 3D position fell from 1,402 to 728 mm, the neck02 bone error from 175.8 to 4.8 degrees and the finger error from 17.4 to 8.3 degrees. On the 3DPW test set, measured against my converted labels, the final model reaches 89.0 mm PVE, against 126.9 mm for NAVER's released Anny model and 121.8 mm for the starting model; most of this gain comes from training on real images at all. On real photos with age and gender labels (Relative Human), gender accuracy rose from 46.5 to 64.2 % and the separation of children from adults (AUC) from 0.61 to 0.72. For a smaller model, a ViT-B student distilled from the final model has 3.14 times fewer parameters and 3.19 times fewer FLOPs; its training was running when this text was written.

Three things remain open. Children are a measured improvement, not solved. Near and far people still receive similar depths: the fix moved the depth offset and the body size, not the depth slope (0.42, where 1 is correct). Every result comes from one training run per model, without a fixed seed.

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1. Introduction

1.1. The task

Human mesh recovery (HMR) estimates the 3D body surface of the people in an image. Methods rarely predict the thousands of mesh vertices directly. They predict the parameters of a parametric body model, a function that turns a pose and a few shape values into a closed mesh. The body model supplies the anatomy, so the network only has to learn a compact target, and the result can be animated.

Multi-HMR [Bar+24] does this for every person in an image in one forward pass. A Vision Transformer cuts the image into patches and turns each patch into a feature vector, a *token*. For each token the network scores whether a person's anchor point lies in that patch. For each detected person, a small cross-attention head reads the image tokens and outputs the body's pose, its shape and its distance from the camera. Detection and body regression share one backbone, and the body model appears only in the last layer and in the loss.

In this thesis the output is a body of Anny [Bré+26], a body model released in 2025. For one photo the network returns, for every person in it, the rotations of Anny's 163 bones, the 11 shape values of Anny, and where the body stands in 3D. I start from the public ViT-L checkpoint that NAVER released for Anny, keep its backbone, train new heads on synthetic images and then fine-tune on real ones. Figure 4.1 shows the network.

When I began the work reported here, the model I had trained (called Base from Chapter 4 on) showed three failures that anyone could see in its renders. Necks were thin, twisted stalks under the head (Figure 5.1). Fingers curled into claws, so every real-photo demo set them to the rest pose until the fixes (Figure 6.4 shows predicted fingers before and after). A boy in one of our own photos received an adult body. The usual error measure, the mean vertex error after the pelvis is aligned (PVE), was 70.5 mm on held-out synthetic images and showed none of this. Behind the visible failures lay one that no render showed: the network placed people far behind their true position and drew them about a third too small. PVE could not see it, because it removes the body's position before it compares.

A large part of this thesis is therefore about finding out why the model failed before changing it. I tested each cause that had been assumed (a property of the rig for the neck, the image resolution for the fingers, an age that was never learned for the children) on its own. The rig and the age explanations were wrong, and resolution explains little of the finger error. The fixes that followed are small: training targets built at the right joint, and a constant rotation of

four bones at the output. What made them possible were measurements that can see each failure. The thesis reports these measurements, the fixes, and what they do and do not achieve on synthetic and real images.

1.2. Why Anny

Almost every HMR system of the last decade predicts a member of the SMPL family [Lop+15; RTB17; Pav+19]. Two properties of that family motivate a change. SMPL is released under a research-only licence that forbids commercial use. Its shape space is learned by principal component analysis from scans of adults, so its ten shape coefficients have no names, and children need a separate infant model or a blend with one [Hes+18; Pat+21; Sun+22].

Anny addresses both points. Its code is under the Apache 2.0 licence and its MakeHuman-derived assets under CC0 [NAV26a]. Its shape is set by 11 named *phenotype* values between 0 and 1: gender, age, muscle, weight, height, proportions, cup size, firmness and three ethnicity weights. Age is one of these values, so a baby, a child, an adult and an old person come from the same model: age 0 is the baby anchor, 0.333 the child, 0.667 the young adult and 1 the old adult. Gender 0 is the male end and 1 the female end. The full rig has 163 bones, of which 88 are helper bones that the rest of the skeleton drives. Because the values have names, a predicted body can be read as "older" or "heavier", and a supervisor can ask whether the body visibly changes with the person.

Anny comes with AnnyOne [Bré+26], a large set of synthetic images with exact labels. I use 579,960 of its images, rendered at 1280 px and resized to 672 px for the network. NAVER has also released Multi-HMR-Anny, a Multi-HMR checkpoint trained on AnnyOne, so a multi-person HMR system for Anny already exists. NAVER's model is where my backbone comes from and the baseline I compare against. What did not exist were real images with Anny labels. I therefore converted the SMPL ground truth of 3DPW [Mar+18], a real video data set with 3D labels, to Anny. The AnnyOne images and NAVER's checkpoints come with licence terms of their own, which matter for a model release (Chapter 8).

1.3. What the thesis had to deliver

My supervisor set the goals of the thesis as a list of requirements in three tiers. Table 1.1 gives them in short form. The first tier is a condition for submission: a model whose pose looks correct on the 3DPW test set and on any real photo, whose shape visibly changes with the person (1a); a model that is not exactly Multi-HMR, either a new architecture that beats it or a much smaller model that keeps at least 95 % of its accuracy on both test sets (1b); and an evaluation in Anny space on real images (1c). The second tier is recommended for an acceptable grade: 3DPW converted to Anny and used in training (2a), and a model that uses the motion in video instead of treating each frame on its own (2b). The third tier, for the best grade, is a well-argued

Req.	Tier	What it asks for	Main evidence
1a-pose	1	pose of all body parts looks correct on the 3DPW test set and on any real photo	Sections 6.3–6.6
1a-shape	1	size, proportions, age, muscle and the other shape values visibly change with the person	Sections 6.7, 6.8
1b	1	not exactly Multi-HMR: new and better on both test sets, or much smaller with $\geq 95\,\%$ of the accuracy	Sections 6.9, 6.10
1c	1	evaluation with Anny parameters on a real data set	Chapter 3, Section 6.6
2a	2	3DPW train and test converted to Anny; the train set used in training	Chapter 3, Section 6.6
2b	2	the model uses motion in video instead of single frames	Chapter 8 (plan only)
3	3	a well-written report; a clean codebase and verifiable models	this thesis, Appendix A

Table 1.1.: *The supervisor's requirements in short form. Tier 1 is required for submission, tier 2 recommended, tier 3 for the best grade. Status words and the full evidence are in Table 7.1 (Section 7.1).*

report and a clean codebase with models that can be checked independently (3).

These requirements are the frame of the thesis. Chapters 3 to 6 report the work in the order it was done, and Section 7.1 returns to the list and gives each requirement a status word, its evidence and what is missing. I state here only the outcome on 18 October 2026: 2a is met, 1c is met with caveats, 1a is partly met, 1b depends on a student model whose training was running, and 2b had not been started.

1.4. Contributions

Every comparison below is between runs trained once, without a fixed seed. Two identical runs differed by 7.6 mm PVE (Section 5.7), so I compare each change with a matched control and call smaller gaps "within run-to-run variation". The model names are those of Table 4.3: Base is the model I started from, Final the thesis model, Control the same training as Final without the two fixes, and Student the distilled model.

1. **A diagnosis of three visible failures.** Before changing the training, I ran six structural checks on Base: label decoding, the body model's response, the gradients that reach every bone and shape output, prediction against truth, a shape swap and a capped overfit. They rule out a rig or decoding problem (Section 5.2). Each failure was then traced to a specific mechanism (Sections 5.3–5.6).

2. **The depth anchor and its fix.** The network places each person by the head, but I had built the training targets at the pelvis. The 2D losses reconciled the two by pushing bodies away and drawing them smaller.

Base put every holdout person too far away, and the pelvis-aligned PVE could not see it (Section 5.3). Building the targets at the anchor joint and raising the depth weight (Section 4.4) cut the root error from 1,402 to 728 mm against Control and raised the drawn size from 0.59 to 0.78 of the true size. The depth slope stayed at 0.42 (Section 6.2, Table 5.3).

3. **The bone correction.** A per-bone dump showed that two neck bones (neck02 and head) and two hand bones are turned by half a turn in every checkpoint of the Base line. The two neck half-turns cancel, so the head points the right way while the skin between is twisted thin (Section 5.4). A constant rotation per bone, estimated once from 200 training images and applied at the output, turns them back and adds no parameters (Section 4.5). In Final the neck02 error is 4.8 degrees against 175.8 in Control, the neck girth 0.996 of the true girth against 0.464, and the finger error 8.3 degrees against 17.4 (Sections 6.3 and 6.4). Why the bones were flipped is not known.

4. **3DPW labels in Anny space.** I wrote a converter that fits Anny to 3DPW's SMPL ground truth, with a joint correspondence checked by rest-pose geometry and virtual joints where the two skeletons differ (Section 3.3). In the first label set I found three defects: the gender convention was reversed, the second actor was missing in 407 of 960 checked two-person images, and labels were missing where image files were absent. I fixed all three, fitted all 87 people again and checked the result independently: the median body-joint error against 3DPW fell from 18.7 to 14.1 mm (Section 3.5, Table 3.7). The corrected set has 16,694 images with 22,749 people (Table 3.6).

5. **Final against a matched control.** Final fine-tunes Base for 10 epochs with both fixes and 30 % real images per batch; Control differs only by the two fixes. On the AnnyOne holdout the PVE gap (73.7 against 77.4 mm) is within run-to-run variation, while PA-PVE (50.3 against 56.4 mm), root error, neck and fingers moved far beyond it. On the 3DPW test set Final has the lowest error on our labels, 89.0 mm PVE against 91.5 for Control and 121.8 for Base; most of the gain comes from the real images, and the fixes add 2.5 mm (Sections 6.5 and 6.6, Table 6.1).

6. **Body shape on real photos.** On the Relative Human photos, which label age group and gender, Final raises gender accuracy from 46.5 to 64.2 % and the child-against-adult AUC from 0.606 to 0.719 (Section 6.7, Table 6.2). A shape mask on the converter's constant phenotypes brings back the person-to-person variety of muscle and weight but loses the gender gain; I kept Final unmasked (Section 6.8).

7. **The baseline, measured with the same code.** I evaluated NAVER's Multi-HMR-Anny with our labels and settings. On 3DPW it reaches 126.9 mm PVE, against 89.0 mm for Final. NAVER's model was trained on the AnnyOne release, which includes the 100 images we hold out, so

its 60.3 mm there is most likely an in-sample number; ours (73.7 mm) are held out. On 3DPW, which neither model saw before our fine-tune, the final model is 37.9 mm better. I also measured parameters, FLOPs and run time of the models (Section 6.9, Table 6.3).

8. **A smaller student for requirement 1b.** A ViT-B student is distilled from Final through a projector into the teacher's heads (Section 4.7.2). It has 3.14× fewer parameters, 3.19× fewer FLOPs and runs 2.14× faster than Final; its accuracy against the 95 % rule was pending when this was written (Section 6.10).

Two smaller contributions support these. Appendix A maps every reported number to its run, log and checkpoint, and Section 7.4 lists the pipeline bugs whose symptoms looked like model failures.

1.5. Outline

Chapter 2 introduces parametric body models with Anny, single- and multi-person HMR, synthetic training data, label conversion between body models, knowledge distillation and temporal models. Chapter 3 describes AnnyOne, 3DPW, the converter and its three defects, the limits of the converted labels, Relative Human and the diagnostic images. Chapter 4 gives the network, the losses in plain words, the training schedule and the model names, the depth fix, the bone correction, training with real images, distillation and the evaluation protocol. Chapter 5 shows how each failure was isolated, including the explanations that turned out wrong and the run-to-run noise. Chapter 6 reports the fixes, Final against Control and Base on AnnyOne and 3DPW, body shape on real photos, the shape-mask trade-off, the baseline and the model cost, the student and the history of the earlier ViT-S students. Chapter 7 goes through the requirements one by one, explains why PVE barely moved, and states the limitations and engineering lessons. Chapter 8 concludes and gives the next steps. The appendices hold the runs and their jobs (Appendix A), the joint correspondence of the converter (Appendix B), the training flags (Appendix C), additional figures (Appendix D) and the pilot results in full (Appendix E).

2. Background and Related Work

This chapter introduces the body models, networks, data and training ideas the thesis builds on, and for each prior work it says which choice of mine it explains or constrains. It ends with what this thesis adds (Section 2.7).

2.1. Parametric body models: the SMPL family and Anny

A parametric body model is a differentiable function $M(\boldsymbol{\theta}, \boldsymbol{\beta}) \to \mathbf{V} \in \mathbb{R}^{N\times 3}$. It maps a pose $\boldsymbol{\theta}$ and a shape $\boldsymbol{\beta}$ to the N vertices of a mesh. The shape first deforms a template mesh in its rest pose. The pose then moves a skeleton of bones, and the skin follows by *linear blend skinning* (LBS): each vertex is moved by a weighted sum of the transforms of the few bones it is attached to,

$$\mathbf{v}'_i \;=\; \sum_k w_{ik}\, \mathbf{G}_k(\boldsymbol{\theta}, \boldsymbol{\beta})\, \mathbf{v}_i(\boldsymbol{\beta}), \tag{2.1}$$

where $\mathbf{v}_i(\boldsymbol{\beta})$ is the shaped rest vertex, w_{ik} the skinning weight of bone k and $\mathbf{G}_k$ the bone's world transform, found by chaining the rotations of all bones from the root down to k. Because the transforms chain, a wrong rotation high in the tree moves everything below it. This matters in Chapter 5, where two neck bones turned by half a turn leave the head pointing the right way but twist the skin between them.

SMPL. SMPL [Lop+15] is the model most human mesh recovery work is built on. It has 6,890 vertices and 24 joints. Its shape space is a set of principal components learned from registered body scans; most methods use the first 10 coefficients $\boldsymbol{\beta}$, and the pose is one axis-angle rotation per joint. Two properties of SMPL shaped my choice of body model. First, its shape coefficients are statistical directions, not named properties: no coefficient means "older" or "more muscular", so a network that predicts SMPL shape cannot be asked directly for a person's age or build. Second, the shape space was learned from adult scans, so children and infants lie outside it. SMPL is also distributed under a licence for non-commercial research; commercial use needs a separate licence. Every SMPL-based network and dataset inherits these terms.

SMPL-H and SMPL-X. SMPL-H [RTB17] adds the MANO hand model with 15 articulated joints per hand. SMPL-X [Pav+19] adds a face with jaw, eyes and expression coefficients, for 55 joints in total on a mesh of 10,475 vertices. Multi-HMR, the network this thesis starts from, predicts SMPL-X [Bar+24].

Moving it to Anny therefore meant replacing the whole output layer, not adding a few values to it.

Children in the SMPL family. SMIL [Hes+18] is a separate infant body model learned from RGB-D sequences of infants. AGORA [Pat+21] and BEV [Sun+22] reach children by interpolating between the adult template and an infant template with one extra coefficient. This works, but age is then a blend of two templates grafted onto an adult shape space. Anny instead has age as one of its own shape values, which is the main reason this thesis uses it.

Anny. Anny [Bré+26] is a body model by NAVER LABS Europe built from the open MakeHuman project [Mak] instead of from scans. Its code is under the Apache 2.0 licence and the MakeHuman-derived assets are under CC0 [NAV26a], so it carries no restriction from scan data. I use version 0.6.1 of the package with the full MakeHuman rig of 163 bones and a mesh of 13,718 vertices. This is the rig of the AnnyOne labels and of NAVER's Anny checkpoint. (Since version 0.6 the package defaults to a pruned rig of 104 bones with a different pose convention; nothing in this thesis uses it.) Pose is one rigid transform per bone, applied by LBS as in Equation (2.1).

Shape in Anny is set by 11 *phenotype* values, each between 0 and 1: `gender`, `age`, `muscle`, `weight`, `height`, `proportions`, `cupsize`, `firmness`, and three blend weights `african`, `asian` and `caucasian`. Each value drives artist-made MakeHuman blend shapes, which are vertex offsets of the rest mesh. A blend shape belongs to a combination of values, for example a young, muscular, male body, and its weight is the product of the interpolation weights of those values. Unlike SMPL coefficients, the values have names, so a predicted body can be read as "older" or "heavier". The authors report that HMR networks trained on Anny's synthetic data reach accuracy comparable to training with scan-based models [Bré+26].

Two conventions matter throughout the thesis. Gender 0 is the male end and 1 the female end: the rest body is 1.753 m tall at gender 0 and 1.632 m at gender 1. An earlier version of my 3DPW converter used the opposite convention (Chapter 3). Age is a blend between five anchor bodies, newborn, baby, child, young and old, placed at $\mathrm{linspace}(-\frac{1}{3}, 1, 5)$. Age 0 is the baby, 0.333 the child, 0.667 the young adult and 1 the old adult. Values are clamped to the anchor range, so the newborn would need an age below 0, which a network output squashed into $(0, 1)$ never gives. The height and proportions blend shapes exist separately for each age, so "height" scales a body that already has an age.

Figure 2.1 shows what the age value does. At the mean shape of the AnnyOne data, in rest pose, moving age from 0 to 1 raises the skeletal height (head joint to mean ankle) from 0.53 m to 1.54 m, with 1.37 m at age 0.5. Most of the change happens below an age of about 0.4 to 0.5; between 0.667 and 1 stature and body ratios barely move. A small height value does not make a child: at age 0.6 the smallest reachable body is 1.22 m tall and keeps adult proportions. What marks a child in Anny is narrow shoulders relative to the head. The ratio of shoulder width to head width is 2.24 at age 0.1 and 3.00 at age 0.6,



Figure 2.1.: *Anny at the mean AnnyOne shape in rest pose, with only the age value changed from 0 to 1. The skeletal height (head joint to mean ankle) grows from 0.53 m to 1.54 m. Body model only, no network; one camera for all panels, so the relative size is visible. The approximate years printed in the panel titles come from one of three age scales that disagree (Section 2.1) and are not used in this thesis.*

both at height 0.5. Only a low age value produces it. This is why Chapter 5 treats children as an age problem and not as a height problem.

Anny's package also ships a calibration from age values to years. I never convert ages to years, because three scales are in use and disagree: the value read as 11 years is 0.415 in Anny's calibration, 0.333 in the conversion my code inherited, and 0.19 on the MakeHuman slider. I call a person *child-shaped* if the age value is below about 0.45, which selects child-like bodies on any of the three scales.

Two properties of the rig matter later. Of the 163 bones, 88 are helper bones (toes, breasts, face micro-bones and similar) whose rotation an image cannot reveal. I keep them at rest in training, which leaves 75 predicted bones. The neck is modelled by three stacked bones, `neck01`, `neck02` and `neck03`, followed by `head`.

AnnyOne. AnnyOne is NAVER's synthetic dataset for Anny: rendered scenes of several people with exact Anny labels for every person (all bone transforms, all 11 phenotypes and the camera). Its authors describe more than 780,000 images [Bra+26]; the release copy I train on holds 579,960 images of 1280×1280 pixels, which I resize to 672 pixels for the network. Chapter 3 describes its content, its few children and the 100 images I hold out. AnnyOne is the reason Anny is usable for mesh recovery at all: it is the only source of complete Anny labels, including shape, that I have.

2.2. Single-person and multi-person human mesh recovery

HMR. Human mesh recovery (HMR) regresses body-model parameters from an image. HMR [Kan+18] did this for one cropped person with a convolutional encoder and an iterative regressor that refines its own estimate, and it used an adversarial prior to keep poses plausible when only 2D keypoints supervise them. My model follows the same principle: a network outputs the parameters directly, with no optimisation at test time.

SMPLify and SPIN. SMPLify [Bog+16] goes the other way. It fits SMPL to detected 2D keypoints by optimisation, with pose and shape priors.

SPIN [Kol+19] puts such a fit inside the training loop: the network's guess starts the fit, and the fitted result becomes a better training target. My 3DPW converter (Chapter 3) is an optimisation of the SMPLify kind, but in 3D and between two body models: it fits Anny's joints to the joints of 3DPW's SMPL ground truth. Unlike SPIN, the fit runs once, offline, and its output is a fixed label set.

HMR 2.0 and 4DHumans. HMR 2.0 [Goe+23] replaced the convolutional encoder by a Vision Transformer (ViT) [Dos+21] with a transformer decoder and showed that a large ViT trained on a broad data mixture is a strong and simple recipe. The same work tracks people through video (4DHumans, building on PHALP [Raj+22]). Its single-image recipe supports the choice of a large ViT backbone, which Multi-HMR shares.

ROMP. Multi-person methods must also find the people. Two-stage methods run a detector and then a single-person network on each crop; the cost grows with the number of people, and each crop loses the context of the others. ROMP [Sun+21] does both in one pass: it predicts a heatmap of body centres and, at every location, a map of SMPL parameters, and reads the parameters at the detected centres. Multi-HMR keeps this single-shot idea: one forward pass per image whatever the number of people.

BEV and Relative Human. BEV [Sun+22] extends ROMP with an explicit reasoning about depth, so that people are ordered correctly from near to far, and with an age-aware body: an age offset blends the adult SMPL template towards an infant one. To train and test this, the authors collected Relative Human, real photos in which every person carries an age group (adult, teenager, kid, baby), a depth layer and keypoints. BEV relates to two choices of mine. Its depth focus mirrors the depth defect I found (Chapter 5): a mesh error measured after centring the pelvis cannot see where a person is placed. And Relative Human is the only real data with age labels I could use, so I use it, for evaluation only, to test whether predicted ages and genders follow the people on real photos (Chapter 6). Relative Human has no 3D bodies, so it can test age groups and gender but not shape values.

Multi-HMR. Multi-HMR [Bar+24] is the architecture of this thesis. A ViT backbone initialised from DINOv2 [ODM+23] turns the image into one token per 14×14-pixel patch; at 672 pixels this is a 48×48 grid. For each token a small head predicts a score that a person's primary keypoint lies in this patch, and a 2D offset inside the patch. In the released code the primary keypoint is the head joint [NAV26b]. Tokens with a high score become the queries of the *Human Perception Head* (HPH): a few blocks in which the queries attend to each other and, by cross-attention, to all image tokens. The image tokens are joined with a Fourier encoding of each patch's viewing ray, computed from the camera intrinsics, so the head knows the camera. For every query the HPH regresses the body-model parameters and the person's distance from the

camera; the 3D position follows by back-projecting the detected head pixel to that distance. I kept this design and changed what comes out of it (Chapter 4). The distance belongs to the head joint, and this anchor is at the centre of the depth defect of Chapter 5.

The authors trained Multi-HMR on synthetic data, mainly BEDLAM [Bla+23] and AGORA [Pat+21], together with their own close-up set CUFFS. The project page lists the released SMPL-X checkpoints with their per-vertex error (PVE) on the 3DPW test set and their run time on a V100 GPU [NAV26b]. At 672 pixels, the ViT-L checkpoint reaches 94.1 mm in 74 ms, the ViT-B checkpoint 94.0 mm in 43 ms and the ViT-S checkpoint 102.4 mm in 29 ms; a ViT-L at 896 pixels reaches 89.9 mm in 126 ms. On BEDLAM validation the gap between ViT-L and ViT-B is larger, 58.6 against 67.2 mm. Two things follow for this thesis. A ViT-B backbone loses little on real 3DPW images and runs in 43 instead of 74 ms, which is why my smaller model is a ViT-B student and not a ViT-S one (Section 2.5). And these numbers are SMPL-X errors against 3DPW's own SMPL ground truth; they cannot be compared with the Anny-space errors of this thesis.

Multi-HMR-Anny, NAVER's Anny checkpoint. In February 2026 NAVER released a Multi-HMR checkpoint that predicts Anny (`multiHMR_672_L_anny`): a ViT-L backbone at 672 pixels, trained on BEDLAM, AnnyOne and further real images [NAV26b]. A multi-person HMR network that predicts Anny therefore already exists, and this thesis does not claim to be the first. This checkpoint, called Multi-HMR-Anny in this thesis, matters in three ways. It is where my model comes from: the backbone weights of my whole model line were taken from it (Chapter 4). It is the baseline I compare against, evaluated with my code on my labels (Chapter 6, Section 6.9). And it differs from my model in its head: its decoder has 8 cross-attention blocks and 76.57 M parameters, against 2 blocks and 14.01 M in mine, on the same backbone. In its released inference code the predicted 3D position is applied at the head joint, the same joint the detection refers to. The checkpoint has its own licence file, separate from Anny's. NAVER trained it on AnnyOne, which includes the images I hold out, so its AnnyOne error is most likely an in-sample number; Chapter 6 states this next to every comparison.

Anny-Fit. Anny-Fit [Bra+26], by the same group, refines Multi-HMR's Anny predictions per image by optimisation, guided by off-the-shelf semantic, depth, keypoint and segmentation cues, and adapts adult-only models to children without retraining. It targets the weakness I measure on real photos, children with adult bodies, but at test time, while my model stays a single forward pass. I did not use it; it is the natural comparison for future work on children.

2.3. Synthetic training data

AGORA [Pat+21] renders scanned people with SMPL-X fits into realistic scenes, children included. BEDLAM [Bla+23] renders animated SMPL-X bod-

ies with simulated clothing and showed that synthetic images alone can train competitive networks. AnnyOne does the same for Anny. Synthetic labels are exact and complete, shape included, but leave open how well a network transfers to real photos. In this thesis the question is concrete: on the synthetic holdout my shape predictions follow the labels, on real photos they collapse towards the training average (Chapter 5). This is why the final model mixes real images into every batch (30 %), and why those images need Anny labels.

2.4. Real-data labels for a new body model

Why 3DPW needs a conversion. 3DPW [Mar+18] is the standard real-world test set for mesh recovery. It holds 60 outdoor video sequences filmed with a moving phone camera; body-worn inertial sensors and the video together give SMPL ground truth for the one or two actors of each sequence. People walking by are not labelled. The ground truth exists only in SMPL. SMPL and Anny share neither their skeleton nor their shape space: the joint positions differ, the bone hierarchy differs, and SMPL's statistical coefficients have no mapping onto Anny's named phenotypes. To train or test an Anny network on 3DPW, the labels must therefore be re-expressed in Anny, by fitting.

Existing tools. The SMPL-X code includes a tool that transfers fits between SMPL-family models through a fixed vertex correspondence [Pav+19], and the Anny package offers SMPL-X-compatible topologies and an inverter that fits Anny to a target mesh [NAV26a]. None turns a 3DPW label into an Anny label directly, and I did not evaluate the inverter. I wrote my own converter: it fits Anny's joints to the 3D joints of the 3DPW ground truth frame by frame, then fixes one body per person and fits the pose again (Chapter 3, with the three defects I later found and the independent check of the corrected labels).

What converted labels can and cannot supervise. A fit to joint positions recovers what joint positions determine and nothing more. The converted labels supervise the root position and rotation, the bones between the mapped joints, gender (from 3DPW's metadata) and one height and one proportions value per person. They cannot supervise muscle, weight, cup size, firmness or the three ethnicity blends, which joint positions do not reveal. Age is not fitted either; 3DPW shows only adults. In my labels age is the constant 0.55 and the seven others are 0.5, so eight of the 11 phenotypes are constants. Of the 163 bones, 144 stay at rest by design, including `neck02`, `neck03`, `head` and every finger bone: SMPL's 24 joints constrain only the bones mapped onto them, and 3DPW's SMPL ground truth has no finger joints. The fit itself leaves an error floor of about 14 mm on the body joints. Three consequences run through the thesis. On 3DPW only pose, gender, height and proportions are tested, not full shape. Training on the converted labels pulls the eight constant phenotypes towards their constants on real images, which Chapter 6 measures. And errors on these labels are Anny-space errors over Anny joints with a conversion floor, not comparable with published 3DPW results.

Attribute labels as an alternative. SHAPY [Cho+22] supervises body shape on real images with measurements and crowd-sourced word attributes instead of 3D ground truth. Anny's named phenotypes would suit such labels; I collected none and rely on the age groups and genders of Relative Human, two of the 11 values.

2.5. Knowledge distillation

Output distillation. Knowledge distillation [HVD15] trains a small *student* network to reproduce a large, trained *teacher*. In the original form the student matches the teacher's class probabilities, both softened by a temperature T, with a Kullback–Leibler (KL) divergence:

$$\mathcal{L}_{\mathrm{KD}} \;=\; T^2\,\mathrm{KL}\big(\mathrm{softmax}(\mathbf{z}_t/T)\,\big\|\,\mathrm{softmax}(\mathbf{z}_s/T)\big)\,, \tag{2.2}$$

where $\mathbf{z}_t$ and $\mathbf{z}_s$ are the teacher's and the student's outputs before the softmax. A higher T spreads the probabilities and exposes how the teacher ranks the wrong answers; the factor T^2 keeps the gradient size independent of T. Mesh recovery regresses numbers rather than classes, so the output part of my distillation is a plain L1 distance between the student's and the teacher's bone rotations, phenotypes and depth.

Feature distillation. FitNets [Rom+15] let the student also match an intermediate representation of the teacher, through a small learned regressor that bridges the different widths. My student does the same at the end of the backbone: a linear projector maps its 768-wide ViT-B tokens to the teacher's 1024-wide tokens, and a KL term at $T = 4$, in the form of Equation (2.2) applied to the projected tokens, pulls them towards the teacher's. Because the projected tokens have the teacher's width, the teacher's trained detection and HPH heads can be copied into the student unchanged and fine-tuned, instead of being trained from scratch. Chapter 4 gives the recipe.

DINOv2. DINOv2 [ODM+23] trained its largest ViT with self-supervision and distilled the smaller ViTs from it, so a DINOv2 ViT-B is a good starting point for a student. The question for this thesis is whether a ViT-B student of my final ViT-L model keeps at least 95 % of its accuracy on both test sets, the bar my supervisor set for a smaller model. The student has 101.38 M parameters against 318.38 M, a factor of 3.14 fewer.

2.6. Temporal models for video

My model works on single frames. The temporal requirement of the thesis asks whether motion in video can be used. This work has not started, so this section only names the methods a frame-wise model like mine could build on. Most refine per-frame features of a frozen image network, the form that would sit on top of my per-person HPH tokens.

HMMR. HMMR [Kan+19] runs a temporal convolution over per-frame features and predicts the current, past and future poses. It showed that a short window already smooths the motion.

VIBE. VIBE [KAB20] adds a recurrent network over per-frame features and a motion discriminator trained on motion capture. The discriminator is tied to SMPL; for Anny it would need Anny motion data.

TCMR. TCMR [Cho+21] found that a strong per-frame feature lets the temporal model ignore its neighbours, which causes jitter. It adds branches that predict the current frame from past and future frames alone and lowers the acceleration error, the jitter measure I would use as well.

MPS-Net and GLoT. MPS-Net [Wei+22] replaces the recurrence by attention over frames. GLoT [She+23] combines a global transformer over a long window, trained by masking poses, with a local one over neighbouring frames. Both refine per-frame features, as a refiner on my model would.

SmoothNet. SmoothNet [Zen+22] removes jitter from the output poses of any estimator without image features. It needs no change to my model, so it is the baseline a learned refiner would have to beat.

PHALP, 4DHumans and WHAM. PHALP [Raj+22] tracks people by predicting their 3D appearance, location and pose into the next frame; 4DHumans [Goe+23] combines it with HMR 2.0. Tracking is the step any multi-person video method needs first. WHAM [Shi+24] places the motion in world coordinates using the camera's rotation and foot contact, which addresses a problem my model shares: per-frame depth that is not consistent over time.

2.7. What this thesis adds

NAVER's Multi-HMR-Anny already predicts Anny for several people in one image. What was missing is knowledge of how such a model fails on real images, why, and how far it can be repaired, together with real labels to measure it in Anny space and a smaller model of comparable accuracy. This thesis contributes the following. Each item is stated against that baseline and names the chapter with the evidence.

A measured diagnosis of three failures. The model I started from showed thin, twisted necks, claw-like fingers and children with adult bodies. Before changing anything I tested what is not the cause (labels, rig, gradients, optimiser), and then traced each failure to a specific mechanism (Chapter 5).

The depth-anchor fix. The network places each person by the head, but my training targets were built at the pelvis. The 2D losses reconciled the two by pushing bodies away and drawing them too small, which the pelvis-centred mesh error could not see. Building the targets at the anchor joint nearly halves the error of the body's 3D position against a matched control (Chapters 4 and 6).

The bone correction. Two neck bones and two hand bones were turned by half a turn in every checkpoint of my model line. A constant rotation applied at the output, estimated once and with no parameters, turns them back and restores the neck's thickness; its origin is unknown (Chapters 5 and 4).

Converted 3DPW labels with a verified check. I converted 3DPW's SMPL ground truth to Anny for all 87 actors, found three defects in the first label set (reversed gender, a missing actor in two-person frames, missing images), fitted all actors again and checked the new labels with an independent test (Chapter 3). This is what makes training on real images and Anny-space evaluation on 3DPW possible.

The final model and its control. The final model, called Final from Chapter 4 on, combines both fixes with 30 % real images in every batch. A control run with the same data, steps and start but without the two fixes separates what the fixes do from what the extra training does (Chapter 6).

The baseline comparison. I evaluated Multi-HMR-Anny with my own code, settings and labels, so that both models are measured the same way on Anny-One and on 3DPW (Section 6.9).

The smaller student. A ViT-B student (Student) distilled from the final model, with 3.14 times fewer parameters, is the route to the requirement of a model that is not exactly Multi-HMR (Section 6.10).

3. Data

This chapter describes the images and labels behind every number in the thesis (Table 3.1). AnnyOne is synthetic, carries exact Anny labels and is the main training set. 3DPW is real video with SMPL ground truth; I converted it to Anny so that the model can train and be tested on real people in Anny space (requirements 1c and 2a). Relative Human is a set of real photos with age-group and gender labels, used only to evaluate children and gender. A diagnostic manifest and twelve demo images serve the checks of Chapter 5 and the by-eye comparisons of Chapter 6. Most of the chapter is about the 3DPW conversion, because the converted labels decide what the real-image numbers can show.

dataset	type	labels	used for	size
AnnyOne	synthetic	full Anny labels: 163 bone rotations, 11 phenotypes, camera; 1280 px renders, 2.73 people per image	training (70% of each mixed batch); last 100 images = holdout	579,960 images = 579,860 + 100
3DPW-Anny, v23 labels	real video	3DPW's SMPL ground truth converted to Anny by my fit; every third frame; 1.36 people per image	training (30%, train split); validation (300 images per epoch); test	train 5,882 / 7,584; val 2,622 / 3,311; test 8,190 / 11,854
Relative Human, validation split	real photos	per person: box, age group (adult, teen, kid, baby), gender; no 3D	evaluation only (children, gender)	635 / 2,130 (whole set 8,741 images)
diagnostic manifest	synthetic	AnnyOne labels of fixed (image, person) entries	the checks of Chapter 5; excluded from fine-tune sampling	40 training + 15 holdout = 55
demo images	real	none used	by-eye comparison of models	4 photos + 8 3DPW test frames

Table 3.1.: *The datasets of this thesis. Sizes are images / labelled people where both are given. The 3DPW-Anny counts are those of the v23 labels (Section 3.5); numbers on 3DPW are always against these converted labels, never against the original SMPL meshes.*

3.1. AnnyOne

AnnyOne [Bré+26] is the synthetic dataset released with the Anny body model. It holds 579,960 rendered images of 1280×1280 pixels, with 2.73 people per image on average. For every visible person the label gives the exact Anny pose (163 bone transforms), the 11 phenotype values and the camera intrinsics. The network sees each image resized to 672 pixels, a grid of 48×48 tokens of 14 pixels. The field of view changes from image to image (51° to 110° on a sample of five images), so training and evaluation use each image's own intrinsics.

Split. Indices 0 to 579,859 are the training set. The last 100 images, indices 579,860 to 579,959, are the holdout; no model trained in this thesis has seen them. The holdout is my own choice, not an official test split. NAVER's model was trained on the AnnyOne release, which includes these 100 images, so for that model the holdout is most likely in-sample (Chapter 6). The 3DPW test script also scores the last 100 AnnyOne images, most likely the same ones, but in setting V0 (helper bones unmasked, Chapter 4). Its "AnnyOne test-100" value is therefore not the holdout number, which is always given in setting V1.

Few children. In Anny the child shape appears below an age of about 0.4 to 0.5 (Chapter 2). I never convert these values to years, because three age scales in the code and the data disagree. Only 2.53% of AnnyOne's people are younger than 0.3 (1.06% below 0.2, 4.45% below 0.4); the median age is 0.601, and 41% of all people lie between 0.5 and 0.6. In a normal training batch, 2.1% to 2.3% of the supervised people are younger than 0.3. The holdout has nobody younger than 0.513. It tests adults only, which is why I added Relative Human (Section 3.7).

3.2. 3DPW

3DPW [Mar+18] is a real video dataset of 60 sequences filmed with a hand-held phone camera, outdoors and indoors, at 1080×1920 pixels (portrait) and 30 frames per second. Inertial sensors on the actors give SMPL ground truth. The standard release has 24 training, 12 validation and 24 test sequences, with one or two labelled actors each: 87 person-sequences in total (62 men, 25 women). Bystanders have no labels.

SMPL differs from Anny in skeleton and shape parameters, so I wrote a converter that fits an Anny body to every labelled person (Section 3.3). The splits stay as 3DPW defines them (uses in Table 3.8). One validation sequence, `outdoors_crosscountry_00`, yields no labels (Section 3.5), which leaves 11 usable validation sequences.

3.3. Converting SMPL to Anny

Figure 3.1 shows the six steps from a 3DPW frame to an Anny label. The converter (`smpl_to_anny.py`, version 2.3) fits each person; a label builder merges

the fits into one file per image.

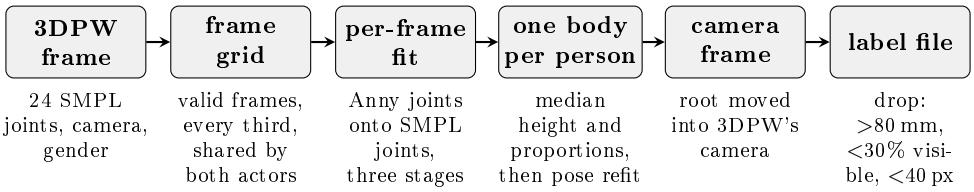


Figure 3.1.: *The six steps from a 3DPW frame to an Anny label: SMPL joints, frame grid, per-frame fit, one body per person, camera frame, label file with the drop rules. Since converter version 2.3 both actors of a sequence share the frame grid.*

Objective. Anny has no analytic inverse of its skinning, so I fit by gradient descent, following NVlabs' `smpl2soma` for the autograd route and `joints2smpl` for the joint objective. For each person and frame I search Anny's 163 rotations $\mathbf{r}$ (6D), the phenotype logits $\boldsymbol{\ell}$ and a translation $\mathbf{t}$ so that Anny's joints land on the 24 SMPL joints $\hat{\mathbf{J}}^S$ of the frame:

$$\begin{aligned}\min_{\mathbf{r},\boldsymbol{\ell},\mathbf{t}} \frac{1}{24}\sum_{k=1}^{24}\left\|\mathbf{p}_{\pi(k)}(\mathbf{r},\sigma(\boldsymbol{\ell}))+\mathbf{t}-\hat{\mathbf{J}}^S_k\right\|_2^2 \\ +\lambda_r\sum_{b\neq\text{root}}\|\mathbf{R}_b-\mathbf{I}\|_F^2+\lambda_\phi\sum_{j\in\text{free}}\left(\sigma(\ell_j)-\tfrac{1}{2}\right)^2,\end{aligned} \tag{3.1}$$

where π is the joint correspondence and $\mathbf{p}_b$ the point attached to bone b (defined below). The first regulariser ($\lambda_r = 10^{-4}$) pulls every non-root rotation towards identity: a twist about a bone's own axis moves no joint but corkscrews the mesh, so without it the solution is not unique. The second ($\lambda_\phi = 10^{-3}$) pulls the free phenotypes towards 0.5, so that one remaining joint gap cannot drag a phenotype to an extreme.

Stages and targets. Adam (learning rate 0.05) first fits the root orientation and translation, then the bone rotations, then the free phenotypes. Shape comes last because it is the weakest signal; fitted earlier, it absorbs pose error. Each frame starts from the previous frame's solution (150 / 400 / 250 iterations per stage for a person's first frame, 20 / 80 / 60 afterwards). The targets are 3DPW's own per-frame `jointPositions`. A forward pass of the neutral SMPL model is wrong here, because 3DPW's shape coefficients belong to the gendered models; on the first frame I checked, it moved the joints by 37 mm on average and 93 mm at the feet. Only joint positions are matched, not SMPL's rotations, shape or surface.

Correspondence checked by geometry. My first converter chose the Anny bones by name, and two choices were wrong in a way no name reveals. Anny's MakeHuman rig numbers the spine top-down (`spine01` below the neck), SMPL

bottom-up, so SMPL's lumbar `spine1` had been mapped to a bone 25 cm above it. SMPL's hips had been mapped to `pelvis.L/R`, whose heads lie 4 cm from the hip joint, instead of `upperleg01.L/R`. The converter now places both models in their rest poses and picks the spine bones by height along the body axis and the hip bones by distance from the pelvis (Appendix B).

Virtual joints. Even with the right bones, the rigs define their torso joints at different anatomical points. In the rest pose, SMPL's pelvis joint sits 9 cm above its hips, while Anny's root sits at hip height and 7 cm in front; SMPL's `spine3` lies 6.5 cm from the nearest Anny bone head and its collars 6 cm further out. For nine torso joints I therefore measure once, in the rest pose, the offset $\boldsymbol{\delta}_b$ from the Anny bone head to the SMPL joint, in the bone's own frame. The fit moves the point $\mathbf{p}_b = \mathbf{h}_b + \mathbf{R}_b\boldsymbol{\delta}_b$ onto the SMPL joint, where $\mathbf{h}_b$ and $\mathbf{R}_b$ are the posed bone head and frame. The offsets are 11.6 cm (pelvis), 4.3 cm (hips), 3 to 7 cm (spine) and 5 cm (collars); limb joints use the bone head.

Bones and shape. Only the root and 18 of the 23 mapped bones may rotate. The other bones have no target; in the first version about 130 of them drifted freely. In the labels 144 of 163 bones stay at rest, among them `neck02`, `neck03`, `head`, the jaw and every finger bone. Phenotypes are not mapped from SMPL's shape coefficients, which have no semantic counterpart, and joints cannot determine most of them: weight, muscle, cup size, firmness and the three ethnicity values move no joint, and adult age barely does. Left free, they became slack (Section 3.4). The converter therefore reads gender from 3DPW, fixes age at 0.55 (close to the AnnyOne mean; the 3DPW actors are adults) and the seven others at 0.5, and fits only height and proportions. In Anny, gender 0 is male and 1 is female (rest stature 1.753 m at 0, 1.632 m at 1). Since version 2.3 the converter writes men as 0 and women as 1 (`-gender_convention anny`); earlier versions had it reversed (Section 3.5). After a first pass over all frames, the median of each free phenotype becomes the person's one body, and every frame's pose is fitted again with it (100 iterations). Table 3.2 lists what is fitted, read and fixed.

Frames, camera and label files. I fit every third valid frame (10 frames per second); neighbouring frames are near-duplicates. Since version 2.3 one absolute grid holds for every person of a sequence (frame numbers divisible by 3, `-frame_grid shared`), so both actors are fitted on the same images wherever both are valid. Only the root changes when the body moves into the camera frame: its rotation becomes $\mathbf{R}_{\text{cam}}\mathbf{Q}_{\text{root}}$ and its position $\mathbf{R}_{\text{cam}}(\mathbf{t}+\mathbf{t}_{\text{3DPW}})+\mathbf{t}_{\text{cam}}$. The posed mesh, projected with 3DPW's camera, gives a box and the fraction of vertices inside the image. The builder writes one file per image in the AnnyOne format and drops a person when the fit error exceeds 80 mm, less than 30% of the mesh is inside the image, or the box is narrower or shorter than 40 px. A dataset class returns these labels in AnnyOne's sample format, padding the portrait images to a square and shifting the principal point, so the training and evaluation code run on them unchanged.

fitted to the 3DPW joints	read from 3DPW metadata	fixed
root position and rotation, per frame	gender: men 0, women 1 (Anny's convention, since v2.3)	age 0.55
18 of the 23 mapped bones, per frame		muscle, weight, cup size, firmness, african, asian, caucasian: 0.5
height and proportions, one value per person		144 of 163 bones at rest, among them neck02, neck03, head, jaw and all finger bones
dropped by the label builder: fit error above 80 mm; less than 30% of the mesh inside the image; box narrower or shorter than 40 px		

Table 3.2.: *What the SMPL-to-Anny converter fits, what it reads from 3DPW and what it fixes. Everything in the right column is the same for every 3DPW person, so the labels carry no information about it.*

3.4. Development of the converter

Each step of the converter's development answered a measured symptom.

A common floor. The first converter (v1) gave 34.5 mm on four frames of `downtown_bar_00` (Table 3.3). On four test scenes, two facts in the log showed a defect before I looked at any render. The error was 30 to 37 mm on every frame of every scene, and all four actors came out with the same body (gender 0.02, age 0.67, height 0.33, asian 0.73). A residual independent of the pose and a shape independent of the person mean that the shape absorbed a fixed mismatch between the skeleton definitions. The rest-pose comparison then showed the two wrong bones described above. The renders had looked wrong for a separate reason: the renderer used 3DPW's `img_frame_ids`, an index into the 60 Hz pose arrays, as image numbers.

frame	root only	+ pose	+ shape	final (mm)
0	137.9	40.4	36.2	36.2
50	131.2	43.1	35.1	35.1
100	169.5	39.4	34.1	34.1
199	135.0	39.0	32.6	32.6
mean				34.5

Table 3.3.: *First converter (v1, correspondence by name): mean joint error after each stage on four frames of* `downtown_bar_00`.

Effect of each correction. Table 3.4 isolates the corrections on the hardest scene, `outdoors_fencing_01` (38 frames, including full lunges). The geometric correspondence and 3DPW's own joints (v2) removed the common floor: limbs and head fell to 1 to 10 mm, but the torso stayed at 43 to 73 mm, and the optimiser shifted the pelvis 5 cm and drove `proportions` to 0.06, a value that must not become a label. The virtual joints (v2.1, with the bone mask, the warm start and the shape prior) brought the torso to 13 to 22 mm, the pelvis to 19 mm and `proportions` to 0.27. The per-joint table is then flat, as a correct correspondence should give. Read as an ablation, the correspondence is worth about 10 mm and the virtual joints a further 9 mm, on different joints (limbs and head, then torso). An intermediate version that down-weighted the torso joints only hid the symptom, and I dropped it.

converter	mean (mm)	max	worst joints (mm)
v1: correspondence by name, neutral SMPL targets	34.3	37	all joints, common floor
v2: correspondence by geometry, 3DPW joints, gender from metadata (reversed until v2.3)	24.6	30	spine1 73, collars 55–59, spine3 58, pelvis 53
v2.1: + virtual torso joints, bone mask, warm start	15.8	18.8	shoulder 36, wrists 25–27

Table 3.4.: *Mean joint error over all 24 joints on `outdoors_fencing_01` (38 frames) for three converter versions.*

Age is not identifiable. With v2.1, four varied test scenes convert to 15.8 to 20.1 mm (Table 3.5); only one occluded office frame reaches 43.5 mm, and the bodies now differ between actors. Age did not: it came out at exactly 0.67 for every actor, as in every earlier version. Joints cannot tell one adult age from another, and the optimiser used the age morph to absorb a small torso-length gap. Version 2.2 therefore fixes age at 0.55 and adds boxes from the posed mesh. On all 87 person-sequences it fitted 22,915 frames with a median error of 18.2 mm and a worst frame of 70.9 mm. By every measure the converter prints, these fits looked clean. They still held two of the three defects of the next section.

scene	frames	mean (mm)	max	height	proportions	age
`downtown_walking_00`	40	17.5	19.0	0.52	0.34	0.67
`outdoors_fencing_01`	38	15.8	18.8	0.51	0.27	0.67
`office_phoneCall_00`	32	20.1	43.5	0.49	0.42	0.67
`flat_guitar_01`	30	16.2	18.0	0.52	0.23	0.67

Table 3.5.: *Converter v2.1 on four 3DPW test scenes (stride 25, one body per scene, all 24 joints). Age was still free.*

3.5. Three label defects and the v2.3 re-fit

The converter grades its own fit, so it cannot see errors in what it is asked to fit or in what happens to the fits afterwards. I therefore wrote an independent check that reads the finished label files. On sampled images from every sequence it compares the labels with 3DPW's joints, surface and 2D keypoints, confirms that the loader passes them on unchanged, and tests their consistency: gender, a label for every valid actor, head turn. On the first label set ("old labels", v2.2 fits, about 2,000 images) its grade was FAIL, with three defects.

Defect 1: gender reversed. The converter wrote 1.0 for men and 0.0 for women, MakeHuman's convention and the reverse of Anny's. Every 3DPW man was fitted and labelled with a female body. Refitting with the flipped value lowered the joint error for 72 of 73 men (median 19.2 to 15.3 mm). For women it raised the joint error (13.7 to 16.3 mm) but brought the surface closer for 27 of 31, so the fix was then confirmed for men only. With 3DPW at 30% of each batch, a model trained on these labels keeps learning a female body for every real man.

Defect 2: missing actor. In 407 of 960 sampled two-person images, one valid actor had no label because it had not been fitted on that frame. Each person's every-third-frame grid started at that person's own first valid frame, so after an invalid gap the two actors sat on different phases (frames 0, 3, 6 against 1, 4, 7). The detection loss then treats a visible, real person as background.

Defect 3: missing labels. The training split had 4,451 label images where the fits implied 6,437 (31% missing), validation 2,458 of 2,777. A first rebuild gave the same counts. The image files of 8 sequences were missing or incomplete on the cluster (`courtyard_captureSelfies_00`: 480 of 750 files), most likely from an incomplete unzip, and the builder skips frames without an image file without counting them. After I re-extracted the images, a second rebuild recovered every missing label (Table 3.6, "rebuilt"). That set still had defects 1 and 2, so I never trained on it.

The re-fit. Defects 1 and 2 arise inside the fit. Version 2.3 adds the two switches named above, `-gender_convention anny` and `-frame_grid shared`; their defaults reproduce v2.2 exactly, so the old labels of the history models stay reproducible. I re-fitted all 87 people with both switches on. The re-fit's post-check passed on all 60 sequences (Table 3.7, middle block). Every man now has gender 0 and every woman 1. The men's mean joint error fell from 19.0 to 14.7 mm, the women's stayed level (15.0 to 15.2 mm). 9 single-person outdoor sequences became 1 to 3 mm worse; the likely reason is a different warm start, which I did not test. Two-person frames with one valid actor unfitted went from 4,457 of 8,274 (54%) to none.

The v23 labels. I call the labels built from the v2.3 fits the v23 labels: 16,694 images and 22,749 people (Table 3.6). They have fewer images than the rebuilt set but the same people, because both actors now share each frame. No person was dropped for fit error or box size. 163 frames were dropped for visibility, all from `outdoors_crosscountry_00`. With 3DPW's own camera, the ground-truth joints of that sequence sit at a depth of -0.3 m, behind the camera, and project far outside the image. The old fits show the same (162 frames), and no label set ever contained this sequence. The 3D fit is fine; image labels for it cannot exist with this camera.

	old labels	rebuilt	v23 labels
converter fits	v2.2	v2.2	v2.3
image files of 8 sequences	missing	re-extracted	re-extracted
train, images / people	4,451 / 5,183	6,437 / 7,589	5,882 / 7,584
validation	2,458 / 2,995	2,777 / 3,314	2,622 / 3,311
test	9,722 / 11,850	9,722 / 11,850	8,190 / 11,854
total	16,631 / 20,028	18,936 / 22,753	16,694 / 22,749
gender reversed	yes	yes	no
actor missing in two-person frames	yes	yes	no
labels missing (no image file)	yes	no	no
used for training	history models only	never	Final, Control, Student

Table 3.6.: *The three 3DPW-Anny label sets. All are built at every third frame with the same drop rules. The v23 labels have fewer images than the rebuilt set but the same people, because both actors of a two-person sequence now share each frame.* `outdoors_crosscountry_00` *yields no labels in any set (Section 3.5). Numbers measured on the old labels are retired and never appear next to numbers on the v23 labels.*

The check on the v23 labels. I ran the same check on the v23 labels (3,314 people in 2,360 images, 40 per sequence, 59 sequences; Table 3.7). The grade moved from FAIL to WARN. The median body-joint error fell from 18.7 to 14.1 mm. Flipping the gender now makes the fit worse (men 13.9 to 20.0 mm), and no sampled two-person image misses an actor. The head turn relative to the chest is now kept (19.6° in the labels against 19.3° in SMPL); in the old labels it was mostly lost, above all in the test split (6.3° against 20.0°). I did not check why. Two rows stay at WARN: the limb-length ratios (forearm 0.948, thigh 1.064) and the 3.2% of people with a joint more than 100 mm off, mostly at the hip. The likely reason for both is that Anny's and SMPL's limb proportions and hip joints differ; I did not test it further. The three defects are gone, and the independent check confirms it. Figure 3.2 shows panels of the check.

check	old labels	v23 labels
overall grade of the check	FAIL	WARN
body joints vs 3DPW, median / p95 (mm)	18.7 / 24.5	14.1 / 21.1
surface, Anny→SMPL / SMPL→Anny (mm)	18.9 / 18.3	18.9 / 16.1
2D error over raw 3DPW (px)	+1.1	+0.4
gender labels	reversed	men 0, women 1
refit with flipped gender	better for 72 of 73 men	worse: men 13.9 → 20.0 mm
two-person images with an unlabelled actor	407 of 960	0 of 1,080
fitted frames; per-frame error, median (mm)	22,915; 18.2	22,912; 14.0
head turn vs chest, labels / SMPL, median (deg)	9.5 / 20.2	19.6 / 19.3
post-check of the re-fit, all fitted frames, 87 people		
mean joint error, men (62) / women (25) (mm)	19.0 / 15.0	14.7 / 15.2
two-person frames with one valid actor unfitted	4,457 of 8,274	0 of 6,055
still WARN on the v23 labels		
limb-length ratio Anny / SMPL, forearm / thigh		0.948 / 1.064
people with a joint more than 100 mm off		3.2%

Table 3.7.: *Independent conversion check of the 3DPW-Anny labels before (old labels, v2.2 fits) and after the re-fit (v23 labels, v2.3 fits). The check samples images from every sequence and compares the labels with 3DPW's own joints, surface and 2D keypoints. The middle block is the re-fit's own post-check over all fitted frames; old and new fits fitted different frames, so it is not a paired comparison.*

Figure pending: `F14_label_check.jpg`, three panels from the conversion check of the v23 labels, one of them a two-person frame.

Figure 3.2.: *Conversion check on the v23 labels: the 3DPW photo, 3DPW's SMPL ground truth with its 2D keypoints, and our Anny label drawn with 3DPW's camera. Real 3DPW frames. By design the labels have rest-pose hands and a head that turns only through* `neck01` *(Section 3.6).*

3.6. What the 3DPW labels can and cannot test

The v23 labels are free of the three defects, but they are converted labels, not ground truth. Five limits follow from the converter's design, and each bounds a later claim.

Eight constant phenotypes. The labels test three phenotypes: gender (from metadata), height and proportions (fitted; medians 0.47 and 0.51). The other 8 are the same for every person: age 0.55 and seven values at 0.5. A model trained on 3DPW is pulled towards these constants for every real person, by the shape loss and by the mesh terms, because a 3DPW person's ground-truth mesh is built from the same values. By my estimate 3DPW people are about 16% of the supervised people in a 70/30 batch. The 3DPW error is also measured against these neutral-body meshes, so it likely rewards a neutral prediction; I did not test this. Chapter 6 reports the option that removes the shape loss on the eight constants for 3DPW people (Final-masked).

The fixed age suits the 3DPW actors, who are all adults, but carries no age information. Figure 2.1 (Chapter 2) shows what age does in Anny: the skeletal height rises from 0.53 m at age 0 to 1.37 m at 0.5 and 1.55 m at 0.8, and the child cue is the shoulder width relative to the head (2.24 at age 0.1, 3.00 at 0.6) more than stature. At 0.55 every 3DPW label is a full adult. 3DPW has no children, so it can neither teach nor test them.

Head and hands at rest. `neck02`, `neck03` and `head` stay at identity in every label, so the head turns only through `neck01`, and the labels cannot test the half-turned neck bones of Chapter 5. 3DPW's SMPL ground truth has no finger joints, so every label has rest-pose hands. For 3DPW people the training masks the rotation terms of the 38 hand bones (Chapter 4); the mesh terms still carry the rest-pose hands.

A floor of about 14 mm. The labels sit 14.1 mm (median) and 21.1 mm (p95) from 3DPW's own joints. Every Anny-space error on 3DPW contains this residual.

Not comparable with published 3DPW numbers. I never evaluated against the original SMPL meshes. Our MPJPE averages all 163 Anny joints after pelvis alignment, and PVE and MPJPE both use the converted labels. I therefore write "lowest error on our 3DPW-Anny labels" and never set these numbers against published ones.

Bystanders. 3DPW labels only its actors. A correct detection of a bystander counts as a false positive, so precision on 3DPW is a lower bound. Recall counts labelled actors only and is not affected.

3.7. Relative Human

Neither the holdout nor 3DPW contains children, so neither shows whether a model reads children or gender on real photos. Relative Human, introduced with BEV [Sun+22], fills this gap. It holds 8,741 real photos (5,218 train, 635 validation, 1,836 test), many crowded and with children. Each person has a box, an age group (adult, teen, kid, baby) and a gender coded 0 for male, as in Anny; there are no 3D labels. I use only the validation split and only for evaluation, since the dataset ships without a licence file: 635 images with 2,130 labelled people (1,680 adults, 101 teens, 209 kids, 117 babies, 23 of unknown age).

The model runs once on each whole photo, as in the demo, with a field of view of 55°. Each detection's box is the extent of its projected mesh, and detections are paired one to one with labelled boxes, greedily, at an intersection over union of at least 0.3. A predicted age below 0.45 counts as child-shaped, a body-shape threshold rather than an age in years, and a predicted gender below 0.5 as male. I report recall and mean predicted age per age group, the child-vs-adult AUC of the predicted age (kids and babies against adults) and the gender accuracy. Every number covers detected people only, so recall always stands next to it.

3.8. The diagnostic manifest, the holdout and the demo images

Diagnostic manifest. The checks of Chapter 5 need the same people every time. A seeded script picks them from the AnnyOne labels alone and stores 55 fixed (image, person) entries. 40 come from the training range (adults, the youngest children, clearly visible hands, the hardest head turns), 15 from the holdout, which has no children. Base saw the training entries; every later fine-tune (Final, Control, Student) gives their images a sampling weight of zero, so they remain the same in-sample reference for all models. The manifest children are therefore in-sample: they show whether a model can fit a child it has seen, not whether it generalises.

Holdout. The 100 holdout images (Section 3.1) are the headline synthetic test set: small, adult-only and synthetic. Every holdout comparison in Chapter 6 is read against the run-to-run variation measured in Chapter 5.

Demo images. For by-eye comparisons I use 12 real images: 4 photos and 8 3DPW test frames, one middle frame from each of 8 test sequences. No model trained on them and no metric is computed on them; every statement about them is a by-eye reading and says so.

3.9. Summary: the splits and their use

Table 3.8 lists every split with its size and its role. Two rules hold throughout: no test image is trained on, and numbers on the old 3DPW labels never appear next to numbers on the v23 labels.

split	size (images / people)	used for	chapters
AnnyOne training	579,860 images	training of every model; 70% of each mixed batch	4, 6
AnnyOne holdout	100 images	headline synthetic evaluation (V1); never trained on	5, 6
diagnostic manifest	40 + 15 entries	per-person diagnostics; excluded from every fine-tune	5
3DPW-Anny train, v23	5,882 / 7,584	30% of each mixed batch	4, 6
3DPW-Anny val., v23	2,622 / 3,311	300 images scored after each epoch; monitoring only	6
3DPW-Anny test, v23	8,190 / 11,854	reported 3DPW numbers; never trained on	6
3DPW-Anny old labels	16,631 / 20,028	training of the history models only; retired	4
Relative Human val.	635 / 2,130	children and gender on real photos; evaluation only	5, 6
demo images	12 images	by-eye comparison; no metric	5, 6

Table 3.8.: *Every split of the thesis, its size and its use. 3DPW-Anny sizes are images / labelled people after the drop rules. The final model is the last epoch of its run, so the 3DPW validation split does not select it.*

4. Method

This chapter describes the network, its losses, the training runs and the evaluation. Section 4.3 defines the model names used in the rest of the thesis. The two changes that matter most are in Sections 4.4 and 4.5: the training targets now describe the joint the network anchors, and a constant rotation at the output turns four flipped bones back. Chapter 5 shows how the defects were found; Chapter 6 measures what the changes did.

4.1. Multi-HMR for Anny: task, architecture, inputs and outputs

4.1.1. The task

The input is one RGB image with camera intrinsics $\mathbf{K}$ (taken from an assumed field of view when they are unknown). It shows an unknown number of people. For each person i the network predicts the parent-relative rotations $\mathbf{R}_i \in SO(3)^{163}$ of Anny's bones, the phenotypes $\boldsymbol{\phi}_i \in [0,1]^{11}$, and the image position (u_i, v_i) and depth z_i of one anchor joint a, the head. The Anny body model M [Bré+26] turns $(\mathbf{R}_i, \boldsymbol{\phi}_i)$ into vertices and joints, and the mesh is moved so that its anchor joint sits at the back-projected anchor pixel:

$$\mathbf{V}_i = M_V(\mathbf{R}_i, \boldsymbol{\phi}_i) - M_{J,a}(\mathbf{R}_i, \boldsymbol{\phi}_i) + \mathbf{t}_i, \qquad \mathbf{t}_i = z_i\, \mathbf{K}^{-1}\,(u_i, v_i, 1)^\top. \tag{4.1}$$

The goal is a mesh with the right pose, shape and place in the camera frame, from one pass over the whole image, with no person crops and no other video frames.

4.1.2. Architecture

The network is Multi-HMR [Bar+24] with its output resized for Anny (Figure 4.1). I did not change the backbone, the detection or the cross-attention head, so for requirement 1b (a model that is not exactly Multi-HMR) the architecture is no answer; the route to 1b is the smaller student of Section 4.7.

Backbone and detection. A DINOv2 ViT-L with patch size 14 [ODM+23] maps the 672×672 input to 48×48 tokens of width 1024; AnnyOne renders (1280×1280 pixels) are resized to this input. The backbone weights come from NAVER's Anny checkpoint (Section 4.3). A two-layer MLP gives every token the probability that a person's anchor joint lies in it; at inference a 3×3 non-maximum suppression and a threshold keep the detected tokens. A second MLP predicts a 2D offset in token units, so the anchor pixel is

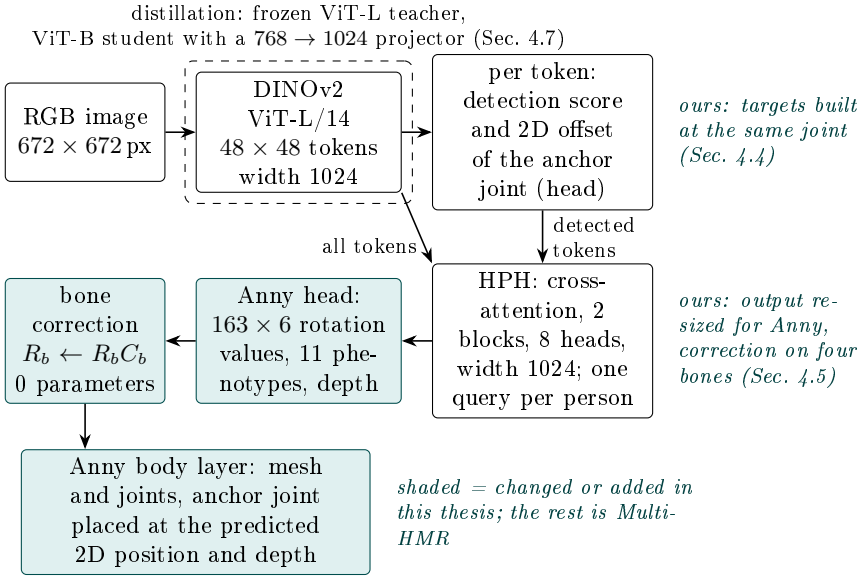


Figure 4.1.: *The network. A 672-pixel image passes through a DINOv2 ViT-L backbone (48×48 tokens). Every token gets a detection score and a 2D offset for the anchor joint, the head. For each detected person a cross-attention head (HPH) predicts 163×6 rotation values, 11 phenotypes and a depth; a constant correction turns four flipped bones back; the Anny layer builds the body and places it at the predicted distance. Shaded boxes are changed or added in this thesis; the dashed box marks the part that distillation replaces.*

	Tensor	Shape
Input	RGB image (AnnyOne: from 1280^2)	$3 \times 672 \times 672$
	intrinsics $\mathbf{K}$	3×3
Output	detection scores, one per token	48×48
Per person	anchor joint (head): token, offset, depth z	2, 2, 1
	translation $\mathbf{t}$ of the anchor joint	3
	bone rotations $\mathbf{R}$	$163 \times 3 \times 3$
	phenotypes $\boldsymbol{\phi}$	11
	mesh vertices $\mathbf{V}$	$13{,}718 \times 3$
	joints $\mathbf{J}$	163×3

Table 4.1.: *Inputs and outputs of the network. The rotations are parent-relative; 88 of them are fixed to the identity by the helper-bone mask.*

(token index $+ 0.5 +$ offset) $\cdot 14$. In every model of this thesis the anchor joint is the head, Anny's bone at the base of the skull (`--person_center head`).

Human Perception Head (HPH). Each detected token becomes one query, built from the token, a Fourier embedding of its viewing ray through $\mathbf{K}$ (99 channels), learned row and column embeddings, and constant start values for pose, shape and camera. A transformer decoder with 2 blocks, 8 heads of size 32, width 1024 and MLP width 1024 processes all queries of an image together: in each block the people attend to each other and then to all image tokens. These are the values in `model.py`; the draft of this thesis gave width 512 and 16 heads, which was wrong.

Anny head. Three linear layers read the decoder's output token; they are what differs from the SMPL-X version of Multi-HMR. *Pose:* 163×6 values are added to the identity in the continuous 6D representation [Zho+19] and orthonormalised by Gram–Schmidt into rotation matrices. *Shape:* 11 logits pass through a sigmoid, so every phenotype lies in $[0, 1]$. *Depth:* as in Multi-HMR, a log-depth normalised by the focal length, $z = \exp(d\, f / f_n)$, with f_n the focal length of a 60° field of view, clamped to 50 m. A fixed mask then forces 88 helper bones (toes, breasts, face, subdivision bones) to the identity and keeps 75 predicted; the same mask is applied to the ground truth in training. Right after Gram–Schmidt and before the mask, four bones are multiplied by one constant rotation each (Section 4.5).

Body-model layer and size. The Anny model runs inside the network in PyTorch, with linear blend skinning, and places the body with Equation (4.1); the 3D and 2D losses are back-propagated through it. The network has 318.38 M parameters (heads 14.01 M; NAVER's released Anny model has the same backbone and a 76.57 M head) and needs 1,927.4 GFLOPs per image with one person (one multiply-add = two FLOPs). Table 4.1 lists inputs and outputs.

4.2. Losses in plain words

Anny runs on each person's true parameters to give target vertices and joints; the anchor targets are built from one joint (Section 4.4). The total loss is

$$\begin{aligned}\mathcal{L} = {} & \alpha_{\text{det}}\mathcal{L}_{\text{det}} + \alpha_{\text{off}}\mathcal{L}_{\text{off}} + \alpha_{\text{rot}}\mathcal{L}_{\text{rot}} + \alpha_{\text{shape}}\mathcal{L}_{\text{shape}} \\ & + \alpha_{\text{dist}}\mathcal{L}_{\text{dist}} + \alpha_{\text{t}}\mathcal{L}_{\text{t}} + \alpha_{\text{j3d}}\mathcal{L}_{\text{j3d}} + \alpha_{\text{v3d}}\mathcal{L}_{\text{v3d}} \\ & + \mathbb{1}[e \geq e_{\text{2D}}]\,(\alpha_{\text{j2d}}\mathcal{L}_{\text{j2d}} + \alpha_{\text{v2d}}\mathcal{L}_{\text{v2d}}) + \lambda_{\text{kd}}\mathcal{L}_{\text{kd}} + \lambda_{\text{kd-out}}\mathcal{L}_{\text{kd-out}},\end{aligned} \tag{4.2}$$

with e the epoch; the distillation terms exist only for a student. Table 4.2 says what each term compares and gives its weights.

Detection, rotations, shape. The detection term is the focal loss of CornerNet [LD18] in CenterNet's form; the log calls it `bce`, but it is not a binary cross-entropy. A second person whose anchor falls into an occupied token is dropped from the targets. The Base recipe boosts neck, head and hand bones by 4 and rescales all rotation weights to a mean of 1, so the boosted bones weigh $\times 2.41$ and the rest $\times 0.60$: the boost moves loss, it does not add any. The shape L1 is smallest at the median of the plausible answers, so weak image cues pull predictions toward the training median, and the flat ends of the sigmoid make extreme values slow to learn (Chapter 5).

Which terms see depth. The 3D terms align both skeletons at the pelvis and carry no placement information. Only distance and translation see placement directly, and the 2D terms indirectly, through the projection. Before the depth fix, distance and translation made up about 2.5% of the total loss, against 25.7% for shape.

Schedule of the 2D terms, masks, numerics. The code default switches the 2D terms on at epoch 10; Final, Control and Student start from trained heads and use them from epoch 0 (the draft said otherwise). For 3DPW people some terms are masked (Section 4.6). Training runs in fp32 with TF32 products (mixed precision gave NaN values with the Anny layer); steps with NaN or Inf gradients are skipped, 16 of 10,000 in Final. No weight was tuned in this thesis.

4.3. Training schedule and model names

The Base lineage. Training began from NAVER's public ViT-L checkpoint, Multi-HMR-Anny (`multiHMR_672_L_anny`). Its head has a different structure, so only the backbone was transferred. Four stages followed, L-stage 1 to L-stage 4 (Table 4.3): fresh heads with the whole backbone trained, a side branch that leads nowhere; fresh heads with the backbone frozen (learning rate 10^{-4}, then 3×10^{-5} after a resume, per the training scripts, which were edited after the runs); the last 4 of 24 blocks unfrozen at 10^{-5} with the helper mask; and the neck and hand boost. A fifth run raised the shape weight to 20; its

Term (log name; `loss.py`)	What it compares, in plain words	Default	Final	Control	Student
Detection (`bce`; 314)	CornerNet focal loss on the per-token score "an anchor joint lies in this token"	10	10	10	10
Offset (`offset`; 315)	L1 on the anchor's position inside its token	1	1	1	1
Rotations (`rotmat`; 319–335)	L1 over the 9 entries of every bone's parent-relative rotation matrix	0.1	0.1	0.1	0.1
neck and hand boost	multiplier on neck, head and hand bones, then renormalised to mean 1 ($\times$2.41 for them, $\times$0.60 for the rest)	1	4	4	4
Shape (`shape`; 336–343)	L1 on the 11 phenotypes after the sigmoid	1	20	20	20
Distance (`dist`; 344)	L1 on the anchor's log-depth, normalised by the focal length	1	5	1	5
Translation (`transl`; 345)	L1 on the anchor's 3D position in the camera frame	1	5	1	5
3D joints (`j3d`; 367–377)	L1 on the 163 joints, each skeleton moved to its own pelvis	100	100	100	100
3D vertices (`v3d`; 378)	the same for the mesh vertices	100	100	100	100
2D joints (`j2d`; 383–384)	L1 in pixels on the projected joints, placed by the predicted anchor and depth	1	0.1	0.1	0.1
2D vertices (`v2d`; 381–382)	the same for the projected vertices	1	0.1	0.1	0.1
first epoch of the 2D terms	`--start_2d_epoch`	10	0	0	0
Global rotation (`global_rot`; 350–363)	L1 on the camera-frame orientation of head, neck03 and both wrists	0	0	0	0
Feature distillation (`kd`)	KL divergence between the softened channel distributions of student and teacher tokens, $T = 4$	–	–	–	1.0
Output distillation (`kd_out`)	L1 to the teacher's rotations, phenotypes and depth on the same people	–	–	–	1.0
Joint the targets describe	`--gt_center_joint`: transl, dist, detection token and offset targets	pelvis	head	pelvis	head
Bone correction	constant output rotation on neck02, head, metacarpal1.L, finger4-1.R	none	on	none	on

Table 4.2.: *Loss terms and their weights. "Default" is the value in `loss.py`; Final, Control and Student are the values of those runs (model names in Table 4.3). The total loss is the weighted sum of all rows above the rule. For people from 3DPW the rotation and global-rotation terms skip the 38 finger, thumb and metacarpal bones in all three runs; the mesh terms are not masked. The shape mask for 3DPW people is off in all three runs and on only in Final-masked (Section 4.6). The Student keeps the teacher's bone correction during training and is evaluated with its own, re-estimated after training (Section 4.7.2).*

epoch 99 is *Base*: synthetic images only, and the start of every fine-tune here. Continuations used a learning rate ten times below the cold start, which on a partly trained model had caused hundreds of NaN gradient events per epoch.

Final and Control. Final starts from Base and trains 10 epochs of 1,000 iterations at batch 4 (40,000 images) at a constant learning rate of 10^{-5}; the last 4 backbone blocks and the heads train. Three changes come on together: the depth fix with distance and translation weight 5 (Section 4.4), the bone correction (Section 4.5), and 30% real images per batch with the hand mask (Section 4.6). The 55 images of the diagnostic manifest (Chapter 3) are never sampled. Training took 4,292 s on one A100 (10.9 images per second). Control is identical but keeps the pelvis targets with weight 1 and has no bone correction, so Control against Final isolates the two fixes, and Base against Control shows the real images plus the restart cost. Each fix was tested alone only in short pilots (Chapter 5).

Model names. Table 4.3 defines the names used from here on. Each name is one run; "L" marks the ViT-L line, "L-stage" its transfer schedule, and Student-S the ViT-S students by their old version number. Table A.1 in Appendix A is the full definition: it maps each name to the run and checkpoint it had during the project, its starting weights, data, schedule and flags.

4.4. The depth anchor and its fix

Anchor and targets. The network places each person by the head: the detection token, the offset and the predicted depth describe the head, and the mesh hangs from it (Equation (4.1)). For AnnyOne images, however, the training code built the translation, distance, token and offset targets at joint 0, the pelvis. Every checkpoint before this thesis, Base included, was trained on these targets, so prediction and target described two joints about half a body apart.

What the mismatch does. The network must put its head joint where the pelvis target is. The offset head tried to close the gap: in a one-batch debug print its offsets were -2 to -19 tokens, where a correct offset lies within ± 0.5. A mesh hanging from a head placed at the pelvis is drawn too low. The 2D terms penalise this misfit, and the network can shrink it by moving the body away, where it is drawn smaller; the pelvis-aligned 3D terms cannot object and the direct depth terms were weak. The mismatch is certain; the mechanism is likely but was not isolated (Figure 4.2). On the AnnyOne holdout Base predicts the pelvis depth as $0.33 \times \text{true} + 2.36\,\text{m}$: all 222 matched people are too far (pelvis error 1.21 m), drawn at 0.63 of their size although their 3D height is right (1.405 against 1.409 m). PVE aligns the pelvises and the real-photo demos place bodies from their boxes, so neither showed it (Chapter 5).

The fix. A single training switch builds all four anchor targets at the joint the model anchors (in the code, the flag `--gt_center_joint` set to

Name	Meaning
Multi-HMR-Anny	NAVER's released ViT-L checkpoint for Anny; the source of every backbone here and the baseline of requirement 1b
L-stage 1 to 4	the four steps from Multi-HMR-Anny's backbone to Base: fresh heads with the whole backbone trained (a side branch), fresh heads with the backbone frozen, the last 4 blocks trained, the neck and hand loss weights raised
Base	the end of the schedule (shape weight 20), AnnyOne only, epoch 99; the start of every fine-tune
Mixed-L	Base fine-tuned with 30% 3DPW on the labels before their correction (Chapter 3); the previous best on 3DPW
Fresh-heads-L	fresh heads on Multi-HMR-Anny's backbone, trained with 30% 3DPW (old labels) from the first step
Control	Base, 10 epochs, 30% 3DPW on the corrected labels, hand mask, neither fix; the reference for Final
Final	Control's recipe plus the depth fix and the bone correction; the thesis model and the Student's teacher
Final-masked	Final plus the shape mask on the 8 converter constants; a tested option
Final-partial-mask	Final plus the shape mask on the 7 constants other than age; pending
Student-S/1 to /6c, /mixed, /final	the ViT-S students of Base (/mixed and /final also saw 3DPW on the old labels; /final's teacher is Mixed-L), before both fixes; history
Student	the ViT-B student distilled from Final; the answer to requirement 1b

Table 4.3.: *Model names used in this thesis. Every name is one run; no run was repeated with another seed. Runs, checkpoints, data, schedules and flags: Table A.1 (Appendix A).*

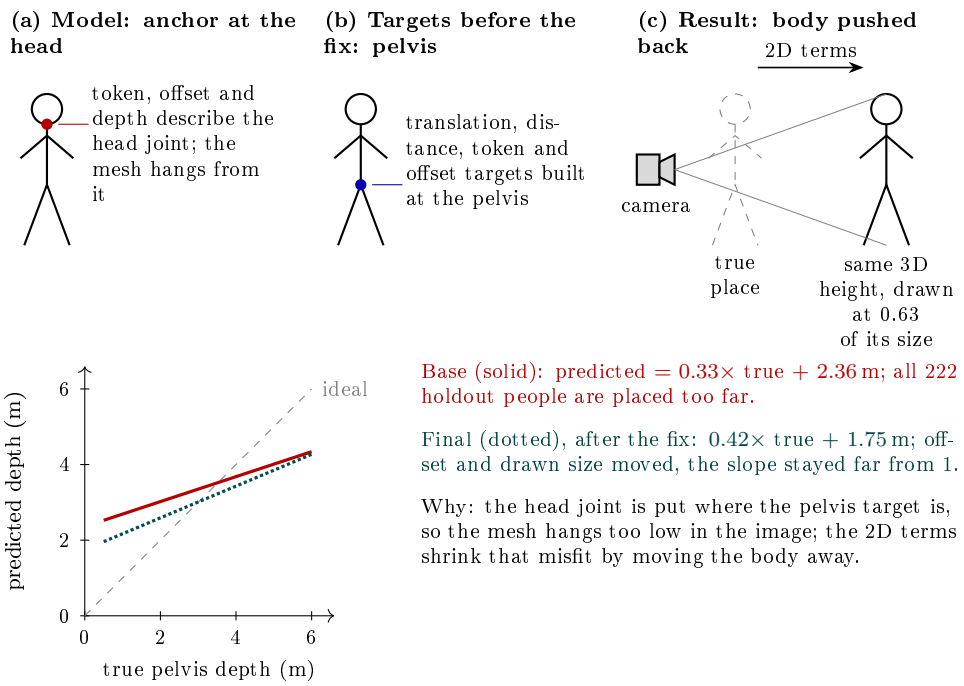


Figure 4.2.: *The depth-anchor bug. (a) The model places each body by its head; (b) the training targets were built at the pelvis; (c) the 2D terms reconcile the two by moving the body away, where the same 3D body is drawn smaller. Bottom: the fitted line of predicted against true pelvis depth for the 222 matched people of the AnnyOne holdout (setting V1), Base and Final. Schematic; the measured scatter is in Chapter 5.*

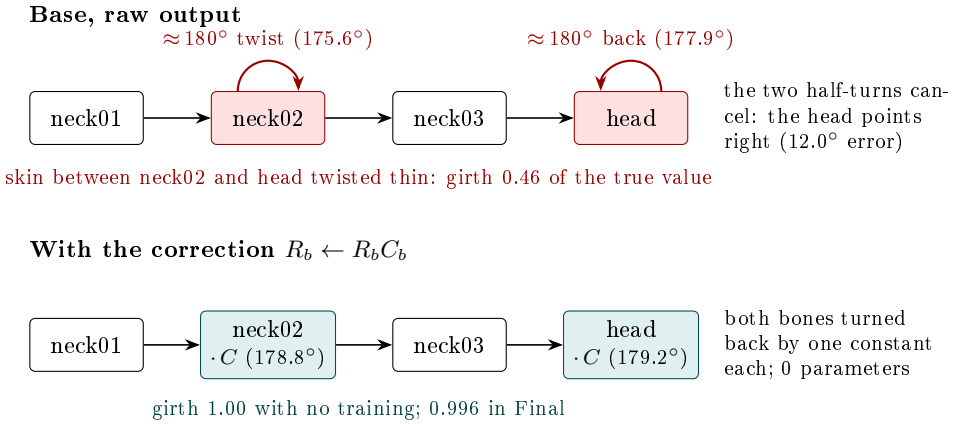


Figure 4.3.: *The neck chain neck01–neck02–neck03–head. Top: in Base, neck02 and the head bone are each turned about half a turn; the turns cancel, so the head points right, but the skin between them is twisted to 0.46 of its true girth. Bottom: one constant rotation per bone turns both back. Girth on the AnnyOne holdout, setting V1.*

`person_center`); the visibility filter and the one-person-per-token rule move with them. The network does not change. I also raised the distance and translation weights from 1 to 5, because they were the only direct depth terms. In the 10-epoch pilots the switch alone reached a pelvis error of 0.73 m, and the switch with weight 5 reached 0.54 m; weight 20 was only tried for 2 epochs. Final uses the switch with weight 5; Control keeps the pelvis targets with weight 1. The fix moved the offset and the drawn size, not the slope: Final's line is $0.42 \times \text{true} + 1.75$ m (Chapter 6).

4.5. The bone correction

What it corrects. A per-bone dump on the holdout (Chapter 5) showed two bones of the neck chain turned half a turn in all 8 checkpoints of the Base lineage: neck02 by 175.6° and the head bone by 177.9° (median error of the parent-relative rotation). The half-turns cancel, so the head points the right way (12.0° error) and PVE never showed the flip, but the neck skin between them is twisted to 0.46 of its true girth (Figure 4.3).

How it works. The parent-relative rotation of each flipped bone b is multiplied on the right by a constant, $\mathbf{R}_b \leftarrow \mathbf{R}_b\mathbf{C}_b$. If the prediction is $\hat{\mathbf{R}}_b\mathbf{F}_b$ for the true rotation $\hat{\mathbf{R}}_b$ and a constant offset $\mathbf{F}_b$, then $\mathbf{C}_b = \mathbf{F}_b^\top$ restores the truth. I estimate $\mathbf{C}_b$ as the rotation closest to the mean of $\mathbf{R}_{b,k}^\top\hat{\mathbf{R}}_{b,k}$ over the people k of 200 random training images, never the holdout; the script selects every bone whose median error exceeds 90° with a constant offset. For Base it selected neck02 (178.8°), head (179.2°), metacarpal1.L, the base of the left index finger (179.8°), and finger4-1.R, a bone of the right ring finger (179.7°). The product comes before the helper mask, the kinematics and the losses, so training,

evaluation and demos all see corrected rotations. The constants are saved with the checkpoint and reapplied on loading; the correction adds no parameters. A constant suffices because the HPH runs once from constant start values and never sees its own output.

Why a loss cannot do it. The L1 distance over matrix entries has a barrier along a twist: at 135° it is larger than at 180° (4.83 against 4.00), so a partial turn back raises the loss, and neck02 and head would have to turn back together. A pilot with a geodesic loss, which has no barrier, moved neck02 only from 175.6° to 175.0°, and the neck boost cannot undo a half-turn either.

Limits. With the correction and no training the global head direction gets worse (12.0° to 19.1°): the head bone's offset varies between people, so one constant per bone leaves the link between the two bones slightly wrong. Final therefore fine-tunes with the correction in place; a two-sided correction was not implemented. The origin of the flip is unknown: the singular 6D start value would give a tilt, which fits metacarpal1.L but not neck02's twist, and another ViT-L model with fresh heads flipped arm and hand bones instead.

4.6. Training with real images

The mix. A weighted sampler draws on average 30% of each batch from 3DPW-train converted to Anny (Chapter 3; corrected labels, 5,882 images, 7,584 people) and 70% from AnnyOne. With about 1.2 labelled people per 3DPW image against 2.73 in AnnyOne, roughly 16% of the supervised people are real (an estimate, not a count). After each epoch 300 3DPW-validation images are evaluated; the test split is never used for training. The mix answers requirement 2a.

The hand mask. Every person carries a flag, 1 for AnnyOne and 0 for 3DPW. The converter fits only the root and the bones that carry a mapped SMPL joint, so every finger, thumb and metacarpal bone of a 3DPW label is at rest. With `--mask_3dpw_hands 1` the rotation and global-rotation terms skip these 38 bones for 3DPW people, and the fingers are learned from AnnyOne only. The hand mask removes the rotation terms only; the mesh terms still carry the rest-pose hands.

The shape mask, tested. The converter fits height and proportions, takes gender from the 3DPW metadata, and leaves 8 phenotypes at constants: age 0.55, and muscle, weight, cup size, firmness and the three ethnicity values at 0.5. With `--mask_3dpw_shape 1` the shape term ignores these constants for 3DPW people; gender, height and proportions stay supervised. The mesh terms still carry the neutral body, which is also what the 3DPW test PVE compares against. The partial variant (`--mask_3dpw_shape_names`) masks only the 7 values at 0.5 and keeps age supervised. Final-masked uses the full mask;

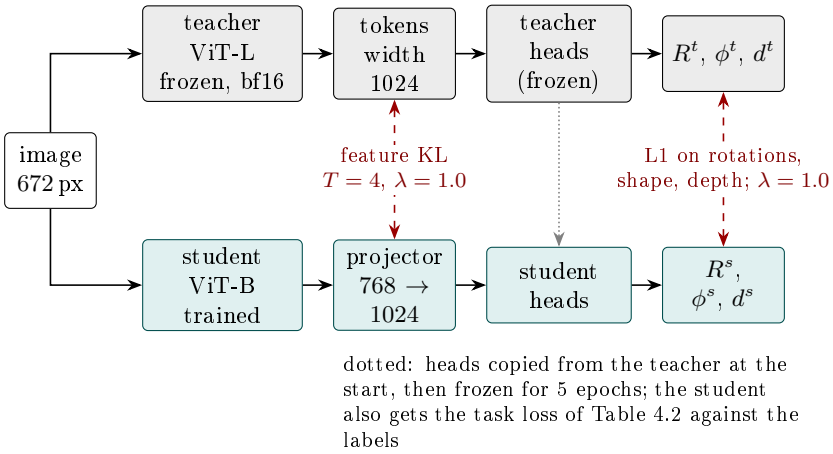


Figure 4.4.: *Distillation of the Student. The frozen ViT-L teacher (Final, backbone in bf16) and the ViT-B student see the same image. A projector maps the student's tokens from width 768 to 1024, and heads copied from the teacher read them. The feature loss compares the projected tokens with the teacher's ($T = 4$); the output loss compares rotations, phenotypes and depth of the same people. Shaded boxes train.*

Final-partial-mask, with the partial one, had not finished when this chapter was written. I kept the unmasked model as Final and teacher: the masked one regained body variety but lost the gender gain and part of the child gain on real photos (Chapter 6).

4.7. Knowledge distillation

A frozen teacher supervises a smaller student on the same image, on top of the task loss (Figure 4.4). Three mechanisms carry over from the first students to the final one.

Feature distillation. With teacher tokens $\mathbf{z}_n^t$ and the tokens $\mathbf{z}_n^s$ that the student's heads see, both of width C_t,

$$\mathcal{L}_{\text{kd}} = T^2 \cdot \frac{1}{N} \sum_{n=1}^{N} \text{KL}\left(\sigma\left(\frac{\mathbf{z}_n^t}{T}\right) \,\middle\|\, \sigma\left(\frac{\mathbf{z}_n^s}{T}\right)\right), \tag{4.3}$$

where σ is the softmax over the channels of one token, T the temperature [HVD15], and T^2 keeps the gradient size comparable across temperatures. The loss sits between backbone and heads, so the student copies the teacher's representation, not only its predictions, as in FitNets [Rom+15].

Head transfer and output distillation. Teacher and student are the same model class, so the student copies every head tensor whose name and shape match. Both run on the true person locations during training, so person i of

one is person i of the other without matching, and

$$\mathcal{L}_{\text{kd-out}} = \alpha_{\text{rot}} \left\| \mathbf{R}^s - \mathbf{R}^t \right\|_1 + \alpha_{\text{shape}} \left\| \boldsymbol{\phi}^s - \boldsymbol{\phi}^t \right\|_1 + \alpha_{\text{dist}} \left| d^s - d^t \right| \tag{4.4}$$

pulls the student's outputs toward the teacher's detached outputs, with the task weights of Table 4.2.

Warm-up freeze and projector-fed heads. Copied heads that read through random input layers give confident wrong outputs, and their gradients damage the copied weights before the input layers adapt; the copied tensors therefore stay frozen for the first epochs. The final design removes the random input layers: a linear projector maps the student's tokens to the teacher's width once, every head is built at that width (`--head_dim`), all head tensors transfer, and Equation (4.3) trains the projector toward exactly the input the heads expect. The deployed student is backbone, projector and heads. TF32 products and the teacher's backbone in bfloat16 (tokens cast back to fp32) cut a ViT-S student's step from 1.24 to 0.47 s on an A100, with unchanged evaluation numbers.

4.7.1. History: the ViT-S students

The first answer to the size requirement was a line of ViT-S students, Student-S/1 to Student-S/final (the number after the slash is the old version number, Table A.1), trained before both fixes and the label correction; their teacher Base reached 74.3 mm holdout PVE in setting V0. Feature distillation alone (Student-S/1, Student-S/2) left the student at 193.4 mm, because nothing supervised its random heads. Copied heads (Student-S/3) kept their 13 width-dependent input tensors random and needed about 20 epochs to recover. A backbone trained for mesh recovery (Student-S/4) gave the largest gain (193 to 156 mm), and projector-fed heads (Student-S/6) and longer training at twice the batch (Student-S/6c) reached 114.5 mm, still improving: the student was limited by the images it had seen, not by its 37 M parameters. Real images (Student-S/mixed, Student-S/final) did not lower the holdout error. These numbers are V0 and their 3DPW numbers used the retired labels; Chapter 6 keeps the tables as history.

4.7.2. The ViT-B student of Final

Recipe. The Student follows the Student-S/6 design with a ViT-B backbone and the corrected training of this chapter (Table 4.4). Its backbone starts from NAVER's Multi-HMR ViT-B, trained for mesh recovery, and trains fully; the copied heads stay frozen for 5 epochs; the targets are exactly the teacher's. The job script's header mentions colour jitter, but no code in the repository reads such a setting, so I claim no augmentation. The copied heads output the half-turn that the teacher's bone correction undoes, so the student trains with that correction; afterwards its own is estimated on 200 training images (Section 4.5) and used in every evaluation. Its heads keep the teacher's width and are a larger share of the student.

A. Recipe		
	Final (the teacher)	Student
Backbone	ViT-L/14 (width 1024) from Base; last 4 of 24 blocks trained; frozen in bf16 when teaching	ViT-B/14 (width 768) from NAVER's HMR-trained Multi-HMR ViT-B; fully trained
Heads	HPH at width 1024 (14.01 M)	copied from the teacher, frozen 5 epochs, fed by a projector 768→1024 (0.79 M)
Params / GFLOPs / ms	318.38 M / 1,927.4 / 110	101.38 M / 603.3 / 51.4 (3.14× / 3.19× / 2.14× less)
Schedule	10 epochs × 1,000 iterations × batch 4 = 40,000 images	30 epochs × 1,000 iterations × batch 8 = 240,000 images
Learning rate	10^{-5}, constant	3×10^{-5}, halved every 12 epochs
Targets	head anchor, depth weight 5, 30% 3DPW (v23), hand mask	identical
Distillation	–	feature KL $\lambda = 1.0$, $T = 4$; output L1 $\lambda = 1.0$
Bone correction	four bones, estimated on Base	teacher's in training; own one afterwards
B. The 95% bars: student error ≤ reference error / 0.95		
	against Final	against Multi-HMR-Anny
AnnyOne holdout PVE (V1)	73.7 → **77.6**	60.3 → 63.5 (likely in-sample)
AnnyOne holdout MPJPE (V1)	**[TODO: Final V1 MPJPE, re-measured in the student job]**	63.3 → 66.6
3DPW test PVE (v23)	89.0 → **93.7**	126.9 → 133.6
3DPW test MPJPE (v23)	88.3 → 92.9	129.4 → 136.2

Table 4.4.: *The Student (ViT-B) next to its teacher Final. Part A: recipe and measured cost. Part B: the error bars of the 95% rule in mm; the job computes the bars from the teacher re-measured in the same job, so these values are for orientation. Cost at 672 px, one person; ms on an A40, batch 1. The rule also asks for fewer parameters, fewer FLOPs and a faster forward pass. Read as "error at most 1.05 times the teacher's", the bars against Final would be 77.4 and 93.4 mm. The Student's own errors are in Chapter 6.*

The 95% rule. Requirement 1b asks for a smaller model within 95% of the large one. I read it as: student error at most the teacher's error divided by 0.95, for PVE and MPJPE on both the AnnyOne holdout (setting V1) and the 3DPW test set, with fewer parameters, fewer FLOPs and a faster forward pass. Against Final the bars are 77.6 mm holdout PVE and 93.7 mm 3DPW test PVE; the job re-measures the teacher and computes the bars from those numbers. Against Multi-HMR-Anny they would be 63.5 and 133.6 mm. NAVER's model was trained on the AnnyOne release, which includes the 100 images we hold out, so its 60.3 mm is most likely an in-sample number; ours are held out. A student of a 73.7 mm teacher cannot be expected to reach 63.5 mm on AnnyOne, while the 3DPW bar of 133.6 mm lies above the teacher's own error. Which reference counts is the supervisor's decision; Chapter 6 reports the student against both.

4.8. Evaluation protocol

Table 4.5 lists the metrics. Errors are means over matched people: a greedy 2D matcher pairs each labelled person with at most one detection, and recall is the share of labelled people that got one.

AnnyOne holdout. The last 100 AnnyOne images (indices 579,860 to 579,959), never trained on and without children (Chapter 3). The headline setting V1 masks the helper bones, as the demos do; V0 does not and reads 3.8 mm higher (the draft's headline was V0). A trace script adds placement metrics from the model's own depth and every bone's rotation error.

3DPW test. All 8,190 test images (11,854 people) of the corrected labels, with the same script for every model, in setting V0. MPJPE averages all 163 Anny joints and the conversion itself errs by about 14 mm (Chapter 3), so these numbers are not comparable with published 3DPW numbers. 3DPW labels only the actors; a correct detection of a bystander counts against precision.

Relative Human and demo images. Relative Human gives real photos with an age group, a gender and a box per person, but no 3D body. Its validation split (635 images, 2,130 people) serves for evaluation only: boxes from the projected vertices, one-to-one matching at IoU ≥ 0.3, child-shaped meaning a predicted age below 0.45, detected people only. The 4 demo photos and 8 3DPW test frames are rendered with the fingers at rest (`--relax_hands 1`) and each body placed from its detection box, which hides depth errors; they are judged by eye, with no metric.

One run, no seed. The training code sets no random seed, and every model was trained once. Two identical 2-epoch runs differed by 7.6 mm PVE, 1.9° global head direction and 0.8 recall points, so one run resolves about 10 mm, 3° and 2 recall points; smaller gaps are within run-to-run variation. Restarting

Metric	What it measures	Ideal value	Blind to
PVE (mm)	mean distance between predicted and true mesh vertices after both pelvises are put at the same point	0	placement, depth, drawn size
PA-PVE (mm)	PVE after the best rotation, scale and shift (Procrustes)	0	placement, size, global orientation
MPJPE (mm)	PVE on the 163 Anny joints instead of the vertices, pelvis-aligned	0	as PVE; not comparable with published 3DPW numbers
Recall (%)	share of labelled people matched by a detection	100	false detections
Precision (%)	share of detections matched to a labelled person	100	unlabelled bystanders count as errors
Root error (mm)	3D distance between predicted and true pelvis joint, no alignment	0	pose and shape
Size ratio	predicted head-to-pelvis distance in the image over the true one	1	3D body size
Depth slope / intercept	line fitted to predicted against true pelvis depth	1 / 0 m	per-person errors around the line
Bone angle error (deg)	angle of $\hat{R}_b^\top R_b$ for one bone's parent-relative rotation; about 180 means a flipped bone	0	bones that compensate each other
Global head direction (deg)	angle between predicted and true camera-frame head orientation	0	the chain below the head
Neck girth ratio	radius of the neck skin, predicted over true	1	neck length
$\mathrm{PVE}_{\mathrm{neck}}$ (mm)	vertex error over the neck skin, centred at neck01	0	the rest of the body
Child AUC	chance that a random child gets a lower predicted age than a random adult (Relative Human)	1 (0.5 = chance)	undetected people; how far ages are off
Gender accuracy (%)	predicted gender on the right side of 0.5 (Relative Human)	100 (50 = chance)	undetected people

Table 4.5.: *Metrics used in this thesis. All errors are means over matched people. On the AnnyOne holdout the headline setting is V1 (helper bones masked); V0 leaves them unmasked and reads 3.8 mm higher in PVE on the same people. 3DPW numbers are against our converted labels only (Chapter 3).*

a fine-tune of Base with nothing changed already costs about 2.4 mm PVE, so every fine-tune is compared with its own control, never with Base alone. The pelvis error has no measured noise floor, and no bootstrap intervals were computed.

5. Diagnostics: How Each Failure Was Isolated

Base showed three failures in its renders: thin, twisted necks, claw-like fingers and adult bodies on children. Before changing any training setting, I tested each belief about their causes on its own. This chapter reports those tests, including the explanations that turned out wrong; Chapter 6 reports the models built on them. Unless stated otherwise, every number is from Base on the 100-image AnnyOne holdout in setting V1 (helper bones masked, matched people; Chapter 4). Every comparison is one run per arm without a fixed seed and is read against the run-to-run variation of Section 5.7.

5.1. The three visible failures

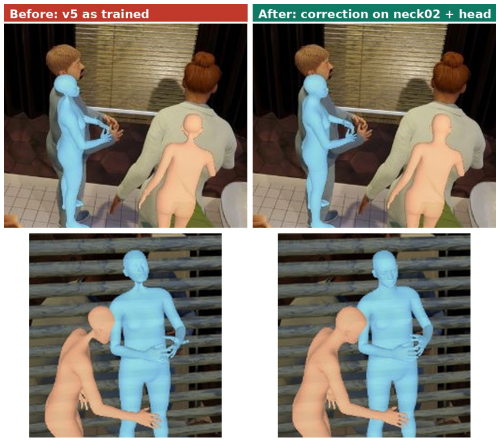


Figure 5.1.: *Holdout images 579862 (top) and 579863 (bottom). Left: Base as trained. Right: the same checkpoint with the output correction of Section 5.4, no training. Synthetic, bodies at the model's own depth, so the depth bug of Section 5.3 still draws them too small. Necks judged by eye.*

Figure 5.1 shows the first failure as it was seen: in the left column Base draws each neck as a thin stalk pinched under the head, on holdout images it never trained on. The second failure was the fingers, which curled into claws; from then on every real-photo demo set them to rest pose. The third was children: a standing boy in one of our own photos received an adult body (Figure 6.9).

The draft gave each failure a cause without testing an alternative. For the neck it concluded that linear blend skinning stretches the skin between

Anny's stacked neck bones, "a property of the rig, not a training failure"; a later reading blamed weak supervision on a redundant four-bone chain. For the fingers the belief was resolution. For children it was that age had not been learned, and the evidence was five small people in our own photos predicted at adult ages of 0.57 to 0.67. This chapter shows that the neck explanation is wrong, that resolution explains little of the finger error, and that the child evidence was wrong. Looked at again, four of the five "small people" were adults who crouched, sat or were cut off by the frame: their skeletal height is measured on the posed body, and their height phenotype (0.47–0.56) gives an adult rest body. Only the boy was a child, and he received age 0.60 and height 0.48: an adult.

The figure shows neither fingers, children nor a real photo; the neck widths are judged by eye, and the number behind them is the neck girth of Table 5.4.

5.2. What is not the cause: six structural checks

check	sample	result
1 labels decode onto the people (no network)	manifest, 55 people	meshes sit on the people; 84.5 % of joints inside the image, 84.9 % inside the box (the rest is truncation); dataset camera right on 100 % of images
2 body model, age sweep	body model	skeletal height 0.53 m at age 0, 1.55 m at age 0.8; head/height 0.200 to 0.106
3 single-joint moves	body model	all 16 pass (96–100 % of moved vertices in the intended part)
4 gradients	4 holdout images	zero-gradient bones are exactly the 88 helpers; all 75 active bones and 11 shape outputs get gradient; two of three camera outputs get none
5 prediction vs. truth, shape swap	holdout, manifest	neck length 99.3 % of truth; true shape lowers the 10 manifest children's PVE 47.8 → 41.6 mm
6 capped overfit	40 manifest training images	training PVE −36 % (80.7 → 51.8 mm), holdout +45 %; two warnings (text)

Table 5.1.: *Six checks on Base before any training change. The manifest holds 40 training images (10 adults, 10 children, 10 with visible hands, 10 hard neck poses) and 15 holdout images without children; it is excluded from every later fine-tune. Check 6 trains a throw-away copy for 400 steps and reports V0.*

Table 5.1 lists the checks and Figure 5.2 shows the first. The joints missing from the image are also missing from the box, so they are truncation, not a decoding error. The body model makes a child with child proportions, and each single-joint move, neck and fingers included, moves the intended part. Gradients range from 598 at the root to 4.7 at the fingers; shape dominates the loss (25.7 %), translation and distance take 1.9 % and 0.6 %.

So the rig, the label decoding and the gradient path are not the cause; the draft's four neck tests (rest-pose render normal, all neck bones active, thin neck on synthetic images, proportions accurate) stay valid as eliminations of



(a) *holdout 579888, a close-up with hands* **(b)** *training image 462901, low camera*

Figure 5.2.: *Check 1 on two manifest images, holdout 579888 (a) and training image 462901 (b): each as image | ground-truth mesh | ground-truth joints and box. The AnnyOne label is decoded through the training body model and drawn with the dataset camera; no network is involved. By eye the mesh lies on the person, down to the spread fingers in (a) and the raised hand and the hand at the mouth in (b), and the joints fall on the body. In (a) the frame cuts the head and the legs, so part of the joints lie outside the image and the box: truncation, not a decoding error (the black corner is part of the dataset image). Over all 55 manifest people 84.5 % of the joints lie inside the image and 84.9 % inside the box. Synthetic images; two of the four rendered cases.*

the same kind. The cause had to be in what training asked of the network. The overfit run gave two warnings. The summed rotation loss barely moved while the images were memorised (44.6 → 40.8). And the translation and 2D terms rose (1.67 → 1.89, 58 → 75, first against last five epochs) while the pelvis-centred 3D terms fell: possibly a first sign of the depth bug. That evidence is weak, since each value is a single-batch print and the half-run averages fall instead (2.10 → 1.96, 69 → 63).

The checks are blind to two faults, and both were present: a target built at a different joint than the model's anchor, because the checks compare pelvis-centred quantities, and a bone flipped the same way in every person, because the rig was tested in isolation and bone errors were averaged by group. The holdout has no children, so every child number below is in-sample or from real photos.

5.3. The hidden depth bug

Table 5.2 is the placement section of the holdout trace. Base places every person too far (Figure 6.2), and the nearest are pushed farthest. The skeleton has the right height, so the body is drawn at 0.63 of its true size; its pelvis lands near the right pixel, its head does not. PVE is blind to all of it (correlation −0.03).

The code gave the cause. The model places each mesh by its head joint and reports the head as its position, but the training code built the translation, distance, detection-cell and offset targets at joint 0, the pelvis. A one-batch debug print showed predicted offsets of −2 to −19 cells where the legal range is ±0.5. The likely mechanism, not tested on its own: the detection and offset terms pull the anchor pixel toward the pelvis while the mesh hangs from it

quantity	truth	Base
pelvis depth, mean (sd)	1.92 m (0.49)	3.00 m (0.32)
fit of predicted on true depth	$1 \times \text{true} + 0$	$0.33 \times \text{true} + 2.36$ m
people placed farther than truth	–	100 %
corr. of true depth with depth error	–	−0.76
skeletal height (3D)	1.409 m	1.405 m ($r = 0.89$)
drawn size (pixels), pred./true	1	0.63
head / pelvis 2D error, median	0	46 / 14 px
corr. of PVE with depth error	–	−0.03

Table 5.2.: *Where Base places people. V1, 100-image AnnyOne holdout, all 222 matched people. Right metric size, wrong distance, so drawn too small; PVE does not see it.*

run (depth loss)	root (m)	size	slope	icpt. (m)	head px	pelvis px	PVE	recall
Base	1.21	0.63	0.33	2.36	64.5	23.4	70.5	90.2
2-epoch fine-tunes of Base, AnnyOne only								
control ×1	1.11	0.66	0.33	2.28	58.1	23.3	72.3	89.8
fix ×1	0.69	0.78	0.30	1.93	9.6	37.3	77.3	87.8
fix ×5	0.66	0.80	0.44	1.62	8.5	38.1	76.8	87.8
fix ×20	0.53	0.87	0.43	1.44	8.8	31.4	75.7	87.4
10-epoch fine-tunes of Base, AnnyOne only								
control ×1	1.44	0.59	0.34	2.59	64.0	24.3	72.2	89.4
fix ×1	0.73	0.78	0.30	1.98	7.6	38.5	75.5	87.0
fix ×5	0.54	0.84	0.38	1.59	7.2	33.1	74.8	87.8
10-epoch fine-tunes of Base with 30% 3DPW (v23 labels)								
Control ×1	1.40	0.59	0.28	2.66	67.1	26.7	77.4	89.0
Final ×5	0.73	0.78	0.42	1.75	7.6	41.1	73.7	87.0
other checkpoints trained with pelvis targets								
Mixed-L	1.39	0.59	0.31	2.60	–	25.1	75.0	88.6
Fresh-heads-L	1.49	0.57	0.28	2.71	–	24.0	74.1	89.0

Table 5.3.: *Placement before and after building the training targets at the head joint. V1, 100-image AnnyOne holdout, matched people. Root error: absolute 3D distance of the pelvis; size: predicted over true head-to-pelvis distance in pixels (1 is right); slope and intercept: linear fit of predicted against true pelvis depth (1 and 0 m are right); head and pelvis px: 2D error of those joints; PVE in mm, recall in %. Control rows build the targets at the pelvis, fix rows at the head; ×1/5/20: weight of the distance and translation terms. Final also has the bone correction. One run per row, no seed. No control was run at ×5 or ×20; each fix row is read against the control of the same length.*

by the head, so every projected joint is displaced. The 2D terms, the only strong terms that see depth (the 3D terms are pelvis-centred), can shrink that displacement only by drawing the body smaller, that is, farther away; the distance and translation terms, 2.5 % of the loss, could not resist. Near people have the largest displacement and were pushed farthest. Building the targets at the head (Chapter 4) is the fix, and Table 5.3 tests it against same-schedule controls.

All five fix runs reach a root error of 0.53–0.73 m, both controls stay at 1.11–1.44 m; the drawn size rises from 0.59–0.66 to 0.78–0.87 and the head lands within 7–10 instead of 58–67 pixels. With real images and the bone correction in training, Final keeps it: 0.73 against 1.40 m for Control, size 0.78 against 0.59. Every fix run lies below every control, far outside run-to-run variation. Two other checkpoints trained with pelvis targets carry the bug too. It stayed hidden because PVE aligns the pelvis first and the real-photo demos place each body from its detection box; only the synthetic demo and the trace panels used the model's own depth.

The table does not show correct depth. The fix moved the offset and the body size, not the slope (Final 0.42; 0.30 with the switch alone, 0.38–0.44 with a stronger depth loss); predictions are still pulled toward 2.5–2.9 m for a reason I have not found. The pelvis 2D error rose from about 23 to 31–41 pixels, likely because the head is now pinned while the body is still too small (not checked). Every fix run is worse than its control by +2.6 to +4.9 mm PVE and −1.6 to −2.4 recall points: each gap within run-to-run variation, all in the same direction, so a likely cost. By the success rule written beforehand (PVE $\leq$ 71.5 mm, recall $\geq$ 89.2 %) every run failed, the controls included.

5.4. The neck: a per-bone dump

A per-bone dump of the neck chain found the fault (Table 5.4, first column). neck02 is 175.6 deg wrong, a half-turn about its own axis (twist 175.7, swing 3.3 deg), and the head bone 177.9 deg. neck03 is right relative to its parent (5.7 deg) but 173.3 deg off in the camera frame. The two half-turns cancel, so the head points the right way (12.0 deg) and PVE never showed it; the skin between is twisted thin, girth 0.46 of the truth and neck-skin error 27.3 mm. The earlier group means (62.7 and 89.5 deg) hid the flip. The flip is in all eight checkpoints of the Base line (L-stage 2 to L-stage 4, Base, Mixed-L, the 10-epoch depth pilot and two geodesic-loss pilots; neck02 175.0–176.0 deg, girth 0.46–0.47), from the first stage on, before the helper mask, the neck boost and the shape weight of 20. The depth pilot keeps it, so depth is not the neck's cause. Fresh-heads-L, trained from the same pretrained backbone with fresh heads, has a normal neck (girth 1.00) but flips other bones: a compensated pair in the left arm and three metacarpals.

This refutes the draft. The rig passes all 16 single-joint moves, and the normal rest-pose render fits the flip just as well, because rest pose sets every rotation to the identity and removes both half-turns; only the dump separates the two explanations. A loss cannot undo the flip: the neck boost (renormalised

	Base	corr., no train.	corr. + 2-ep. FT	2-ep. control	Control 10 ep.	Final 10 ep.
neck02 (deg)	175.6	4.5	4.7	175.7	175.8	4.8
head bone (deg)	177.9	12.4	11.2	177.9	177.7	8.9
metacarpal1.L (deg)	180.0	0.0	0.0	–	180.0	0.0
finger4-1.R (deg)	n.d.	10.2	10.3	–	180.0	9.6
neck girth, pred./true	0.461	1.004	0.997	0.461	0.464	0.996
PVE_{neck} (mm)	27.3	10.8	10.7	27.9	27.7	11.0
body PVE (mm)	70.5	71.3	68.8	76.9	77.4	73.7
PA-PVE (mm)	53.3	52.2	50.9	55.2	56.4	50.3
recall (%)	90.2	90.2	89.4	91.1	89.0	87.0
head direction (deg)	12.0	19.1	14.4	13.8	13.0	13.1
fingers (deg)	17.3	8.2	8.3	17.4	17.4	8.3

Table 5.4.: *The two half-turned neck bones and their correction. V1, 100-image AnnyOne holdout, matched people. Bone errors are local (relative to the parent bone) medians; about 180 deg means the bone is turned half a turn. Girth: radius of the neck skin, predicted over true. PVE_{neck}: vertex error over the 171 neck-skin vertices, centred at neck01. Corr.: one constant rotation per flipped bone (neck02, head, metacarpal1.L, finger4-1.R), estimated on 200 training images. FT: fine-tune. The 2-epoch columns are AnnyOne-only fine-tunes of Base; Control and Final are the 10-epoch runs with 30% 3DPW, Final also with the depth fix. n.d.: not dumped on the holdout. One run per column, no seed; PVE gaps below about 10 mm are within run-to-run variation (Table 5.7).*

to ×2.41 for the boosted bones, ×0.60 for the rest) left it unchanged, and a geodesic loss moved neck02 only from 175.6 to 175.0 deg, likely because neck02 and head must turn back together and any partial turn raises the loss. So I corrected the output: one constant rotation per flipped bone, $R_b \leftarrow R_b C_b$, estimated on 200 training images (Chapter 4); the constants are 178.8, 179.2, 179.8 and 179.7 deg for neck02, head, metacarpal1.L and finger4-1.R.

Without training the correction turns neck02 back to 4.5 deg and restores girth (1.004) and neck-skin error (10.8 mm), but the head direction worsens from 12.0 to 19.1 deg, above the 15-degree limit set beforehand. The likely reason is the compensated chain: the head's error differs per person, so one constant cannot undo it. After a 2-epoch fine-tune with the correction the head direction is 14.4 deg against 13.8 deg for the control, which keeps the flip. Body PVE is 70.5, 71.3 and 68.8 against 76.9 mm for the control, a gap within variation (the control also sits above six earlier arms): no sign of a PVE cost, not a gain. Final keeps the correction through 10 epochs with real images (neck02 4.8 deg, girth 0.996, head direction 13.1 against 13.0 deg for Control). Figure D.2 shows two more holdout pairs.

Not shown: why the bones flipped. The 6D start value of the rotation output is singular and starts a bone tilted about its local x axis, which fits the tilted metacarpal1.L but not neck02's twist; the origin is unknown, and the correction, which adds no parameters, says nothing about it. A two-sided correction for compensated chains was not implemented. Real photos have no

neck ground truth; their pairs (Figure 6.3) are judged by eye and also differ by the fine-tune.

5.5. The fingers

(a) finger rotation error on the holdout setting	fingers (deg)	reference (deg)
Base, predicted fingers	17.3	–
Base, fingers set to rest pose	15.1	–
Base, test-time person crop (PVE 124.8 mm)	17.2	17.3 (paired)
region-normalised rotation loss, 2 ep.	17.5	17.4 (control)
Base + bone correction, no training	8.2	17.3 (Base)
Base + bone correction, 2-ep. fine-tune	8.3	17.4 (control)
Final (10 ep., 30% 3DPW)	8.3	17.4 (Control)
Fresh-heads-L, three metacarpals corrected	17.8	32.0 (uncorr.)
(b) hand size in tokens (14-px patches of the 672-px input)	median length	under 2 tokens
full image, all hands	2.25	41%
person crop at test time	4.2	0%
children's hands, full image	0.7	80%

Table 5.5.: *Fingers. (a) Finger rotation error, V1, 100-image AnnyOne holdout, matched people; the reference is the matching run without the change. Rest pose keeps the metacarpals predicted, so the 15.1 deg still contains the flipped metacarpal1.L. (b) Hand length in input tokens. One run per row, no seed.*

In Table 5.5 predicted fingers are 17.3 deg off and fingers at rest 15.1 deg: the prediction was worse than none. Hands are small, 2.25 tokens long at the median and under two tokens for 41 %. A person crop at test time raises the median to 4.2 tokens, but the finger error stays at 17.2 deg while body PVE nearly doubles (124.8 mm), since the model never saw such a camera. A region-normalised rotation loss gave 17.5 against 17.4 deg. The per-bone search on training images then found two hand bones half-turned, metacarpal1.L (180 deg) and finger4-1.R (178.8 deg); correcting them gives 8.2 deg without training, 8.3 after the fine-tune, and 8.3 for Final against 17.4 deg for Control.

About half the finger error was two flipped bones, not resolution. Resolution was tested only by test-time zoom, which did not help; a hand-crop branch was never tried. The region-normalised loss cut the hands' share of the rotation loss from about 70 % (with the boost) to 33 %, a rebalance, not an upweighting. The fingers also have the smallest gradients of any bone group; that they are under-trained is a guess I did not test.

Until the last demo every real-photo render set the fingers to rest pose; predicted fingers on real images, judged by eye, are in Figure 6.4 of Chapter 6. Rest pose keeps the metacarpals, so the 15.1 deg contains the metacarpal flip; a rest-pose floor with the correction, and the split of the 9.1-degree drop between the two bones, were not measured.

5.6. Children and body shape

(a) predicted minus true phenotype, 10 holdout images (draft measurement)

phenotype	true mean	true std	weight 1	weight 20 (Base)	reading
age	0.553	0.036	+0.199	−0.002	near the mode
height	0.579	0.026	−0.154	−0.026	near the mode
proportions	0.763	0.070	−0.017	−0.001	near the mode
firmness	0.757	0.039	−0.167	−0.190	unchanged
gender	0.694	–	−0.042	−0.063	≈
weight	0.382	–	+0.056	+0.064	≈

(b) Base on the full holdout and in the young tail

holdout, all 222 people (nobody younger than 0.513)	age bias −0.015 (abs. 0.037); gender −0.008 (0.068)
holdout, true age 0.6–0.8 (n = 53, mean 0.67)	predicted age 0.57
50 youngest training people (in-sample, true age 0.015)	predicted age 0.10; height −0.25
training medians (AnnyOne)	gender 0.588, age 0.601, height 0.500

Table 5.6.: *Shape head. (a) The draft's comparison of the stage before Base (shape loss weight 1) with Base (weight 20) on the same 10 AnnyOne holdout images, in phenotype units* $[0, 1]$*; it holds near the mode of the data only. (b) Base, V1, on the full 100-image holdout and on the 50 youngest training people (in-sample: the holdout has no children). Predictions in both tails move toward the training median.*

Table 5.6(a) is the draft's shape-weight test on 10 holdout images: raising the shape weight from 1 to 20 moved the age bias from +0.199 to −0.002 and the height bias from −0.154 to −0.026. Part (b) shows that this holds near the mode only. On all holdout people the age bias is −0.015, but nobody there is younger than 0.513. In both tails predictions move toward the training median: true ages 0.6–0.8 come out at 0.57, and the 50 youngest training people (true 0.015) at 0.10. On real photos Base reads children as short adults. On Relative Human (Table 6.2) kids get a mean age of 0.61, babies 0.62 and adults 0.63; the child-versus-adult AUC is 0.61, and none of 89 kids and 1 of 38 babies fall below the child threshold 0.45. The smallness goes into height instead (adults 0.51, kids 0.45, babies 0.39). Gender carries no signal (men 0.535, women 0.551, accuracy 46.5 %). On our demo photos the medians of all 299 shape lines (gender 0.59, age 0.61, height 0.50) equal the training medians (0.588, 0.601, 0.500). Their narrow age spread (standard deviation 0.050 against 0.161) is no evidence, since the lines are about 30 adults counted several times; only the gender spread is (14 % of lines below 0.5 though men dominate).

Children and female-looking men are one failure: with weak image cues the shape head returns the training median. Two mechanisms fit, and the evidence cannot rank them: the gap between synthetic and real images, or the loss. An L1 on sigmoid outputs regresses to the median, the sigmoid's gradient $s(1-s)$ is 0.25 at $s = 0.5$ but 0.015 at $s = 0.015$, and children are rare (2.53 % of training people below age 0.3, about 2.1 % of supervised people per batch);

the 3DPW labels pin age at 0.55. Anny explains why the failure looks like a short adult. Age blends five shapes at anchors $-1/3$, 0, $1/3$, $2/3$, 1 (newborn to old), and the child shape appears below about 0.4–0.5. Its cue is narrow shoulders relative to the head: shoulder over head width is 2.24 at age 0.1 and 3.00 at 0.6. A low height at age 0.6 gives a short, broad-shouldered adult (at least 1.22 m), and proportions barely matter (leg over stature -0.016); only a lower age gives a child.

Not shown: children on held-out synthetic images (there are none), which mechanism dominates, and ages in years, which I never quote because the code uses three age scales that disagree. Part (a) has $n = 10$. Final improves on Relative Human (Chapter 6): a measured improvement, not a solution.

5.7. Pilot noise and restart cost

quantity	measured	one run resolves
two identical 2-epoch controls: PVE	76.3 vs. 68.7 mm (7.6 mm)	about 10 mm
global head direction	13.6 vs. 11.7 deg (1.9 deg)	about 3 deg
recall	0.8 points apart	about 2 points
six 2-epoch arms of the Base recipe	PVE 68.7–76.3 mm; recall 88.6–90.2 %; head 11.3–13.7 deg	
restart cost (mean of controls vs. Base)	72.9 vs. 70.5 mm (about 2.4 mm)	read pilots against their own control
root error	no measured floor	every fix run (0.53–0.73 m) below every control (1.11–1.44 m)

Table 5.7.: *Run-to-run variation of the 2-epoch pilots. V1, 100-image AnnyOne holdout, matched people. All arms fine-tune Base for 2 epochs at learning rate 10^{-5}; the training code sets no random seed, so two runs of the same configuration differ. The restart cost is what two more epochs of fine-tuning Base cost with nothing changed.*

Two 2-epoch runs of one configuration differed by 7.6 mm PVE (76.3 against 68.7), 1.9 deg head direction and 0.8 recall points (Table 5.7); six arms of the Base recipe span 68.7–76.3 mm. Restarting a fine-tune of Base costs about 2.4 mm (controls 72.9 against 70.5 mm).

One pilot therefore resolves about 10 mm PVE, 3 deg and 2 recall points. That suffices for every change this chapter relies on (root error, bone angles 175 to 5 deg, girth 0.46 to 1.00, fingers 17.3 to 8.3 deg), but not for a PVE change of a few millimetres, so the costs of the depth fix and the neck pilot's PVE stay unresolved. Because of the restart cost a pilot is read against its own control, never against Base alone.

There are no bootstrap intervals, the root error has no measured floor, and the training code sets no seed, so a repeated run is a repeat, not a seeded run.

5.8. What did not work

change	result	reading
age-balanced sampler (children per batch 2.1 → 9.7–11.1 %)	10 manifest children PVE 48.5 vs. 53.1 mm; baby age floor 0.100 → 0.102	inconclusive ($n = 10$, in-sample)
global head orientation loss	head direction 11.3 vs. 11.7 and 13.6 deg (identical controls)	within variation
region-normalised hand loss	fingers 17.5 vs. 17.4 deg	no effect
geodesic rotation loss	neck02 175.6 → 175.0, head 177.9 → 177.1 deg	no movement
neck and hand boost ×4	×2.41 / ×0.60 after renormalisation; neck unchanged	no effect
depth loss ×20	root 0.53 m in 2 epochs vs. 0.54 m for ×5 in 10; slope 0.43 vs. 0.44 (×5, 2 epochs)	no gain over ×5

Table 5.8.: *Changes that did not fix a failure. V1, 100-image AnnyOne holdout; 2-epoch fine-tunes of Base unless stated, one run each, each read against its own control.*

In Table 5.8, the sampler's 4.6 mm on 10 in-sample children is not readable, and the global head loss differs from two identical controls by less than they differ from each other. Each change altered the loss or the sampling around a fault the loss could not see: a half-turned bone, a target at the wrong joint, a shape head pulled to the median. The two fixes that worked, head-joint targets and the bone correction, came from the diagnostics.

None of this shows that an idea cannot work: the sampler is unproven after 8,000 images, and larger hand crops and a 10-epoch run at weight 20 were never tried.

6. Experiments and Results

This chapter reports what the models built on the diagnostics of Chapter 5 achieve. It follows the fixes in the order they were made: depth, the bone correction and the fingers, then the final model against its control on synthetic and real test data, body shape on real photos, the shape-mask option, the comparison with NAVER's own Anny model, the distilled student, and last the history of the earlier students. Each section opens with its figure or table, reads it, interprets it and says what it does not show.

6.1. Evaluation sets and settings

Four sets are used, all described in Chapter 3 and scored as in Section 4.8. The AnnyOne holdout holds the last 100 AnnyOne images (222 people, no children), never trained on; its headline setting V1 masks the helper bones and matches people in 2D, and a trace script adds the placement metrics (root error, depth fit, drawn size) with each body at the model's own predicted depth. The 3DPW test set is all 8,190 test images of the corrected labels, scored in setting V0 with the same script for every model; its numbers hold only against our converted labels, which carry a conversion error of about 14 mm, and are not comparable with published 3DPW numbers. The Relative Human validation split gives real photos with an age group, a gender and a box per person, but no 3D body. The 12 demo images (4 photos and 8 3DPW test frames) have no labels at all and are judged by eye; their bodies are placed from the detection box, so they hide depth errors.

The models are those of Table 4.3. The central comparison is Final against Control: the two runs share the start point (Base), the schedule, the real-image share and the hand mask, and differ only by the depth fix and the bone correction. Base against Control shows what the real images and the restart of training do. Every model was trained once, without a fixed seed. Two identical runs differed by 7.6 mm PVE, 1.9 deg head direction and about 2 recall points (Section 5.7); I call gaps below these "within run-to-run variation" and compare every fine-tune with its own control, because restarting a fine-tune of Base already costs about 2.4 mm PVE. Setting V0 reads about 3.8 mm higher than V1 on the same people; every caption names the setting.

6.2. Depth: the anchor fix on the holdout

Figure 6.1 shows the depth bug and the fix on one image. Base draws the man far behind his true position and too small; the pilot with head-joint targets draws him close to his true size. Figure 6.2 shows the same for all people.

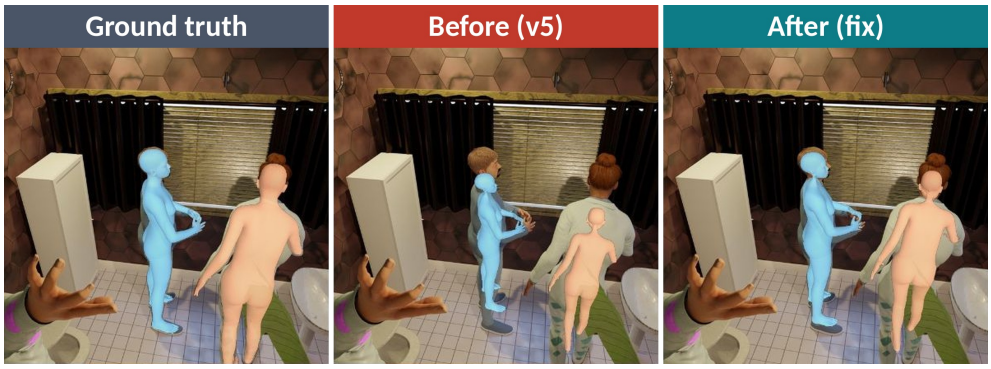


Figure 6.1.: *Holdout image 579862 rendered at each model's own predicted depth: ground truth, Base, and the depth pilot (head-joint targets, depth weight ×5, 10 epochs, AnnyOne only). The predicted body goes from 0.76 to 0.95 of its true height. Synthetic image; chosen because no person is missed, so a good case, not a typical one. Image 579860 is in Figure D.1.*

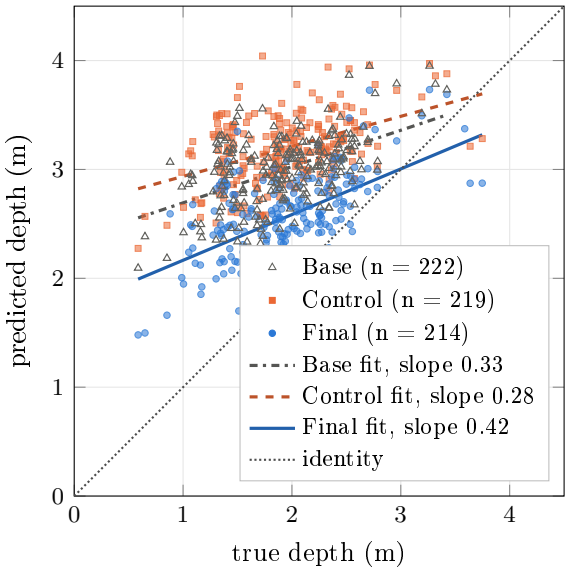


Figure 6.2.: *Predicted against true root depth for every matched person of the AnnyOne holdout (100 images, setting V1, each body at its own predicted depth): Base (222 people), Control (219) and Final (214). Lines: least-squares fits and the identity. Final's fit is* $0.42 \times true + 1.75\,m$*, Control's* $0.28 \times true + 2.66\,m$*, Base's* $0.33 \times true + 2.36\,m$*. One run per model; synthetic images. Statistics in* `data/plot_stats.md`*.*

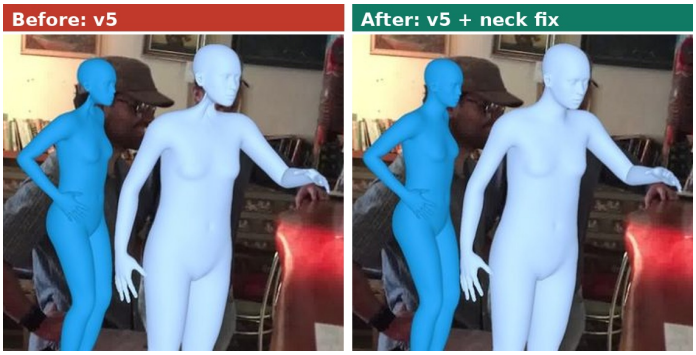


Figure 6.3.: *Real 3DPW test frame* `downtown_bar_00` *701: Base (left) and the neck pilot, fine-tuned 2 epochs with the correction (right); necks ringed. Bodies placed from the detector box; judged by eye, no metric. The two columns differ by the correction and by the 2-epoch fine-tune. A second frame is in Figure D.3.*

Nearly every point of Control lies above the identity line, 99.1 % of its people are placed too far, and its mean root error is 1,402 mm. Final pulls the cloud down: mean root error 728 mm (−48 %; Base 1,208), drawn size 0.78 against 0.59 (Base 0.63), and the correlation of true and predicted depth rises from 0.46 to 0.56. The head now lands within 7.6 pixels of its true position against 67.1 for Control (Table 6.1a). The pilots of Table 5.3 reached the same with AnnyOne only; the 10-epoch depth pilot even reached 544 mm.

The fix holds after ten epochs with real images in every batch and the bone correction on. The gap to Control is far outside run-to-run variation: every run with head targets, pilots included, lies below every run without them. What moved is the offset and the drawn size, not the slope. Final's fit crosses the identity at 3.01 m, so people about three metres away are placed about right, while near people are still placed too far (94.4 % of all people). Predicted depth is compressed toward the range of 2.5–2.9 m, as in every run of Chapter 5. The pelvis now lands farther from its pixel (41.1 against 26.7 pixels), likely because the head is pinned while the body is still drawn too small; I did not check this.

Not shown: correct near and far depth (the slope is 0.42, not 1), and why the slope does not move; measuring the depth distribution of the training batches would be the next test. Final's root error is higher than the depth pilot's; the real-image batches or the bone correction may cause this, which was not tested. Depth is measured on synthetic images only: the 3DPW evaluation aligns the pelvis, and the real-photo demos place each body from its box. Figure 6.1 shows the pilot, not Final.

6.3. The bone correction

Figure 6.3 shows the correction on a real photo that no model trained on: Base draws both necks as thin stalks pinched under the head, the corrected model draws them at normal width. The holdout pair of Figure 5.1 shows the same with the correction alone and no training, and Table 5.4 gives the numbers for

the correction without training and after a 2-epoch fine-tune. Final keeps the correction through ten epochs with real images (Table 6.1a): neck02 is 4.8 deg off against 175.8 for Control, the head bone 8.9 against 177.7, the neck girth is 0.996 of the truth against 0.464, and the neck-skin error 11.0 against 27.7 mm. The joint error of the neck region falls from 62.6 to 5.5 deg. The direction of the head, which got worse when the correction was applied without training (19.1 deg; neck pilot 14.4), is 13.1 deg in Final against 13.0 in Control and 12.0 in Base.

The half-turn of the two neck bones does not come back during training: the network learns on top of the correction instead of undoing it. The correction is one constant rotation per bone and adds no parameters; its origin is unknown, and it says nothing about why the bones flipped. Final's head direction equals Control's within run-to-run variation, so the fine-tune absorbed the head error that one constant could not remove. PVE shows no sign of a cost (Section 6.5).

Not shown: a neck number on real photos, which have no neck ground truth. The real frame of Figure 6.3 is the neck pilot, not Final, and its two columns also differ by the fine-tune; on the holdout the same fine-tune without the correction keeps girth 0.461 (Table 5.4), so the correction is the likely cause of the wider necks. Final's own real-photo renders are in Section 6.6. Flips that compensate each other in a chain, as in the left arm of a model trained from fresh heads (Section 5.4), need a two-sided correction, which I did not implement.

6.4. Fingers

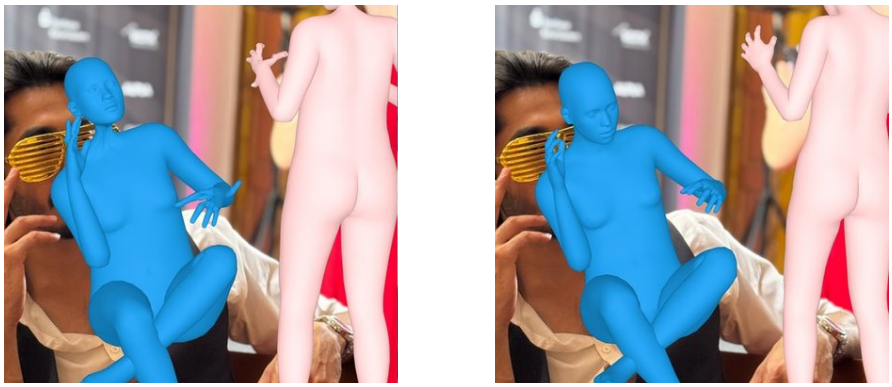
The finger numbers are in Table 5.5 of Chapter 5 and the last rows of Table 6.1a. Predicted fingers are 17.3 deg off in Base and 17.4 deg in Control, worse than fingers left at rest (15.1 deg). Final, with the two flipped hand bones corrected, reaches 8.3 deg; the correction alone, without training, gave 8.2.

About half of the finger error was two flipped bones, metacarpal1.L and finger4-1.R, and ten epochs of training keep the gain. The rest of the error is not explained. Low resolution is a candidate that was tested only by zooming in at test time, which did not help; real images add nothing to the fingers, because the hand mask removes the rotation terms of the 38 hand bones for 3DPW people (their labels hold rest-pose hands), and the mesh terms still carry the rest-pose hands.

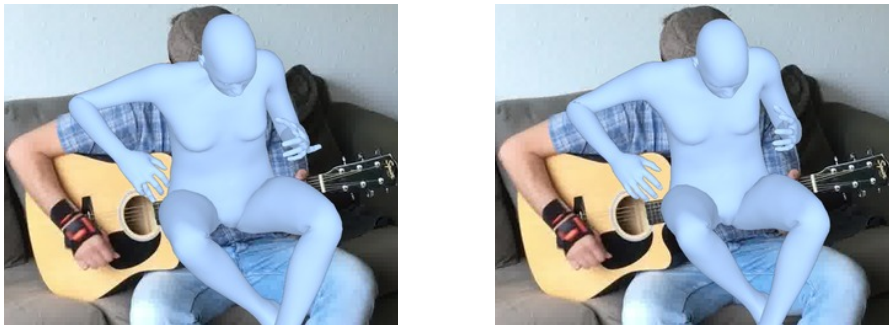
Predicted fingers on real images are shown in Figure 6.4; every earlier real-photo demo set them to rest pose. Judged by eye, the claws of Base are gone in Final: Base spreads the fingers stiffly and lets one finger of the left hand stick out from the others, as the flipped metacarpal1.L would, while Final closes and bends them, draped over an armrest, curled as if fretting, bent over a bar edge, in a way that reads as a hand. Where the hands are small, as in the tennis photo and the street frames of Figure D.6, the difference cannot be judged at print size.

Not shown: a finger number on real images. The finger number is an angle on the synthetic holdout only; the 3DPW labels carry rest-pose hands and the

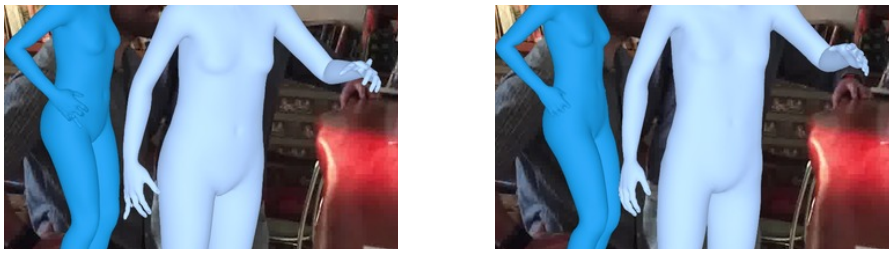
Base, predicted fingers | Final, predicted fingers



(a) party photo: the hand at the face and the hand over the armrest



(b) 3DPW test frame, guitar: strumming hand (left) and fretting hand (right)



(c) 3DPW test frame `downtown_bar`: the hand on the bar and the hanging hand

Figure 6.4.: *Predicted fingers (not set to rest pose) on three demo images, zoomed on the hands: the party photo and the 3DPW test frames guitar and* `downtown_bar`*; Base left, Final right. Demo setting: bodies placed from the detector box, so meshes sit beside their people. Base's renders still carry the two flipped hand bones, the left index-finger base (metacarpal1.L, which bends that finger backwards) and the right ring-finger base; Final has the correction. By eye, Base's fingers are splayed and stiff, with a finger of the left hand sticking out from the others, while Final's are closed and bent and look like hands: draped over the armrest in (a), curled like the fretting hand in (b), bent over the bar edge in (c). Neither model puts the fingers on the real ones. Three images, judged by eye; no finger metric exists for real images, and the two columns differ by the correction and by the fine-tune. All twelve montages, uncropped, are in Figure D.6.*

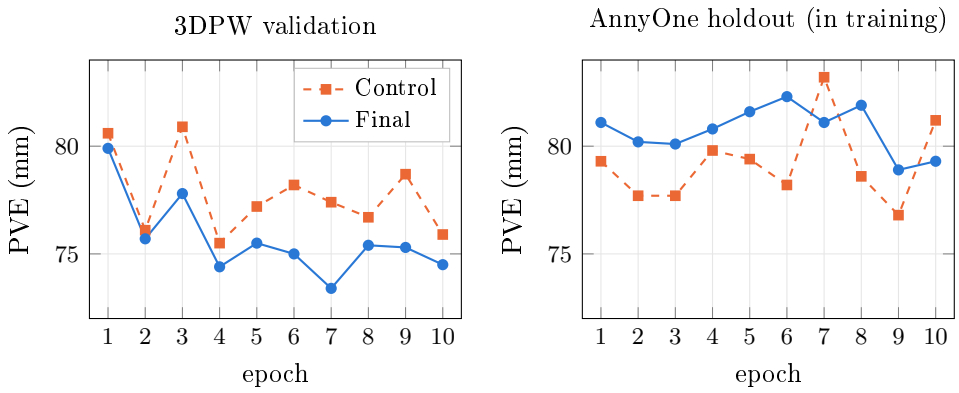


Figure 6.5.: *PVE per epoch during training, Final and Control (one run each, no seed). Left: 300 images of the corrected 3DPW validation labels. Right: the AnnyOne holdout as evaluated during training, setting V0 (helper bones unmasked), so higher than the V1 numbers of Table 6.1a. Values from the training logs.*

demo images have no labels, so the real-image evidence is three images by eye. Base's renders carry both hand flips, so the figure shows the correction and the fine-tune together, not either alone.

6.5. Final against Control and Base on AnnyOne

Table 6.1a puts Final next to Control, Base and the two pilots. Final has the lowest PA-PVE of every model, 50.3 mm against 56.4 for Control (−6.1 mm) and 53.3 for Base. Its PVE is 73.7 mm against 77.4 for Control, a gap of 3.7 mm within run-to-run variation; Base keeps the lowest PVE (70.5 mm), as it did against every fine-tune. Recall is 87.0 % against 89.0 % for Control (−2.0 points, at the limit of the variation) and 90.2 % for Base (−3.2). The root error, the neck and the fingers improve far beyond the variation (Sections 6.2–6.4). The curves of Figure 6.5 agree on 3DPW validation: Final is below Control at all ten epochs (mean 75.7 against 77.7 mm, last epoch 74.5 against 75.9). On the in-training AnnyOne holdout (V0) the order changes from epoch to epoch: Final is higher on average (80.7 against 79.2 mm) and lower at the last epoch (79.3 against 81.2). Figure 6.6 shows six holdout images: Base draws its bodies small and off their people, Final draws them close to the size and place of the ground truth.

The fixes cost nothing in body error that one run can resolve, and they bring the largest PA-PVE gain of the project. PVE aligns the pelvis, so it cannot see the depth gain, and the neck and the fingers are few vertices of a whole body; PA-PVE also removes rotation and scale and is the more sensitive of the two to pose and shape. The sign change of the in-training holdout curves is what a gap within run-to-run variation looks like. The recall cost against Base is larger than the variation and is a real cost. A likely reason is that the detection cell moved from the pelvis to the head, so people whose head is cut

	Base	Depth pilot	Neck pilot	Control	Final
PVE (mm)	70.5	74.8	68.8	77.4	73.7
PA-PVE (mm)	53.3	54.0	50.9	56.4	**50.3**
Recall (%)	90.2	87.8	89.4	89.0	87.0
Root error (mm)	1208	544	1429	1402	**728**
Depth slope	0.33	0.38	0.34	0.28	0.42
Depth intercept (m)	2.36	1.59	2.56	2.66	1.75
Size ratio	0.63	0.84	0.58	0.59	0.78
Neck girth pred/GT	0.461	0.457	0.997	0.464	0.996
PVE_{neck} (mm)	27.3	27.7	10.7	27.7	11.0
Global head direction (deg)	12.0	12.7	14.4	13.0	13.1
Fingers (deg)	17.3	17.4	8.3	17.4	8.3
neck02 bone (deg)	175.6	175.5	4.7	175.8	4.8
head bone (deg)	177.9	178.0	11.2	177.7	8.9

(a) *AnnyOne holdout, 100 images, setting V1 (helper bones masked, matched people); root error, depth fit and size ratio with each body at the model's own predicted depth.*

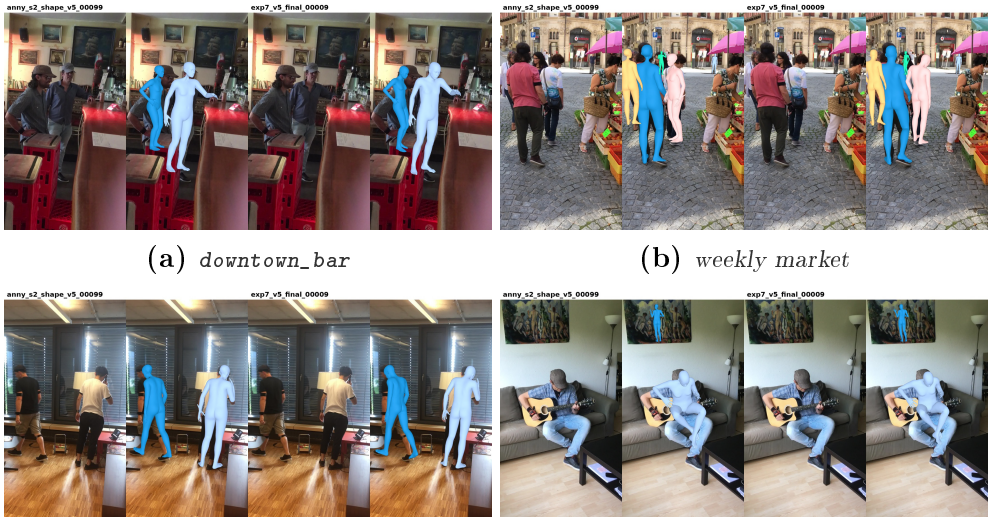
	Base	Mixed-L	Neck pilot	Control	Final	Multi-HMR-Anny
PVE (mm)	121.8	95.5	120.2	91.5	**89.0**	126.9
PA-PVE (mm)	72.0	62.9	68.7	59.7	**57.6**	68.8
MPJPE (mm)[b]	126.9	94.8	124.8	90.9	**88.3**	129.4
Precision (%)	54	73	53	62	64	75
Recall (%)	100	94	100	99	99	97
AnnyOne test-100 PVE[c]	74.3	78.5	72.7	81.2	77.5	(60.3)[a]

(b) *3DPW test, corrected (v23) labels, all 8,190 images, setting V0, same evaluation script for every model. Mixed-L was trained on the labels before their correction and is tested on the corrected ones, like every model here.*

Table 6.1.: *Final against its Control and against Base. One run per model, no seed; two identical runs differed by 7.6mm PVE, so the PVE gap between Control and Final (77.4 vs 73.7) is within run-to-run variation, while PA-PVE, root error, neck and fingers are not. Depth pilot: depth fix, depth weight ×5, 10 epochs; neck pilot: bone correction and a 2-epoch fine-tune. 3DPW numbers are against our converted labels and are not comparable with published 3DPW results.* [a]*Multi-HMR-Anny masks the helper bones by design, so its number is V1 and compares with Final's 73.7 in (a); it was most likely trained on these 100 images.* [b]*Mean over all 163 Anny joints, pelvis-centred.* [c]*The same 100 images as (a) in setting V0 (helper bones unmasked).*



Figure 6.6.: *Six AnnyOne holdout images (index in the first panel), each as image | ground truth | Base | Final. Ground truth drawn with the dataset camera; Base and Final in setting V1, the headline setting of Table 6.1a (both model panels are labelled "V1_ eval_ masked"), each body at its model's own predicted depth. By eye, Base draws its bodies smaller and further back than the ground truth, Final close to its size and place. Mesh colours mark detections, not identities. People without a ground-truth mesh (the foreground arm in 579862) are not counted; people a model misses (the two lying in the foreground of 579865) have no predicted mesh. Synthetic images, one run per model; judged by eye.*



(a) `downtown_bar` (b) *weekly market*

(c) `office_phoneCall` (d) *guitar*

Figure 6.7.: *Four 3DPW test frames, each shown as photo and Base (left half), then photo and Final (right half; the labels in the image name the two checkpoints). Bodies placed from the detector box, fingers at rest pose; judged by eye, no metric. The other four demo frames are in Figure D.4 (Appendix D).*

off or hidden are found less often; this was not tested.

Not shown: PVE paired on the same people (the matched sets differ: 214, 219 and 222 people), recall on people with a visible head, bootstrap intervals, and a second run of Final. The holdout has 100 synthetic images without children, and Figure 6.6 shows six of them.

6.6. 3DPW test on the corrected labels

Table 6.1b scores every model on all 8,190 test images with the same script. Final has the lowest error on our 3DPW-Anny labels: PVE 89.0 mm, PA-PVE 57.6 mm. Control follows at 91.5, Mixed-L, the previous best, at 95.5, the neck pilot at 120.2, Base at 121.8 and Multi-HMR-Anny at 126.9 mm. Final finds 99 % of the labelled people against 94 % for Mixed-L. Precision is low for every model (64 % for Final). Judged by eye, in Figure 6.7 most men get male bodies in Final and female-looking ones in Base, the bodies of Final are a little larger, and the seated guitar player is poor in both.

Most of the gain over Base comes from training on 3DPW at all: Control, without either fix, is 30.3 mm better than Base. The two fixes add 2.5 mm PVE and 2.1 mm PA-PVE on top, within run-to-run variation for PVE; the bone correction alone, in the neck pilot, leaves 3DPW PVE at Base's level. Final is 6.5 mm better than Mixed-L, which saw the same kind of images but with reversed gender labels and missing actors; whether the corrected labels cause part of this gap was not tested, since Mixed-L also differs in schedule and length. The low precision is expected: 3DPW labels only the one or two

	Base	Final	Final-masked
Recall (%)	47.1	64.7	57.2
Gender accuracy (%)	46.5	64.2	49.3
mean gender, men (male = 0)	0.535	0.551	0.717
mean gender, women (female = 1)	0.551	0.861	0.860
Child-vs-adult AUC	0.606	0.719	0.708
Mean age, adult	0.631	0.536	0.638
teen	0.597	0.524	0.616
kid	0.613	0.500	0.611
baby	0.624	0.438	0.524
Babies with age < 0.45 (%)	2.6	49.3	16.9
Mean height, adult	0.507	0.500	0.505
kid	0.451	0.466	0.461
baby	0.394	0.464	0.478
Matched children + adults	948	1,305	1,157
Demo images: standard deviation of the prediction			
Muscle	–	0.04	0.06
Weight	–	0.04	0.11
Cup size	–	0.05	0.11
Firmness	–	0.04	0.11
Age	–	0.03	0.08

Table 6.2.: *Body shape on real photos. Upper part: Relative Human validation split (635 images, 2,130 labelled people), detected people only, matched to the labelled boxes at IoU ≥ 0.3; ages and heights are Anny phenotype values in $[0, 1]$, not years; "child-shaped" means a predicted age below 0.45; the AUC ranks kids and babies against adults by predicted age (0.5 = chance). The matched sets differ between models because recall differs. Lower part: spread of the predicted phenotypes over all detected people of the 12 demo images (4 photos and 8 3DPW test frames), not matched to people; not reported for Base. One run per model, no seed. The neck pilot (bone correction and a 2-epoch fine-tune) is the same as Base within run-to-run variation (AUC 0.613).*

actors of each sequence, so every detected bystander counts as a false positive, and some detections are ghost bodies on half-visible people. The split between the two is unknown.

Not shown: a comparison with published 3DPW results. These numbers average all 163 Anny joints and vertices against converted labels that themselves err by about 14 mm, so they are not comparable with SMPL-based 3DPW numbers. The old-label results of the draft are retired and appear nowhere next to these. There is no breakdown by sequence, and the by-eye judgement of Figure 6.7 covers 8 frames.

6.7. Real photos: children and gender

Table 6.2 measures body shape on real photos with labels. Final finds more people (recall 64.7 % against 47.1 %) and gets the gender right for 64.2 % of

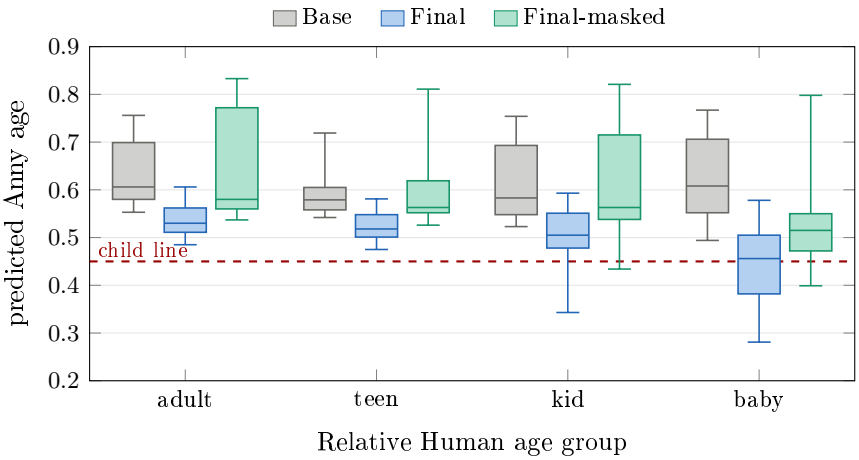


Figure 6.8.: *Predicted Anny age per Relative Human age group for Base, Final and Final-masked, matched (detected) people of the validation split only. Box: quartiles and median; whiskers: 5th and 95th percentile. Dashed: the child line, age 0.45. Ages are Anny phenotype values, not years. One run per model; statistics in* `data/plot_stats.md`.

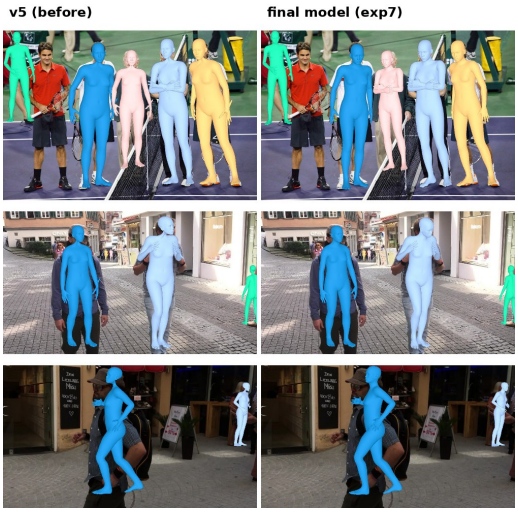


Figure 6.9.: *Three real photos never used in training (a tennis photo and the 3DPW test frames* `downtown_walking_00` *and* `downtown_runForBus_00`*): Base left, Final right, bodies placed from the detector box. Most men get male bodies, women stay female, the boy stays an adult body. Judged by eye; no metric exists for these images.*

them against 46.5 % for Base. Women are recognised (mean predicted gender 0.861, where 1 is female); men stay in the middle (0.551, where 0 is male). Children separate from adults: the child-versus-adult AUC rises from 0.606 to 0.719, and the mean predicted age falls to 0.536 for adults, 0.500 for kids and 0.438 for babies (Base: 0.631, 0.613, 0.624). 49.3 % of the babies are child-shaped against 2.6 % for Base, and 16.8 % of the kids. Figure 6.8 shows the whole distributions: Base's boxes stand at the same height for every group, Final's step down from adults to babies, and only the baby box reaches below the child line (median 0.456). On the demo photos (Figure 6.9 and Figure D.5) most men look male in Final, women stay female, and the boy keeps an adult body.

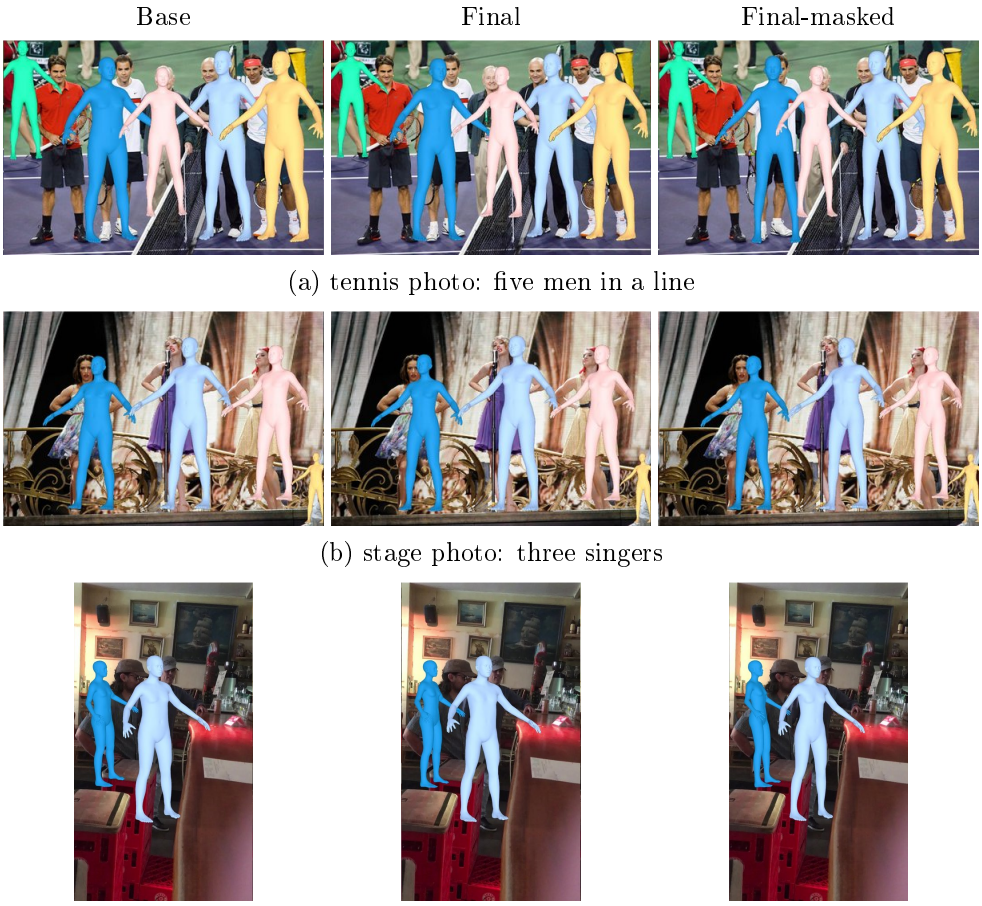
The AUC ranks people by predicted age, so a shift of all ages alone cannot raise it: the step from 0.606 to 0.719 is real separation of children from adults. Part of the lower ages is likely a common shift from the 3DPW age label, which the converter fixes at 0.55 for every real person and which pulls adults from 0.631 to 0.536. The gender gain fits the corrected 3DPW gender labels, which supervise the shape head on real people for the first time; the shape mask of Section 6.8 supports both readings. Children remain a measured improvement, not solved: half of the babies and five of six kids are still above the child line, and men are not yet male.

Not shown: ages in years, which I never quote because the code uses three age scales that disagree; children on held-out synthetic images (the holdout has none); and any shape number beyond age, gender and height, since Relative Human has no 3D bodies. The numbers cover detected people only, and the matched sets differ (192 kids and babies against 1113 adults for Final, 127 against 821 for Base). The demo images hold one child.

The rest-pose line-up of Figure 6.10 shows shape without pose: every detected person is drawn in rest pose, so only the body differs. Judged by eye, Base gives nearly everyone the same slim, flat body; the three singers on the stage look boyish. Final gives the singers a fuller chest and hips and the man in the bar a broader, heavier torso. The men of the tennis photo keep a slim body with a slight chest in all three models, so in rest pose no model draws them clearly male, a weaker result than the posed renders suggest and in line with the men's mid-range gender score in Table 6.2. Final-masked looks like Final on the stage and on the tennis court and draws the man in the bar slimmer again; the variety of muscle and weight that the mask restores in the numbers of Section 6.8 is hard to see at this size. The line-up uses the demo setting (bodies placed from the detector box) and is a by-eye reading of three of the twelve images; the other nine are in Figure D.7.

6.8. The shape-mask trade-off

The masked model is the third column of Table 6.2; its holdout and 3DPW numbers are given here. On the AnnyOne holdout Final-masked reads 71.7 mm PVE, 50.7 mm PA-PVE, 86.2 % recall and 682 mm root error (Final: 73.7, 50.3, 87.0, 728); on the 3DPW test 91.1 mm PVE and 59.0 mm PA-PVE (Final 89.0



(a) tennis photo: five men in a line

(b) stage photo: three singers

(c) 3DPW test frame `downtown_bar`: two men

Figure 6.10.: *Rest-pose line-up on three demo images (the tennis and stage photos and the 3DPW test frame* `downtown_bar`*): every detected person drawn in rest pose over the photo, so only body shape differs; Base, Final and Final-masked from left to right. Demo setting: bodies placed from the detector box. By eye, Base gives everyone a similar slim, flat body; Final gives the singers a fuller female shape and the man in front in (c) a broader torso; the tennis men stay slim with a slight chest in all three; Final-masked is close to Final except for a slimmer man in (c). Judged by eye on three images, one run per model; no shape metric exists for these images (the Relative Human numbers are in Table 6.2). The other nine images are in Figure D.7.*

and 57.6). On Relative Human it loses most of what Final gained: recall 57.2 %, gender accuracy 49.3 % (men 0.717, women 0.860), adults back at age 0.638 and 16.9 % of babies child-shaped; the AUC stays at 0.708. On the demo images it brings back the variety Final had lost: the spread of the predicted muscle rises from 0.04 to 0.06, of weight from 0.04 to 0.11, of cup size from 0.05 to 0.11 and of firmness from 0.04 to 0.11.

The mask stops the shape loss on the 8 values the converter fixes for every 3DPW person: age 0.55 and seven values at 0.5. The body errors stay within run-to-run variation, as expected, since the 3DPW meshes keep the neutral body. The trade-off is in the shape head. Without the mask, Final learned the seven constants and predicts nearly the same muscle, weight, cup size and firmness for everyone on real photos, so shape does not visibly change from person to person for those values. With the mask the values vary again, but gender falls back to chance and the children's gain shrinks. The likely reading is that the age label carries the children and gender gain and the seven constants cause the collapse; it was not tested. Before the run I fixed a rule: the masked model would replace Final if gender accuracy stayed at 60 % or more, the AUC at 0.7 or more, and the spread of muscle and weight rose above 0.08. Gender failed the rule, so Final stays the final model and the Student's teacher.

Not shown: whether the varied values are right. No real data set I have gives muscle or weight, so the spread shows variety, not accuracy. Why gender accuracy falls, although gender itself stays supervised under the mask, is not known. The partial mask, Final-partial-mask, which masks the seven constants and keeps age supervised, decides between the two readings: **[TODO: Final-partial-mask result (holdout, 3DPW, Relative Human, demo spread), job pending]**.

6.9. Baseline comparison and model cost

Table 6.3 sets NAVER's released Anny model, run with our code, labels and settings, next to ours. On the AnnyOne holdout it reads 60.3 mm PVE, 44.6 mm PA-PVE and 89.0 % recall, against 73.7, 50.3 and 87.0 for Final. NAVER's model was trained on the AnnyOne release, which includes the 100 images we hold out, so its 60.3 mm is most likely an in-sample number; ours are held out. On 3DPW, which neither model saw before our fine-tune, the final model is 37.9 mm better (89.0 vs 126.9). Its 3DPW PA-PVE is 68.8 mm against 57.6, about Base's level. Both networks use the same ViT-L backbone (304.37 M parameters, 1,917.2 GFLOPs). The difference is the head: NAVER's 8-layer cross-attention decoder has 76.57 M parameters, ours 14.01 M, so Final has 318.38 M in all against 380.94 M. Final runs in 77.9 ms per image on an A100 and 110 ms on an A40; NAVER's time and its head's FLOPs could not be measured with our tool, which supports only our model class. The Student has 101.38 M parameters and 603.3 GFLOPs and runs in 51.4 ms on the A40: 3.14× fewer parameters, 3.19× fewer FLOPs and 2.14× faster than Final, and 3.76× fewer parameters than NAVER's model (Figure 6.11).

The fair comparison is 3DPW. There, Final is far ahead, but most of its lead

	Cost				AnnyOne holdout (V1)			3DPW test (v23)			
Model	Params (M)	GFLOPs	ms A100	ms A40	PVE	PA-PVE	Recall	PVE	PA-PVE	Recall	Trained on 3DPW
Multi-HMR-Anny (NAVER)	380.94	$\geq 1917.2^b$	n/m	n/m	60.3^a	44.6^a	89.0^a	126.9	68.8	97	no
Final	318.38	1927.4	77.9	110	73.7	50.3	87.0	**89.0**	**57.6**	99	yes
Control	same network as Final[c]				77.4	56.4	89.0	91.5	59.7	99	yes
Base	same network as Final[c]				70.5	53.3	90.2	121.8	72.0	100	no
Student (ViT-B)	101.38	603.3	n/m	51.4	[TODO: job pending]			[TODO: job pending]			yes
Backbone only, analytic count											
DINOv2 ViT-S/14	22.06	196.9									
DINOv2 ViT-B/14	86.58	589.5									
DINOv2 ViT-L/14	304.37	1917.2									

Table 6.3.: *Accuracy and cost; one run per model, no seed. 3DPW numbers are against our converted labels and not comparable with published 3DPW results. GFLOPs count one multiply-add as two FLOPs, at 672 px input with one person; latency is the median forward pass at batch 1 (fp32 with TF32). The two GPUs differ, so only times in the same column compare: the Student is 2.14× faster than Final on the A40 and has 3.14× fewer parameters and 3.19× fewer FLOPs. n/m: not measured. [a]Most likely in-sample for Multi-HMR-Anny, which was trained on the AnnyOne release; ours are held out. [b]Backbone only: the head's FLOPs and the latency cannot be counted for NAVER's model class; its head has 76.57 M parameters against 14.01 M in ours. [c]Base, Control and Final share the architecture; the bone correction adds no parameters.*

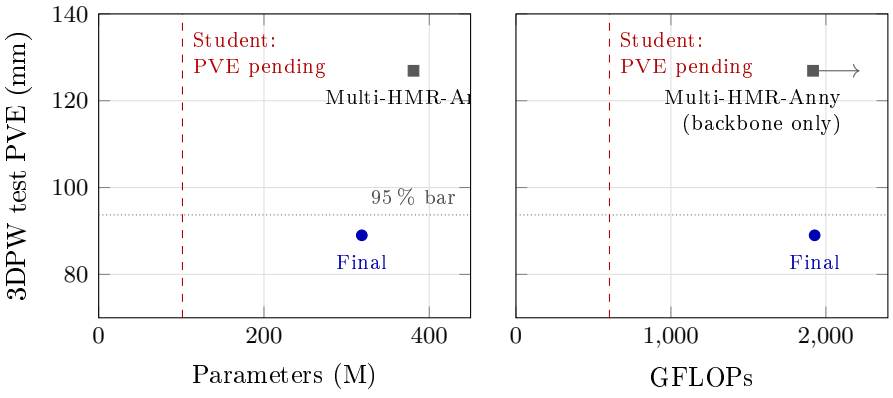


Figure 6.11.: *Model size and compute against 3DPW-test PVE (corrected labels, all 8,190 images; one run per model). Multi-HMR-Anny's GFLOPs count its backbone only (arrow: the true value is higher); the Student's error is pending, so only its cost is drawn. Dotted line: the 95 % bar against Final (93.7 mm).*

Metric	Teacher	Student	Ratio	Bar	Met?
AnnyOne holdout, V1					
PVE (mm)	73.7	**[TODO: job pending]**		$\leq$ 77.6	
PA-PVE (mm)	50.3	**[TODO: job pending]**		$\leq$ 52.9	
MPJPE (mm)	77.9	**[TODO: job pending]**		$\leq$ 82.0	
Recall (%)	87.0	**[TODO: job pending]**		$\geq$ 82.7	
AnnyOne test-100, V0					
PVE (mm)	77.5	**[TODO: job pending]**		$\leq$ 81.6	
PA-PVE (mm)	54.4	**[TODO: job pending]**		$\leq$ 57.3	
Recall (%)	87	**[TODO: job pending]**		$\geq$ 82.7	
3DPW test (v23 labels), V0					
PVE (mm)	89.0	**[TODO: job pending]**		$\leq$ 93.7	
PA-PVE (mm)	57.6	**[TODO: job pending]**		$\leq$ 60.6	
MPJPE (mm)	88.3	**[TODO: job pending]**		$\leq$ 92.9	
Recall (%)	99	**[TODO: job pending]**		$\geq$ 94.1	
Cost					
Params (M)	318.38	101.38	3.14×	fewer	yes
GFLOPs	1927.4	603.3	3.19×	fewer	yes
ms per image (A40)	110	51.4	2.14×	faster	yes
Training images/s	10.9[a]	**[TODO: job pending]**			

Table 6.4.: *The Student against its teacher (Final) and the 95 % rule: every error of the Student at most the teacher's divided by 0.95, every recall at least 0.95 of the teacher's, and fewer parameters, fewer FLOPs and a faster forward pass. One run, no seed. Against Multi-HMR-Anny the PVE bars would be 63.5 mm on AnnyOne (most likely in-sample for it) and 133.6 mm on 3DPW. Cost from the end-to-end test of the distillation script on an A40 (batch 1, one person). [a]Final's training throughput on an A100; the Student's is measured on its own GPU.*

comes from fine-tuning on 3DPW, which NAVER's model never had: Control, without our fixes, is already at 91.5 mm. The comparison therefore shows what the real-image fine-tune buys, not a better network. On AnnyOne the order is reversed, and the in-sample caveat prevents reading it as a gap in skill. In size, Final is 318.38 M against 380.94 M, about one sixth smaller; that alone is not the "significantly smaller" model of requirement 1b, which is why the Student exists.

Not shown: NAVER's model fine-tuned on our 3DPW labels, which would separate the network from the data; its latency and total FLOPs; and its published numbers, which are SMPL-X results on other labels and not comparable. The times of the two GPUs do not mix: only ratios within one column hold.

6.10. The ViT-B student

The Student follows the recipe of Table 4.4 (Section 4.7.2): a DINOv2 ViT-B backbone (86.58 M parameters) started from NAVER's Multi-HMR ViT-B, a projector into the teacher's width, and heads copied from Final; it learns

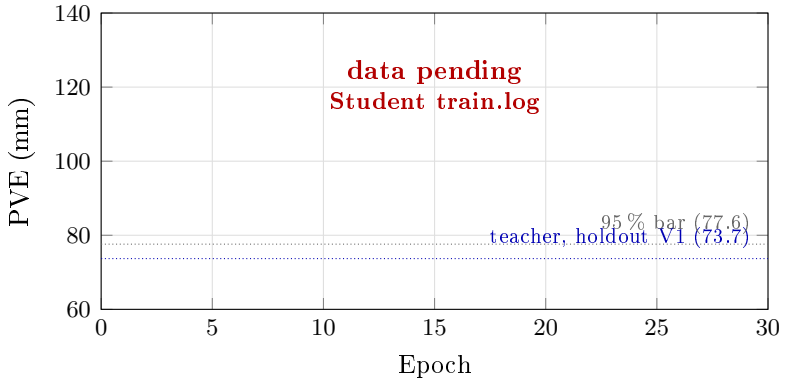


Figure 6.12.: *The Student's holdout (V1) and 3DPW-validation PVE per evaluated epoch, against the teacher's holdout value and the 95 % bar. One run, no seed.* ***[TODO: plot pending: Student training log]***

from the same targets as Final plus the teacher's features and outputs, for 30 epochs, 240,000 images in all. Table 6.4 gives its result against the 95 % rule. On the AnnyOne holdout it reaches **[TODO: student result, job pending]** mm PVE against the teacher's 73.7 (bar 77.6) and **[TODO: student result, job pending]** mm MPJPE against 77.9 (bar 82.0); on the 3DPW test **[TODO: student result, job pending]** mm PVE against 89.0 (bar 93.7) and **[TODO: student result, job pending]** mm MPJPE against 88.3 (bar 92.9). Its recall is **[TODO: student result, job pending]** % on the holdout and **[TODO: student result, job pending]** % on 3DPW. Figure 6.12 shows whether its error was still falling at the last epoch: **[TODO: student result, job pending]**. The cost side is measured and met: 3.14× fewer parameters, 3.19× fewer FLOPs, 2.14× faster.

By the rule against Final the Student **[TODO: meets / misses]** the bar on the holdout and **[TODO: meets / misses]** it on 3DPW. Against Multi-HMR-Anny the bars would be 63.5 mm on AnnyOne, which no student of a 73.7 mm teacher can be expected to reach and which rests on a likely in-sample number, and 133.6 mm on 3DPW, which lies above the teacher's own error; the Student **[TODO: meets / misses]** it. Which reference counts is the supervisor's decision; both are reported. The Student is a Multi-HMR-style network with a smaller backbone, so whether it counts as "not exactly Multi-HMR" for requirement 1b is also his decision.

Not shown: a second run, a longer schedule and the Student's own Relative Human numbers. The Student trains with the teacher's bone correction and then estimates its own on 200 training images; its fingers and necks on real photos are not rendered. No data augmentation was used: the job script's header mentions colour jitter, but no code reads such a setting.

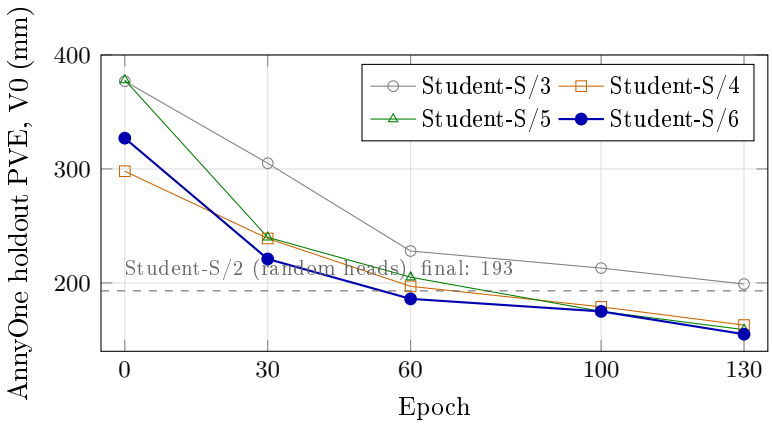


Figure 6.13.: *AnnyOne holdout PVE of the ViT-S head-transfer students at equal epochs (100 images, setting V0; one run each, no seed; history, before the depth and bone fixes and the corrected 3DPW labels). Each student adds one change to the previous one: Student-S/4 a student backbone trained for mesh recovery, Student-S/5 a 5-epoch head freeze, Student-S/6 projector-fed heads. Their teacher Base reads 74.3 mm in the same setting.*

6.11. History: the ViT-S students and the schedule to Base

Table 6.5 keeps the draft's holdout measurements, all in setting V0 and all from before the depth fix, the bone correction and the corrected labels. The staged schedule went from 145 mm with the backbone frozen to 73.6 mm once the last four blocks trained and the helper bones were masked, and ended at 74.3 mm for Base (Table 6.5a). The ViT-S students started at 193.4 mm with feature distillation alone and reached 114.5 mm with projector-fed heads at twice the batch, still improving at the end, 1.54 times their teacher's error (Table 6.5b). Adding 30 % 3DPW images on the old labels (Student-S/mixed, Student-S/final) gave 120.6 mm on the holdout, 1.54 times Student-S/final's own teacher. Figure 6.13 shows the head-transfer students: each starts far above the final error of the student with random heads (the dashed line), Student-S/4 to Student-S/6 fall below it between epochs 60 and 100, Student-S/3 never does, and from epoch 60 on Student-S/6 has the lowest error at every epoch.

Three lessons from this line shaped the Student. A backbone already trained for mesh recovery gave the largest single gain, so the Student starts from NAVER's ViT-B. Heads fed through a projector at the teacher's width beat heads cut to the student's width, so the Student keeps Final's heads. And the students were limited by the images they had seen rather than by their size, so the Student trains on six times the images of Final's own fine-tune. The schedule shows that helper-bone masking and partial unfreezing together halved the error, but the two changes were made in one step and are not separated.

Run	Change against the previous run	PVE	PA-PVE	MPJPE	Rec.
L-stage 1	fresh heads, all blocks trained (side branch)	–	–	–	–
L-stage 2	fresh heads, backbone frozen	145	–	–	–
L-stage 3	+ last 4 of 24 blocks, helper mask	73.6	–	–	–
L-stage 4	+ neck and hand loss ×4	73.6	56.5	73.2	–
Base	+ shape loss weight 20	74.3	57.5	74.4	90

(a) *The transfer schedule that produced Base (names as in Table 4.3); recall (Rec.) in %. L-stages 2–4: evaluation during training; L-stage 1 was never evaluated with this script; Base: the standalone script (the same numbers as the AnnyOne column of Table 6.1b).*

			AnnyOne holdout, V0					3DPW test, v23[c]	
Model	Teacher	Params	PVE	PA-PVE	MPJPE	Prec.	Rec.	PVE	PA-PVE
Base (ViT-L)	–	318.4	74.3	57.5	74.4	81	90	121.8	72.0
Mixed-L	–	318.4	78.5	57.4	78.6	83	89	95.5	62.9
Student-S/2	Base	32.8	193.4	128.2	210.7	–	86	–	–
Student-S/6b[a]	Base	37	126.9	92.1	132.0	77	88	–	–
Student-S/6c	Base	37	**114.5**	**84.6**	**118.4**	81	89	142.6	91.0
Student-S/mixed[b]	Base	37	120.6	88.2	125.2	78	87	–	–
Student-S/final[b]	Mixed-L	37	120.6	86.9	125.0	76	87	121.6	83.7

(b) *The ViT-S students (Student-S, ViT-S backbone) against their ViT-L teachers; parameters in millions, precision (Prec.) and recall (Rec.) in %. Student-S/2: feature distillation only; Student-S/6b: heads fed through a projector at the teacher's width; Student-S/6c: Student-S/6b continued at batch 16; Student-S/mixed and Student-S/final: Student-S/6b and Student-S/6c continued with 3DPW images. Standalone script except [a]Student-S/6b (evaluation during training). [b]30 % 3DPW images per batch, old labels. [c]Corrected labels, all 8,190 test images, the script and setting of Table 6.1b; the only Student-S numbers on these labels (Student-S/6c and Student-S/final were re-scored, the other students were not). Final on the same images: 89.0 / 57.6 mm.*

Table 6.5.: *History, before the depth fix, the bone correction and the corrected 3DPW labels. AnnyOne holdout (100 images), setting V0 (helper bones unmasked), errors in mm; one run per model, no seed. "–": not measured with the same script. The 3DPW columns of (b) are on the corrected labels; the students' earlier 3DPW numbers were measured on the old labels, which are retired, and are not reported.*

The two 3DPW columns of Table 6.5b are the only Student-S numbers on the corrected labels: Student-S/6c and Student-S/final re-scored with the script of Table 6.1b; the students' old-label numbers are retired and not shown. Student-S/final, which saw 3DPW training images with the old labels, is at Base's level on the real test images (121.6 against 121.8 mm PVE), and Student-S/6c, trained on synthetic images only, is at 142.6 mm; both are far from Final's 89.0 mm. Neither is the thesis's answer to requirement 1b; that is the ViT-B Student of Section 6.10. The students also predate both fixes and so carry the depth bug and, likely, the neck flip of their teachers, which was not measured for them. Their holdout numbers are V0 and read about 3.8 mm higher than V1 would.

7. Discussion

This chapter reads the results against the requirements, explains why the headline error barely moved, states the limits of each claim and records what the work taught about building such a pipeline.

7.1. The requirements, one by one

The supervisor set three tiers (Section 1.3): requirements without which he would not consider the thesis ready for submission (1a, 1b, 1c), requirements for an acceptable grade (2a, 2b), and a well-written report with verifiable code for the best grade (3). Table 7.1 gives the status of each on 18 October 2026 in one of five words: met, met with caveats, partly, not met, not started. I split 1a into pose and shape, because their evidence differs. The paragraphs follow his order; each names what is missing and where Chapter 8 plans it.

1a-pose. He asked for a model whose "pose of all body parts must look correct" on the 3DPW test set and on any random image from the wild. The status is *partly*. On the 3DPW test set Final has the lowest error on our 3DPW-Anny labels, 89.0 mm PVE and 57.6 mm PA-PVE, and finds 99 % of the labelled actors (Table 6.1b). Two pose faults that the 3DPW labels cannot see were corrected on the AnnyOne holdout: the half-turned neck bones (neck02 175.8 against 4.8 deg for Control and Final) and two flipped hand bones (fingers 17.4 against 8.3 deg; Sections 6.3 and 6.4). On real photos the evidence is by eye: the necks have normal width, while placement offsets, ghost bodies and hard poses such as the seated guitar player remain (Figures 6.7 and 6.9). Predicted fingers on real images are shown in Figure 6.4: by eye, Final's fingers close and bend like hands where Base's are splayed and stiff, on the three images whose hands are large enough to judge. Missing are correct near and far depth (the slope is 0.42; Figure 6.2) and a second run. Section 8.2 plans a repeat of Final; the slope stays an open limitation.

1a-shape. He asked that "mesh size, proportions, age, muscles, and other aspects of the 3D shape must visibly change according to the person", since AnnyOne carries these labels. The status is *partly*. Training on the labels is in place: every model learns all 11 phenotypes from AnnyOne, the shape term at weight 20 (Table 4.2). On real photos it depends on the phenotype. Gender changes with the person: on Relative Human Final gets it right for 64.2 % of the people against 46.5 % for Base, although men still sit mid-range (0.551, where 0 is male; Table 6.2). Age changes partly: children separate from adults (AUC 0.606 to 0.719), but half of the babies keep an adult-shaped body. Muscle,

Req.	Supervisor's wording (short)	Status	Evidence	What is missing
1a-pose	"Pose of all body parts must look correct" on (a) the 3DPW test set and (b) any in-the-wild image	**partly**	3DPW test: PVE 89.0 mm, recall 99 %, precision 64 % (Table 6.1b); neck and finger flips corrected on the holdout (Sections 6.3, 6.4); real photos by eye only (Figures 6.7, 6.9, 6.4)	depth slope 0.42 (Figure 6.2); a second run
1a-shape	"Mesh size, proportions, age, muscles, and other aspects of the 3D shape must visibly change according to the person"	**partly**	Relative Human: gender 64.2 % (Base 46.5), child AUC 0.719 (Base 0.606) (Table 6.2); muscle, weight, cup size, firmness nearly constant on photos (standard deviation 0.04–0.05); the shape mask trades this for gender (Section 6.8)	per-phenotype agreement on AnnyOne; the partial mask
1b	"Must not be exactly the same as the Multi-HMR model": new and better on both test sets, or "significantly smaller" and ≥ 95 % as good	**not met**	Final is Multi-HMR with an Anny head; 318.38 M against 380.94 M parameters (-16 %); Multi-HMR-Anny 60.3 (likely in-sample) / 126.9 mm against 73.7 / 89.0 mm (Table 6.3); Student 101.38 M, accuracy pending (Table 6.4)	the Student's accuracy; the supervisor's choice of the reference model and whether a smaller student counts
1c	Evaluation on the real-world set with Anny parameters, not only SMPL-X	**met with caveats**	3DPW test in Anny space, 8,190 images / 11,854 people; label check WARN (old labels FAIL), body joints 14.1 mm (Table 3.7)	nothing required; caveats of Section 3.6; a body-joint MPJPE
2a	Convert 3DPW train and test to Anny and include the train set in training	**met**	train 5,882 images / 7,584 people, 30 % of each batch (Section 4.6); 3DPW test: Control 91.5 against Base 121.8 mm (Table 6.1b)	nothing (a larger real-image share is optional)
2b	The model understands motion in video instead of predicting every frame on its own	**not started**	no temporal code; every model runs frame by frame	a temporal model and its evaluation (Section 8.2)
3	A well-written report that shows understanding; a clean codebase and trained models he can verify	**partly**	this thesis; Appendix A maps every number to its run, Appendix C lists every setting	released checkpoints, clean paths, licences checked (Section 8.2)

Table 7.1.: *The supervisor's requirements, in his order, and their status on 18 October 2026. AnnyOne holdout in setting V1; 3DPW test on the corrected (v23) labels, setting V0; Relative Human validation split. One run per model, no seed. Section 7.1 discusses each row.*

weight, cup size and firmness do not visibly change; their standard deviation on the demo images is 0.04–0.05, because Final learned the constants the converter writes for every 3DPW person. The shape mask brought the spread back but lost the gender gain, so I kept the unmasked model (Section 6.8). The rest-pose line-up (Figure 6.10) agrees by eye: Final gives the women a fuller female shape and one man a broader torso where Base draws everyone alike, but the men of the tennis photo stay slim with a slight chest in every model. Missing are a measure of how well each phenotype follows its label on AnnyOne and the partial mask, Final-partial-mask, which frees the seven constants at 0.5 and keeps age supervised **[TODO: Final-partial-mask result: update 1a-shape and T1]**. Whether a predicted muscle or weight value is right for a real person cannot be measured with the data I have. Section 8.2 plans the partial mask and the per-phenotype measure.

1b. His wording: "The model must not be exactly the same as the Multi-HMR (or any other existing) model." A new architecture must beat the baseline on both the AnnyOne and the 3DPW test; a "significantly smaller" model must stay within 95 % of the larger model's accuracy on both. The status is *not met*. Final is Multi-HMR with its output resized to Anny and a correction that adds no parameters (Chapter 4). Against NAVER's own Anny model (Table 6.3): NAVER's model was trained on the AnnyOne release, which includes the 100 images we hold out, so its 60.3 mm is most likely an in-sample number; ours are held out. On 3DPW, which neither model saw before our fine-tune, the final model is 37.9 mm better (89.0 vs 126.9). Final thus trails the baseline by 13.4 mm on a comparison that is probably not fair and beats it on the other. In size Final has 318.38 M parameters against 380.94 M, 16 % fewer, all in the head; that is not "significantly smaller". The second route is the Student (Section 6.10). Its cost side is measured, 3.14× fewer parameters, 3.19× fewer FLOPs and 2.14× faster than Final (Figure 6.11); its accuracy is pending (Table 6.4) **[TODO: Student result: update 1b, T1 and the status word]**. Missing are that accuracy, his decision on which model sets the bar (with NAVER's model the AnnyOne bar is 63.5 mm, which no student of a 73.7 mm teacher can be expected to reach), and his view on whether a Multi-HMR-style network with a smaller backbone counts. If it misses while its error still falls, Section 8.2 plans to continue its training.

1c. He asked for evaluation on a real-world set with Anny parameters, not only SMPL-X, at least for the 3DPW test set. The status is *met with caveats*. All 8,190 test images (11,854 people) have Anny labels from the converter of Chapter 3, and every model is scored against them with the same script (Table 6.1b). The independent check grades these labels WARN where the old ones failed, with body joints 14.1 mm from 3DPW's own at the median (Table 3.7). The caveats are those of Section 3.6: the labels test pose, gender, height and proportions but not body build, and no number compares with published 3DPW results. Nothing further is required; a body-joint MPJPE next to the 163-joint one would make the numbers easier to read, but it is not

yet written.

2a. He asked to convert the 3DPW train and test sets to Anny and to train on the train set, and expected tier 1 to be out of reach without it. The status is *met*. The corrected training labels hold 5,882 images and 7,584 people, and Final and Control draw 30 % of each batch from them (Section 4.6). His expectation held: on the 3DPW test set Control, which differs from Base by the real images and the fine-tune, is 30.3 mm better than Base, and the two fixes add 2.5 mm on top (Table 6.1b). Nothing is missing.

2b. He asked that the model understand motion in video instead of predicting each frame on its own. The status is *not started*: no temporal code exists, and every model here runs on single images. Section 8.2 plans a small refiner over windows of per-frame person tokens from the frozen Final model, trained on 3DPW-train tracks and judged against simple smoothing. Whether a refiner on frozen features counts as the temporal model he means is a question for him.

3. He asked for a well-written report that shows understanding and argued choices, and for a clean codebase with trained models he can verify independently. The status is *partly*. The report is this thesis; Appendix A maps every number to its run, log and checkpoint, and Appendix C lists every setting. The code side is not done: no checkpoint is released, the scripts still carry the cluster's paths, and the licences have not been checked for a release (Section 7.3). Section 8.2 lists the release steps.

7.2. Why PVE barely moved while everything else did

From Control to Final the root error fell by 48 %, neck02 from 175.8 to 4.8 deg, the neck girth from 0.464 to 0.996 of the truth and the finger error from 17.4 to 8.3 deg. PVE fell only from 77.4 to 73.7 mm, within run-to-run variation, and Base keeps the lowest PVE of all (70.5 mm; Table 6.1a). Three facts explain most of this, and one error measure did move.

PVE cannot see placement. PVE moves both meshes so that their pelvises coincide before it measures the vertex distances, so root error, depth and drawn size are removed by construction, and the largest of the fixes is invisible to it. On Base, which placed every person too far, the correlation between a person's PVE and their depth error was −0.03 (Table 5.2). The 3DPW test aligns the pelvis in the same way.

The bone flips cancelled. The two half-turns of neck02 and the head bone cancel, so Base's head pointed the right way (12.0 deg) and the head vertices sat almost where they should; only the neck skin between them was twisted. That skin is 171 of the 13,718 vertices (1.2 %). Its own error fell from 27.7 to

11.0 mm, which spread over the whole mesh is about 0.2 mm, an estimate from the two numbers, not a measurement. The correction applied to Base without training shows the same at a fixed checkpoint, free of run-to-run variation: it halved the finger error (17.3 to 8.2 deg) and restored the neck girth, while PVE rose by 0.8 mm, a change that also contains the worse head direction of that setting (Table 5.4).

Noise floor and restart cost. One run resolves about 10 mm PVE, since two identical runs differed by 7.6 mm (Section 5.7), so the gap of 3.7 mm supports neither a gain nor a cost. Base's lower PVE is no evidence against the fixes either: restarting a fine-tune of Base with nothing changed already cost about 2.4 mm, and Control, which adds 30 % real images to a model trained on AnnyOne only, sits 6.9 mm above Base on these synthetic images. The likely reason is that every fine-tune moves the model away from the synthetic domain of the holdout; this was not tested. Every fix is therefore judged against its own control.

What did move. PA-PVE, which also removes global rotation and scale, fell from 56.4 to 50.3 mm, 6.1 mm below Control and 3.0 mm below Base, the lowest of every model. The pilots tie part of this to the bone correction: applied without training it lowered PA-PVE from 53.3 to 52.2 mm, and the neck pilot reached 50.9 against 55.2 mm for its control (Table 5.4). The likely reason PA-PVE sees what PVE does not is that the error in global rotation, large and different from run to run, dominates PVE and hides the articulated part where the flips sit; a per-region breakdown that would test this was not made. On 3DPW the fixes add 2.5 mm PVE and 2.1 mm PA-PVE to what the real images gave, and Final's validation curve lies below Control's at all ten epochs (Figure 6.5).

I draw one consequence. Each fix is reported with the measure that can see it: root error and drawn size for depth, bone angles and neck girth for the correction, finger angles for the hands. PVE stays in every table as the check that a fix costs nothing a single run can resolve. A headline PVE alone would have hidden all three fixes.

7.3. Limitations

Children: a measured improvement, not solved. Final separates children from adults on Relative Human better than Base (AUC 0.606 to 0.719; mean age of babies 0.438 against adults 0.536), but only 49.3 % of the babies and 16.8 % of the kids fall below the child line, and the boy of the demo photos still gets an adult body (Section 6.7). On real photos Base's shape head returned the training median, and on synthetic images it regressed toward the median in both tails (Section 5.6). Children are rare in AnnyOne, 2.53 % of the people having an age below 0.3, and the holdout has none, so no held-out synthetic number for children exists. The 3DPW labels pin age at 0.55 for every real person; part of Final's lower ages is likely that common shift, although a

common shift alone cannot raise the AUC. Whether the gap between synthetic and real images or an L1 loss through a sigmoid on a rare value dominates, the evidence cannot rank. I quote no ages in years, because the code uses three age scales that disagree (Section 2.1).

Depth: the slope did not move. The fix moved the offset and the body size, not the slope (0.42). Final's fitted line crosses the identity at 3.01 m, so people about three metres away are placed about right, while 94.4 % of all people, the near ones most, are still placed too far (Figure 6.2). Predictions stay compressed toward 2.5–2.9 m, while the holdout people stand at 1.92 m on average. The cause is unknown, and Section 8.2 leaves it open; measuring the depth distribution of the training batches would be the next test. All depth numbers are synthetic: the 3DPW evaluation aligns the pelvis and the demos place each body from its box, so the fix has not been measured on a real image.

One run per arm, no seed. The training code sets no random seed, and every model was trained once. Two identical runs differed by 7.6 mm PVE, 1.9 deg head direction and about 2 recall points (Table 5.7). The claims of this thesis rest on gaps far above that (root error, bone angles, girth, fingers, PA-PVE); the PVE gap, the 3DPW gain of the fixes and the recall cost lie within or at the edge of it. There are no bootstrap intervals, the root error has no measured noise floor, and the holdout holds 100 images with 222 people and no children. A repeat of Final (Section 8.2) would turn "within run-to-run variation" into a measured spread.

Fingers on real photos. Predicted fingers on real images are shown in Figure 6.4 and judged by eye only: Final's fingers look like hands where Base's are splayed, on three images whose hands are large enough to see; on the smaller hands of the other demo images nothing can be judged. The finger number (8.3 deg) is an angle on the synthetic holdout, and the 3DPW labels carry rest-pose hands, so no real-image number exists. Base's renders carry both flipped hand bones, so the figure shows the correction and the fine-tune together.

The recall cost. Final finds 87.0 % of the holdout people, 2.0 points fewer than Control, at the edge of run-to-run variation, and 3.2 points fewer than Base, beyond it; every depth-fix pilot lost 1.6 to 2.4 points against its control, always in the same direction (Section 5.3). I state it as a real cost. The likely reason is that the fix moved the detection cell from the pelvis to the head, so people whose head is cut off or hidden are found less often; this was not tested, since recall on people with a visible head needs code that does not exist yet.

What the 3DPW labels can teach and test. The labels carry a neutral body: 8 of the 11 phenotypes are constants, and neck02, neck03, the head and all hand bones stay at rest (Section 3.6). They cannot supervise body build or fingers, and they cannot test the neck correction. Training on them pulls body build toward the constants (Section 6.8); the hand mask removes the

rotation terms only, and the mesh terms still carry the rest-pose hands. As a test set they put a floor of about 14 mm under every number and likely reward a neutral prediction, which was not tested. Precision (64 % for Final) is a lower bound, since detected bystanders count as false positives, as do ghost bodies on half-visible people; the split between the two is unknown.

The comparison with NAVER's model. Its AnnyOne number is most likely in-sample (Section 7.1); whether our 100 holdout images were in its training set is not known for certain. On 3DPW the comparison is fairer but not like for like: Final was fine-tuned on 3DPW-train and NAVER's model was not, and Control, without our fixes, already reaches 91.5 mm. The gap shows what the real-image fine-tune buys, not a better network; NAVER's model fine-tuned on our labels would separate the two and was not run.

The validation split. 3DPW's validation sequences share actors and locations with the training sequences; the test split shows other people in other scenes. The draft found, on the old labels, that the gain from real images on validation was about twice that on test. I report only test numbers, and the validation split selects nothing, since Final is the last epoch of its run (Table 3.8).

Licences. SMPL, and with it 3DPW's ground truth, is licensed for non-commercial research (Section 2.1), and 3DPW has its own licence. Relative Human ships without a licence file, so I used only its validation split, for evaluation. AnnyOne and NAVER's checkpoints, from whose backbones every model here starts, carry their own terms, separate from Anny's Apache 2.0 code. I have not yet checked whether those terms allow releasing weights derived from them; the release of Section 8.2 depends on that check.

7.4. Engineering lessons

The faults that cost the most time were silent: each produced plausible output and showed only when one stage was measured on its own, against a control, or by its geometry.

Measure each stage alone. The six checks of Section 5.2 decoded the labels without a network, moved single joints of the body model and traced the gradients, and so excluded the rig, the labels and the gradient path before any training change. The converter graded its own fit; only an independent check that read the finished label files found the three label defects (Section 3.5). The holdout trace measured placement at each body's own predicted depth, which neither PVE nor the box-placed demos did. The draft's list of pipeline bugs, from a demo that silently loaded no weights to a learning-rate schedule that froze a long run, shows the same pattern. The label builder skipped frames without an image file and lost 31 % of the 3DPW training labels without a warning. The adapter for NAVER's checkpoint hard-coded a decoder width of

4096 where the checkpoint has 2048; it failed loudly on the first load, which made it the cheapest fault of all.

Read every change against a control. Each fix was read against a run that differed only by that fix, and two identical runs showed how much a run varies. Without the controls, three wrong readings would have stood: the neck pilot's 68.8 mm as a gain over Base, Final's higher PVE as a cost of the fixes, and the 3DPW gain as a gain of the fixes. The restart cost was itself found only because the controls of all pilots sat above Base.

Check by geometry, not by names. Three of the worst faults were faults of naming or convention that only a geometric check exposed. The converter first chose Anny bones by name; the two rigs number the spine in opposite directions, so SMPL's lumbar spine joint was mapped to a bone 25 cm above it, and the fit still looked plausible until the two rest skeletons were compared (Section 3.3). The gender value followed MakeHuman's convention, where 1 is male, while in Anny 1 is female, so every 3DPW man was labelled with a female body; the fit error gave it away, falling for 72 of 73 men when the value was flipped (Section 3.5). The training targets were built at joint 0, the pelvis, while the model places each body by its head; only the drawn size and the depth line of the trace revealed that prediction and target described two joints half a body apart (Section 4.4). The half-turned neck bones belong to the same family: per-group means of 62.7 and 89.5 deg hid them, a per-bone dump found them.

Keep every change as a switch. Each change entered the code as a switch whose default reproduces the old behaviour: the anchor joint of the targets, the bone correction (saved with the checkpoint and reapplied on loading), the hand and shape masks, and the two switches of converter version 2.3. This kept the history models reproducible, made every control a change of one setting, and let the masked arm run from the same code as Final (Appendix C).

In this pipeline silent failure was far more common than loud failure. A check that fails loudly, such as a load percentage or a geometric comparison of two skeletons, costs minutes to write. The silent faults stayed hidden until a measurement aimed at them: the depth bug sat in every checkpoint trained before this thesis.

8. Conclusion and Future Work

8.1. Conclusion

I moved Multi-HMR to the Anny body model and trained it on synthetic AnnyOne images and on real 3DPW images that I converted to Anny. The model I started from had three visible failures. I first tested what was not their cause and then traced each one to a mechanism: training targets built at the pelvis while the network anchors each person at the head, and four bones turned by half a turn at the output. The fixes are small, and their effect is large where it can be measured. Against a control with the same training, Final cut the root error from 1,402 to 728 mm, the neck02 error from 175.8 to 4.8 degrees and the finger error from 17.4 to 8.3 degrees, while the holdout PVE (73.7 against 77.4 mm) stayed within run-to-run variation. The main lesson is about measurement: the standard error measure could not see the worst defect, and each failure became tractable only once a number for it existed.

In the order of the supervisor's requirements: for the pose (1a), Final has 89.0 mm PVE on the 3DPW test set against our labels, with necks and fingers corrected on the holdout; on real photos the pose is judged by eye, and predicted fingers, shown in Figure 6.4, look like hands in Final and are splayed and stiff in Base on the images where the hands are large enough to judge. For the shape (1a), Final recognises gender in 64.2 % of the Relative Human people (Base 46.5 %) and separates children from adults with an AUC of 0.719 (Base 0.606); children are a measured improvement, not solved, and muscle and weight barely vary between people on real photos. For 1b, the Student has 3.14× fewer parameters and 3.19× fewer FLOPs than Final, but its accuracy was not yet measured, so 1b is not met at the time of writing; NAVER's Multi-HMR-Anny, measured with our code, reaches 126.9 mm on 3DPW. For 1c and 2a, the converted 3DPW labels exist with three defects fixed and an independent check, and training on them brought the 3DPW error from 121.8 (Base) to 91.5 mm (Control) before any fix. Motion in video (2b) was not started. For 3, every number traces to a run, a log and a checkpoint (Appendix A); the code is not yet cleaned up or released. Every model was trained once without a fixed seed, and the 3DPW numbers are not comparable with published 3DPW results; Section 7.3 lists the limitations.

8.2. Future work

The steps below are planned, each with its state on 18 October 2026.

The Student's result (1b). The Student was training on that date, in a single job of 30 epochs. Its result is expected within days of the job's start and fills the table of Section 6.10. If it misses the 95 % bar while its error is still falling, I will continue its training from the last checkpoint and report the ratio as measured. Which model sets the bar, Final or Multi-HMR-Anny, is the supervisor's decision.

The partial shape mask (1a). Final-partial-mask masks the seven constant phenotypes of the converted labels and keeps age supervised. It was submitted on 18 October and had not finished. It tests the reading of Section 6.8 that the age label carries the gains on children and gender while the seven constants cause the collapse of muscle and weight. Its Relative Human evaluation and a demo follow.

Evidence that shape follows the person (1a). The rest-pose line-up (Figure 6.10) and the demo with predicted fingers (Figure 6.4) are done, by eye only. Still planned is a number: per-phenotype correlation, slope and R^2 between prediction and label on the AnnyOne holdout, from per-person files that already exist.

A measured spread for Final. A second run of Final with the same recipe would replace "within run-to-run variation" by a measured spread. The training code sets no seed, so this is a repeat, not a seeded run. It takes about two GPU hours and is planned for the fourth week if a slot is free.

Motion in video (2b). In the second month I plan a small temporal refiner on the frozen per-frame model. It reads the per-person tokens of Final over windows of 16 frames (1.6 s at the 10 frames per second of the labels) with a two-layer transformer and pools the body shape over each track. It is trained on the 3DPW training tracks and judged by acceleration error and PVE against the per-frame model, shape averaging and temporal smoothing. If it does not beat smoothing, the thesis will say so.

Code and model release (3). The code will get one script that reproduces the evaluations, paths set by environment variables, an environment file and a rewritten README. Final, Final-masked, Control, Base and the Student will be released with checksums, subject to the licence terms of AnnyOne, NAVER's checkpoints and 3DPW.

Open beyond this plan. Two problems have no plan inside this thesis. Children still receive adult-shaped bodies too often; weak age labels from the Relative Human training split or child-rich synthetic data are candidates. The depth slope of 0.42 means that near and far people still get similar depths; the likely reason, a pull toward the typical training distance, was not tested.

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A. Reproduction: Model Names, Scripts, Runs and Job Ids

Every number in this thesis can be traced in two steps. Each headline number is a macro in one of the files `numbers_*.tex` of the thesis source, and the comment next to it names the Slurm job and the line of its log, for example `3611336:191`. Table A.3 maps that job to the model name used in the text, the script, the log and the checkpoint. Table A.1 below maps every model name to the run name it had during the project. This appendix is the only place where job ids and old run names appear.

A.1. Model names

The chapters use only the thesis names of Table A.1; the old run and checkpoint names appear only here and in Table A.3. The names are built as follows. "L" is our ViT-L line: every model that keeps the ViT-L backbone. "L-stage 1" to "L-stage 4" are the steps of the transfer schedule that turned NAVER's released model into ours: new heads with the whole backbone trained, new heads with the backbone frozen, the last four backbone blocks trained, and the extra loss weight on neck and hand bones. Base is the end of that schedule and the start of every later fine-tune. Mixed-L and Fresh-heads-L are the two older ViT-L runs that also trained on real 3DPW images. Final and Control are the two arms of the last experiment: the same 10-epoch fine-tune of Base, with and without the two fixes. Final-masked and Final-partial-mask add a shape mask for 3DPW people to Final. Student-S is the history of ViT-S students (the number after the slash is the old version number), and Student is the ViT-B model distilled from Final. Multi-HMR-Anny is NAVER's released checkpoint, the starting weights of the L line and the baseline of requirement 1b.

Table A.1.: *Model names, part 1 of 2: the thesis name of every model, the run or checkpoint name it had during the project, the weights it started from, the backbone and the training data. Part 2 (Table A.2) has the schedule, the flags and the checkpoint path of the same rows. "Not recorded" means that no script, log or document in the repository states the detail. 3DPW "old labels" are the labels before their correction (Chapter 3); v23 labels are the corrected ones.*

Thesis name	Old run / checkpoint name	Starting weights	Backbone	Training data
Released starting point and baseline				

Thesis name	Old run / checkpoint name	Starting weights	Backbone	Training data
Multi-HMR-Anny	`multiHMR_672_L_anny` (NAVER's release)	not recorded (trained by NAVER)	ViT-L/14, NAVER's 8-layer cross-attention decoder	AnnyOne, BEDLAM, COCO, MPII (as our first stage script records it); no 3DPW
Our ViT-L line ("L"), the transfer schedule				
L-stage 1	`anny_s1_pretrained_bb_v2` (`train_step1_pretrained_bb.sh` + `_resume`)	Multi-HMR-Anny backbone; fresh heads	ViT-L/14, all blocks trained	AnnyOne
L-stage 2	`anny_s2_frozen_bb_v2`, epoch 299 ("v2 frozen")	Multi-HMR-Anny backbone; fresh heads	ViT-L/14, frozen (heads only)	AnnyOne
L-stage 3	`anny_s2_partialft_v3`, epoch 54 ("v3 partial")	L-stage 2, epoch 299	ViT-L/14, last 4 of 24 blocks trained	AnnyOne
L-stage 4	`anny_s2_boosted_v4`, epoch 299 ("v4 boosted")	L-stage 3, epoch 54	as L-stage 3	AnnyOne
Base	`anny_s2_shape_v5`, epoch 99 ("v5")	L-stage 4, epoch 299	as L-stage 3	AnnyOne (579,860 training images)
Mixed-L	`anny_teacher_v9_mixed`, epoch 62 ("v9", also epoch 149)	Base	as L-stage 3	70 % AnnyOne + 30 % 3DPW train, old labels
Fresh-heads-L	`anny_v10_scratch_mixed`, epoch 290 ("v10"; called "scratch" but not from scratch)	Multi-HMR-Anny backbone; fresh heads	as L-stage 3	70 % AnnyOne + 30 % 3DPW train, old labels, from the first step
The last experiment: two arms and two shape-mask variants				
Control	exp7 control arm `v5_root_dist1_nofix_v23_e10`, epoch 9; job 3611339	Base	as L-stage 3	70 % AnnyOne + 30 % 3DPW train, v23 labels
Final	exp7 fix arm `v5_pc_dist5_bonefix_v23_e10`, epoch 9; job 3611336	Base + Base's bone correction	as L-stage 3	as Control
Final-masked	exp7 `v5_pc_dist5_bonefix_v23_e10_shpmask`, epoch 9; job 3612325	as Final	as L-stage 3	as Control
Final-partial-mask	exp7 `v5_pc_dist5_bonefix_v23_e10_shpmask_noage`; job 3614042 (running on 18 Oct)	as Final	as L-stage 3	as Control
ViT-S students ("Student-S"), history				
Student-S/1	distillation v1, `anny_distill_vits`	ImageNet DINOv2-S; random heads; teacher Base	ViT-S/14, all blocks trained	AnnyOne
Student-S/2	distillation v2, `anny_distill_vits_v2_a`	a Student-S/1 checkpoint, recorded as `anny_distill_vits_v2` epoch 199; teacher Base	as Student-S/1	AnnyOne

Thesis name	Old run / checkpoint name	Starting weights	Backbone	Training data
Student-S/3	distillation v3, `anny_distill_vits_v3`	ImageNet DINOv2-S; 54 of 67 head tensors from Base; teacher Base	as Student-S/1	AnnyOne
Student-S/4	distillation v4, `anny_distill_vits_v4`	NAVER's `multiHMR_672_S` backbone; heads as Student-S/3	as Student-S/1	AnnyOne
Student-S/5	distillation v5, `anny_distill_vits_v5`	as Student-S/4	as Student-S/1	AnnyOne
Student-S/6	distillation v6, `anny_distill_vits_v6`	NAVER's `multiHMR_672_S` backbone; all 67 head tensors from Base through a 384 to 1024 projector	ViT-S/14 + projector	AnnyOne
Student-S/6b	distillation v6b, `anny_distill_vits_v6b`	Student-S/6, epoch 137; teacher Base	as Student-S/6	AnnyOne
Student-S/6c	distillation v6c, `anny_distill_vits_v6c`	Student-S/6b, epoch 140; teacher Base	as Student-S/6	AnnyOne
Student-S/mixed	distillation v7, `anny_distill_vits_v7_mixed`	Student-S/6b, epoch 140; teacher Base	as Student-S/6	70 % AnnyOne + 30 % 3DPW train, old labels
Student-S/final	distillation v8, `anny_distill_vits_v8_final` (evaluated as `evaltest_student_v8_final`)	Student-S/6c, epoch 149; teacher Mixed-L	as Student-S/6	as Student-S/mixed
ViT-B student (requirement 1b)				
Student	exp8 `student_vitb_from_v5_pc_dist5_bonefix_v23_e10_00009_e30`; job 3614191 (running on 18 Oct)	NAVER's `multiHMR_672_B` backbone; every head tensor from Final through a 768 to 1024 projector; teacher Final	ViT-B/14 + projector, all blocks trained	as Control

Table A.2.: *Model names, part 2 of 2: schedule (epochs × iterations per epoch × batch, learning rate), the flags that set the model apart from the row above it, and the checkpoint path relative to the project folder on the cluster (`/netscratch/najib/multi-hmr`). The learning rate halves every 30 epochs unless noted. "Planned" epochs are the script's `-max_iter`; the epoch kept is the one in the name.*

Thesis name	Schedule	Key flags	Checkpoint path
Multi-HMR-Anny	not recorded	original-repo class; evaluated with `-model_class multi_hmr_original`	`models/multiHMR/multiHMR_672_L_anny.pt`
L-stage 1	150 planned × 1,000 × 4, lr 10^{-4}; resumed from its epoch 177 at $3 \cdot 10^{-5}$	`-load_only_backbone 1` `-freeze_backbone 0`, 2D terms from epoch 20 at weight 0.1	not recorded (a side branch: not on the path to Base)

Thesis name	Schedule	Key flags	Checkpoint path
L-stage 2	150 planned × 1,000 × 4, lr 10^{-4}; resumed from its epoch 283 at $3 \cdot 10^{-5}$; epoch 299 kept	`-load_only_backbone 1 -freeze_backbone 1`, 2D terms from epoch 20	`models/multiHMR/anny_s2_frozen_bb_v2_00299.pt`
L-stage 3	300 planned × 1,000 × 4, lr 10^{-5}; epoch 54 kept	`-unfreeze_last_n_blocks 4 -mask_helper_joints_train 1`	`logs/anny_model/anny_s2_partialft_v3/checkpoints/00054.pt`
L-stage 4	300 × 1,000 × 4, lr 10^{-5}	`-boost_neck_weight 4 -boost_hand_weight 4`	`logs/anny_model/anny_s2_boosted_v4/checkpoints/00299.pt`
Base	100 × 1,000 × 4, lr 10^{-5} ($1.25 \cdot 10^{-6}$ in the last 10)	`-alpha_shape 20`, 2D terms from epoch 20	`logs/anny_model/anny_s2_shape_v5/checkpoints/00099.pt` (copy `models/multiHMR/anny_s2_shape_v5_00099.pt`)
Mixed-L	150 planned × 1,000 × 4, lr 10^{-5} halved every 60	`-mix_3dpw_ratio 0.3 -start_2d_epoch 0 -tf32 1`; no hand or shape mask	`models/multiHMR/anny_teacher_v9_mixed_00062.pt` (and `_00149.pt`)
Fresh-heads-L	300 planned × 1,000 × 4, lr 10^{-5}	`-load_only_backbone 1 -unfreeze_last_n_blocks 4 -alpha_shape 20 -mix_3dpw_ratio 0.3`; no neck or hand boost	`logs/anny_model/anny_v10_scratch_mixed/checkpoints/00290.pt` (copy `models/multiHMR/anny_v10_scratch_mixed_00299.pt`)
Control	10 × 1,000 × 4, lr 10^{-5}	Base recipe, `-start_2d_epoch 0 -mask_3dpw_hands 1 -gt_center_joint root -alpha_dist 1 -alpha_transl 1 -bone_fix none` (Appendix C)	`results/experiments/exp7_final_model/v5_root_dist1_nofix_v23_e10/train/checkpoints/00009.pt`
Final	as Control	as Control but `-gt_center_joint person_center -alpha_dist 5 -alpha_transl 5 -bone_fix` (four bones)	`results/experiments/exp7_final_model/v5_pc_dist5_bonefix_v23_e10/train/checkpoints/00009.pt` (copy `models/multiHMR/exp7_v5_final_00009.pt`)
Final-masked	as Control	Final + `-mask_3dpw_shape 1` (8 converter constants)	`results/experiments/exp7_final_model/v5_pc_dist5_bonefix_v23_e10_shpmask/train/checkpoints/00009.pt`
Final-partial-mask	as Control	Final + `-mask_3dpw_shape 1 -mask_3dpw_shape_names` with the 7 constants other than age	`results/experiments/exp7_final_model/v5_pc_dist5_bonefix_v23_e10_shpmask_noage/train/checkpoints/00009.pt` (when finished)
Student-S/1	300 planned × 1,000 × 4, lr 10^{-4}	feature distillation `-lambda_kd 1.0`, 2D terms from epoch 20	not recorded
Student-S/2	200 planned × 1,000 × 4, lr $2 \cdot 10^{-5}$ halved every 60	as Student-S/1, 2D terms from epoch 0	not recorded
Student-S/3	150 planned × 1,000 × 4, lr $5 \cdot 10^{-5}$ halved every 60	`-init_heads_from_teacher 1 -lambda_kd_out 1.0`	not recorded

Thesis name	Schedule	Key flags	Checkpoint path
Student-S/4	150 planned × 1,000 × 8, lr $7 \cdot 10^{-5}$ halved every 60	`-pretrained` NAVER ViT-S `-load_only_backbone 1`	not recorded
Student-S/5	as Student-S/4	`-freeze_heads_epochs 5 -start_2d_epoch 5`	not recorded
Student-S/6	as Student-S/4; stopped at epoch 138 by the time limit	`-head_dim 1024`	`logs/anny_model/anny_distill_vits_v6/checkpoints/00137.pt`
Student-S/6b	150 planned × 1,000 × 8, lr $2 \cdot 10^{-5}$ halved every 60	`-teacher_bf16 1`	`logs/anny_model/anny_distill_vits_v6b/checkpoints/00140.pt`
Student-S/6c	150 × 1,000 × 16, lr $3 \cdot 10^{-5}$ halved every 60	batch 16	`models/multiHMR/anny_distill_vits_v6c_00149.pt`
Student-S/mixed	150 planned × 1,000 × 8, lr $2 \cdot 10^{-5}$ halved every 60; epoch 130 kept	`-mix_3dpw_ratio 0.3`	`models/multiHMR/anny_distill_vits_v7_mixed_00130.pt`
Student-S/final	150 planned × 1,000 × 16, lr $3 \cdot 10^{-5}$ halved every 60; epoch 52 reported	teacher Mixed-L, `-mix_3dpw_ratio 0.3`	`models/multiHMR/anny_distill_vits_v8_final_*.pt` (newest copy)
Student	30 × 1,000 × 8, lr $3 \cdot 10^{-5}$ halved every 12	Final's targets and masks + `-head_dim 1024 -init_heads_from_teacher 1 -freeze_heads_epochs 5 -teacher_bf16 1 -lambda_kd 1.0 -lambda_kd_out 1.0`, teacher's bone correction (Appendix C)	`results/experiments/exp8_distill_vitb/student_vitb_from_v5_pc_dist5_bonefix_v23_e10_00009_e30/train/checkpoints/` (last epoch `00029.pt`, when finished)

A.2. Environment

All training, evaluation and demo jobs ran on the DFKI Pegasus Slurm cluster, from the repository clone in `/netscratch/najib/multi-hmr`. The repository is public on GitHub (`saadnajib/Multi-Person-Whole-Body-Human-Mesh-Recovery-in-a-Single-Shot-with-Annyone-dataset-and-3dpw-dataset`); checkpoints, logs and datasets are not in it. Each job script prints the short git hash of the clone into the run's `settings.txt`, so a run can be matched to a commit.

Every job runs inside the container image `ubuntu+20.04_v4.sqsh` (in `/netscratch/najib/ContainerImages/`), started by `srun` with the option `--container-image` and the mounts `/fscratch`, `/netscratch`, `/home/najib` and `/dev/shm`. Inside it, `slurm/env.sh` is sourced first. It activates the conda environment `multihmr` (Python 3.9, PyTorch 2.0.1 with CUDA 11.7, xformers 0.0.20, torchvision 0.15.2), moves the Hugging Face, Torch and Anny caches to `/netscratch/najib`, sets one thread per data-loader worker, and strips the `X | None` type hints from the downloaded DINOv2 hub code, which Python

3.9 rejects. `slurm/egl.sh` then installs the EGL libraries for the headless renderer; a failure there only disables the mesh overlays.

That PyTorch build has GPU kernels up to compute capability 8.6, so the training and evaluation scripts ask for A100, A40, L40S and RTX A6000 nodes and refuse an H100 or H200 (compute capability 9.0). The V100 nodes `gifu` and `glendale` are excluded: every converter task sent there failed within a minute with an argument-parser error whose cause is unknown. Only the 3DPW converter ran on H100 nodes, through a small virtual environment with PyTorch 2.1.2 for CUDA 12.1 on top of `multihmr` (`PY_VENV=/netscratch/najib/venv_cu121`). Steps that need only NumPy (building the 3DPW label files, the re-fit check) run on the login node. The model-cost script needs a GPU node, because `model.py` imports the renderer and the login node has no EGL.

A.3. Runs, jobs, logs and checkpoints

Table A.3 lists every job behind a number, a table or a figure of the thesis. Jobs submitted on the evening of 18 October (the partial shape mask, the Student and the two figure demos) were still pending or running when this appendix was written; their rows name the log and output folder they write. The Student-S runs on the v23 labels, submitted the same evening, have returned (Table 6.5b).

Table A.3.: *Run names, Slurm jobs, what each produced, and where the log and the checkpoint or output lie. Model names as in Table A.1. Paths are relative to the project folder on the cluster; logs lie in `logs/`.*

Thesis name or item	Job(s)	What	Log; checkpoint or output
Models			
Base	3570222; re-trace 3571596	holdout trace (V1 PVE, placement)	`diag_holdout_3570222.out`; checkpoint `logs/anny_model/anny_s2_shape_v5/checkpoints/00099.pt`
Mixed-L	3571963, 3571964	holdout traces of epochs 62 and 149	`diag_holdout_<job>.out`; `models/multiHMR/anny_teacher_v9_mixed_00062.pt`
Fresh-heads-L	3571794	holdout trace of epoch 290	`diag_holdout_3571794.out`; `logs/anny_model/anny_v10_scratch_mixed/checkpoints/00290.pt`
Final	3611336 (smoke test 3611333)	training, holdout trace, 3DPW test, all in one job	`exp7_final_model_3611336.out`; `results/experiments/exp7_final_model/v5_pc_dist5_bonefix_v23_e10/train/checkpoints/00009.pt`, copy `models/multiHMR/exp7_v5_final_00009.pt`

Thesis name or item	Job(s)	What	Log; checkpoint or output
Control	3611339	the same without the two fixes	`exp7_final_model_3611339.out`; `results/experiments/exp7_final_model/v5_root_dist1_nofix_v23_e10/`
Final-masked	3612325	Final with the shape mask	`exp7_final_model_3612325.out`; `results/experiments/exp7_final_model/v5_pc_dist5_bonefix_v23_e10_shpmask/`
Final-partial-mask	3614042	Final with the seven 0.5 constants masked, age kept (running on 18 Oct)	`exp7_final_model_3614042.out`; `results/experiments/exp7_final_model/v5_pc_dist5_bonefix_v23_e10_shpmask_noage/`
Student	3614191 (smoke test 3613926)	ViT-B student of Final, 30 epochs, with its own evaluation (running on 18 Oct)	`exp8_distill_vitb_3614191.out`; `results/experiments/exp8_distill_vitb/student_vitb_from_v5_pc_dist5_bonefix_v23_e10_00009_e30/` (`final_table.json`)
Multi-HMR-Anny	3614192 (first try 3613928 crashed, FFN width)	NAVER's checkpoint evaluated with our code: holdout and 3DPW test	`eval-test3dpw-3614192_0.out`; `models/multiHMR/multiHMR_672_L_anny.pt`
Model cost	3614947	parameters, GFLOPs, latency of Multi-HMR-Anny and Final (A100)	`model_cost_3614947.out`; `results/diagnostics/model_cost_naver_vs_final.json`
Evaluations and demos			
3DPW test (v23)	3612073, 3612074, 3612075	Base, Mixed-L, neck pilot; Final and Control inside their own jobs	`eval-test3dpw-<job>_0.out`, lines 117–125
3DPW test (v23), Student-S/6c	3615020_2	`anny_distill_vits_v6c_00149.pt` on all 8,190 v23 test images and the AnnyOne test-100 (Table 6.5b)	`eval-test3dpw-3615020_2.out`, line 119
3DPW test (v23), Student-S/final	3615020_4	`anny_distill_vits_v8_final_00052.pt`, the same two sets	`eval-test3dpw-3615020_4.out`, line 118

Thesis name or item	Job(s)	What	Log; checkpoint or output
Relative Human	3611285_0, 3611285_1, 3612131, 3613616; smoke test 3608336	Base, neck pilot, Final, Final-masked	`eval_rh_<job>_<task>.out`; `results/diagnostics/relative_human_val_<tag>/`
Demos	3581312, 3612078, 3613617	neck pilot, Final, Final-masked, each next to Base, on 4 photos + 8 3DPW test frames	`demo-compare-<job>.out`; montages `demo_montage_photos/`, `demo_montage_3dpwtest/`
Demo, fingers	3615018	Base and Final with predicted fingers (`-relax_hands 0`; submitted 18 Oct evening)	`demo-compare-3615018.out`; script `slurm/demo/run_demo_compare_hands.sh` (a copy made on the cluster)
Demo, rest pose	3615019	Base, Final, Final-masked in rest pose, shape only (submitted 18 Oct evening)	`demo-compare-3615019.out`; script `slurm/demo/run_demo_compare_rest.sh` (a copy made on the cluster)
3DPW labels			
Label check	3575369 (old), 3611284 (v23)	conversion check before and after the re-fit	`check_3dpw_conversion_<job>.out`; `results/diagnostics/check_3dpw_conversion_v23/`
Re-fit (v2.3)	3589270 (test), 3590385 + 3602813; crashes 3587852, 3587853	converter v2.3 on all 87 person-sequences	`smpl2anny-all-<job>_<task>.out`; `threedpw_anny_v23/labels/`; `logs/check_refit_all_v23.log`
Label rebuilds	3579266, 3581320	rebuilds from the old fits	`3dpw-labels-<job>.out`; `threedpw_anny_full/labels_rebuilt/`
Pilots and diagnostics			
Phenotypes, youngest 50	3569948, 3569950	AnnyOne phenotype distribution; in-sample youngest people	`diag_pheno_3569948.out`, `diag_young_3569950.out`
Real-photo ages	3569949, 3572419, 3572420	crop-camera read: Base, sampler pilot, its control	`diag_cropK_<job>.out`

Thesis name or item	Job(s)	What	Log; checkpoint or output
Depth pilots (exp1)	3571598, 3571599, 3572418, 3572483; 3571933, 3571934, 3572482	2-epoch fix ×1 / control / ×5 / ×20; 10-epoch fix ×1 / control / ×5	`exp1_depth_anchor_<job>.out`; `results/experiments/exp1_depth_anchor/<run>/`
Sampler pilot (exp2)	3571955, 3571956; measurement 3571601	sampler pilot and control; batch ages only	`exp2_children_<job>.out`; `results/experiments/exp2_children/`
Head loss pilot (exp3)	3571957; controls 3571958, 3571960	loss pilot and two identical controls (noise)	`exp3_neck_global_rot_<job>.out`, `exp4_fingers_3571960.out`
Finger pilot (exp4)	3571959 (control 3571960)	region-normalised loss; crop trace and hand size in the control job	`exp4_fingers_<job>.out`; `results/experiments/exp4_fingers/`
Geodesic pilot (exp5)	3573520, 3573521	geodesic rotation loss and its L1 control	`exp5_neck_fix_<job>.out`; `results/experiments/exp5_neck_fix/`
Per-bone dumps	3573518, 3573519	per-bone angles of eight checkpoints	`diag_neck_bones_<job>.out`; `results/diagnostics/neck_bones/`
Bone correction (exp6)	3579014 (Base), 3579015 (Fresh-heads-L)	correction without training and with a 2-epoch fine-tune (the neck pilot)	`exp6_bone_fix_<job>.out`; `results/experiments/exp6_bone_fix/anny_s2_shape_v5_00099_n200_s0/bone_fix.json`; neck pilot `models/multiHMR/exp6_v5_bonefix_00001.pt`

A.4. Scripts

Table A.4 lists the scripts behind the experiments, one line each. Each Slurm script documents its recipe, its guards and its success rule in its header; the training scripts of the L line and of Student-S lie in `slurm/history/` and `slurm/train/` (Table A.1).

Table A.4.: *Scripts behind the experiments and what each produces. Paths relative to the repository root.*

Script	What it produces
Training experiments (`slurm/experiments/`)	

Script	What it produces
`exp1_depth_anchor.sh`	depth pilots: fine-tune of Base with the targets at the head joint (or at the pelvis, the control), holdout trace with the placement table
`exp2_children.sh`	sampler pilot: children per batch measured, fine-tune with age-balanced sampling, holdout, manifest and youngest-50 traces
`exp3_neck_global_rot.sh`	head-loss pilot: fine-tune with an extra loss on the camera-frame orientation of head, neck03 and wrists
`exp4_fingers.sh`	finger pilot: hand sizes in tokens, a test-time person-crop trace, a fine-tune with the region-normalised rotation loss
`exp5_neck_fix.sh`	geodesic pilot: fine-tune with the geodesic rotation loss, then a per-bone dump
`exp6_bone_fix.sh`	bone correction: estimates the constant per flipped bone, tests it without training and after a 2-epoch fine-tune
`exp7_final_model.sh`	Final (`ARM=fix`), Control (`ARM=control`), Final-masked and Final-partial-mask: label guards, training, holdout trace with per-bone dump, comparison table, 3DPW test
`exp8_distill_vitb.sh`	Student: guards, distillation from Final, the Student's own bone correction, holdout traces of Student and teacher, model cost, 3DPW tests, the final table `final_table.json`
Evaluation (`slurm/eval/`)	
`run_eval_3dpw_test.sh`	`train.py -eval_only 1`: AnnyOne test-100 (V0) and every 3DPW test image of the chosen labels; overrides `EVAL_LABELS`, `MODEL_ENTRY`, `EVAL_BONE_FIX`, `EVAL_MODEL_CLASS`
`run_eval_baseline_naver.sh`	the same evaluation for Multi-HMR-Anny, built with its own model class
`run_eval_3dpw.sh`, `run_eval_3dpw_teacher.sh`	older evaluation of one checkpoint on the AnnyOne holdout and the 3DPW validation split (history)
Diagnostics (`slurm/diagnostics/`)	
`diag_holdout.sh`	holdout trace (`diag_holdout_trace.py`): settings V0 to V3, placement, per-person CSV, panels
`diag_manifest.sh`	diagnostic manifest, ground-truth decode overlays, manifest traces
`diag_bodymodel.sh`	body-model probe: Anny age sweep, 16 single-joint perturbations, the 88 helper bones
`diag_gradients.sh`	gradient check of every head output
`diag_overfit.sh`	capped overfit on the 40 manifest training images
`diag_phenotypes.sh`, `diag_young.sh`	AnnyOne phenotype distribution; trace of the 50 youngest training people
`diag_crop_camera.sh`	predicted ages on the real photos with a corrected crop camera
`diag_neck_bones.sh`	per-bone dump of the neck chain and other bones, one trace per checkpoint
`check_3dpw_conversion.sh`	conversion check of a 3DPW label set against the original 3DPW ground truth

Script	What it produces
`eval_relative_human.sh`	Relative Human evaluation of one checkpoint (`eval_relative_human.py`)
`model_cost.sh`	parameters, GFLOPs and latency of one or more checkpoints on a GPU node (`model_cost.py`)
Demos (`slurm/demo/`)	
`run_demo_compare.sh`	two or more models on the same 4 photos and 8 3DPW test frames, bodies placed from the detector box; model list `DEMO_MODELS`
`run_demo.sh`, `run_demo_3dpw.sh`, `run_demo_syn.sh`, `run_demo_student.sh`	single-model demos on the photos, random 3DPW frames, AnnyOne holdout images, a student checkpoint
Python scripts	
`scripts/analysis/model_cost.py`	analytic backbone FLOPs, hooked head FLOPs, parameters, and with `-bench` the median latency at batch 1
`scripts/diagnostics/eval_relative_human.py`	per Relative Human age group: recall, predicted age and height, child-vs-adult AUC, gender accuracy; `summary.txt` and `per_person.csv`
`scripts/diagnostics/compute_bone_fix.py`	the constant correction per flipped bone, as `bone_fix.json`, with the global guard
`scripts/data/smpl_to_anny.py`	converter v2.3: fits Anny to the SMPL joints of one 3DPW person-sequence, writes one `.npz` per person
`scripts/data/build_threedpw_anny_labels.py`	label files and split indices from the fits (login node)
`scripts/data/check_refit_task.py`	re-fit post-check: gender convention, shared frame grid, error per person (login node)
`slurm/data/run_smpl_to_anny_task.sh`	Slurm array over the 87 person-sequences of `configs/threedpw_tasks.txt`

A.5. Commands

The commands below are the ones that produced the thesis numbers, copied from the script headers and from the project handover. Each runs from the login node in the repository clone, after `git pull` and with the conda environment active (the submit modes run their label checks there before calling `sbatch`).

Final and Control. Guards only, then the two arms (5-hour limit each; one job each). The comparison table is rebuilt on the login node once both have finished:

```
cd /netscratch/najib/multi-hmr && git pull && conda activate multihmr && CHECK_ONLY=1
        bash slurm/experiments/exp7_final_model.sh
cd /netscratch/najib/multi-hmr && bash slurm/experiments/exp7_final_model.sh
cd /netscratch/najib/multi-hmr && ARM=control bash slurm/experiments/exp7_final_model.
        sh
cd /netscratch/najib/multi-hmr && TABLE_ONLY=1 bash slurm/experiments/exp7_final_model.
        sh
```

The smoke test, Final-masked and Final-partial-mask:

```
cd /netscratch/najib/multi-hmr && EPOCHS=1 N_ITERS=20 EVAL_TEST=0 VAL_3DPW_N=20 N_VIS=2
        bash slurm/experiments/exp7_final_model.sh
cd /netscratch/najib/multi-hmr && MASK_3DPW_SHAPE=1 bash slurm/experiments/
       exp7_final_model.sh
cd /netscratch/najib/multi-hmr && MASK_3DPW_SHAPE=1 MASK_3DPW_SHAPE_NAMES=muscle,weight
       ,cupsize,firmness,african,asian,caucasian bash slurm/experiments/
       exp7_final_model.sh
```

3DPW test of the older models. Base, Mixed-L and the neck pilot on the v23 labels, one job each:

```
cd /netscratch/najib/multi-hmr && for M in "logs/anny_model/anny_s2_shape_v5/
       checkpoints/00099.pt|teacher_v5" "models/multiHMR/anny_teacher_v9_mixed_00062.
       pt|teacher_v9_e62" "models/multiHMR/exp6_v5_bonefix_00001.pt|exp6_v5_bonefix";
       do EVAL_LABELS=/netscratch/najib/multi-hmr/threedpw_anny_v23/labels MODEL_ENTRY
       ="/netscratch/najib/multi-hmr/${M%%|*}|0|dinov2_vitl14|${M##*|}" sbatch --array
       =0 slurm/eval/run_eval_3dpw_test.sh; done
```

Student-S/6c and Student-S/final on the same labels (array tasks 2 and 4 of the script's model list):

```
cd /netscratch/najib/multi-hmr && EVAL_LABELS=/netscratch/najib/multi-hmr/
       threedpw_anny_v23/labels sbatch --array=2,4 slurm/eval/run_eval_3dpw_test.sh
```

Multi-HMR-Anny baseline. Checks, then one job (40-minute limit):

```
cd /netscratch/najib/multi-hmr && CHECK_ONLY=1 bash slurm/eval/run_eval_baseline_naver.
       sh && bash slurm/eval/run_eval_baseline_naver.sh
```

Relative Human. Base and the neck pilot (array tasks 0 and 1), Final, and Final-masked; the bone correction comes from each checkpoint's saved arguments:

```
cd /netscratch/najib/multi-hmr && sbatch --array=0-1 slurm/diagnostics/
       eval_relative_human.sh
cd /netscratch/najib/multi-hmr && MODEL=/netscratch/najib/multi-hmr/models/multiHMR/
       exp7_v5_final_00009.pt sbatch --array=0 slurm/diagnostics/eval_relative_human.
       sh
cd /netscratch/najib/multi-hmr && MODEL=/netscratch/najib/multi-hmr/results/experiments
       /exp7_final_model/v5_pc_dist5_bonefix_v23_e10_shpmask/train/checkpoints/00009.
       pt TAG=exp7_shpmask_00009 sbatch --array=0 slurm/diagnostics/
       eval_relative_human.sh
```

The scores can be recomputed from the cached predictions without a GPU, for example with another IoU:

```
cd /netscratch/najib/multi-hmr && conda activate multihmr && python scripts/diagnostics
       /eval_relative_human.py --score_only --out results/diagnostics/
       relative_human_val_exp7_v5_final_00009 --iou 0.5
```

Model cost. Multi-HMR-Anny and Final, with latency (the script submits itself):

```
cd /netscratch/najib/multi-hmr && bash slurm/diagnostics/model_cost.sh
```

The analytic backbone table for ViT-S, ViT-B and ViT-L runs anywhere:

```
python scripts/analysis/model_cost.py --no-build --all
```

Student. Smoke test, then the full run with a 20-hour limit:

```
cd /netscratch/najib/multi-hmr && conda activate multihmr && git pull && SMOKE=1 bash
        slurm/experiments/exp8_distill_vitb.sh
cd /netscratch/najib/multi-hmr && conda activate multihmr && git pull && TIME=20:00:00
        bash slurm/experiments/exp8_distill_vitb.sh
```

A.6. What cannot be reproduced exactly

The training code sets no random seed. The `-seed` of the holdout trace only picks which training images the bone correction is estimated on. The order in which training images are drawn (with replacement) therefore differs between two runs of the same command, and so does the number of optimiser steps skipped for non-finite gradients (16 in Final, 10 in Control, out of 10,000). Two runs of one 2-epoch configuration differed by 7.6 mm PVE, 1.9 deg head direction and 0.8 recall points (Table 5.7); a repeat of Final or Control is expected to land within such a margin, not on the same numbers. The two arms also ran on different GPU types (Final on an A100, Control on an A40, both with TF32), which changes the speed and, at the level of rounding, the arithmetic. Evaluating a fixed checkpoint involves no sampling; that it repeats exactly across GPU types was not tested.

Three further limits apply. The history scripts of the L line and of Student-S were edited after their runs, so their recipes in Table A.1 are as recorded there, not re-run. `train.py` keeps only the last two or three checkpoints of a run, so intermediate epochs are gone. The checkpoints, the AnnyOne and 3DPW data and the Relative Human annotations are not in the repository; 3DPW and Relative Human come with their own licences.

B. SMPL–Anny Joint Correspondence

[TODO: One paragraph: how the correspondence was chosen (rest skeletons compared by geometry) and how the virtual-joint offsets were measured; point to Chapter 3.]

The correspondence used by the final converter (v2.3), with the rest-pose offset of each virtual joint. Offsets are measured on the two rest skeletons after aligning them at the hip midpoint and scaling SMPL to Anny's hip–neck length (factor 0.864); limb joints use the bone head.

SMPL joint	Anny bone	virtual-joint offset (cm)
pelvis	`root`	11.6
left / right hip	`upperleg01.L/R`	4.3
spine1	`spine03`	3–7
spine2	`spine02`	3–7
spine3	`spine01`	3–7
neck	`neck01`	0 (reference)
left / right collar	`clavicle.L/R`	5
head	`head`	—
shoulder	`upperarm01.L/R`	—
elbow	`lowerarm01.L/R`	—
wrist	`wrist.L/R`	—
hand	`finger2-1.L/R`	—
knee	`lowerleg01.L/R`	—
ankle	`foot.L/R`	—
foot	`toe1-1.L/R`	—

Table B.1.: *SMPL→Anny correspondence of the final converter. The fit rotates only the root and 18 bones; the other 144 of Anny's 163 bones stay at their rest rotation, including* `neck02`, `neck03`, `head`, *the jaw and every finger bone, so the head, hand and foot targets are reached through their parent bones. The first converter mapped* `spine1` *to* `spine01`*, 25 cm too high, and the hips to* `pelvis.L/R`*, whose heads lie 4 cm from the hip joints.*

C. Flags and Weights

This appendix lists every command-line flag of the training runs of Final and Control (Table C.1) and of the Student (Table C.2), with the default in the code, the value used and what the flag does. The loss weights appear in Table 4.2 of Chapter 4 and are only named here. Flags that are not passed keep their default. The commands that start the runs are in Section A.5.

Table C.1.: *Training flags of Final and Control. "Value" is Final's; where Control differs, its value follows in brackets. Paths are relative to the project folder on the cluster. Both runs start from Base and see 10 epochs of 1,000 iterations at batch 4, that is 40,000 images.*

Flag	Default	Value (Control)	Meaning
Start and network			
`--pretrained`	none	Base checkpoint	weights the run starts from
`--pretrained_remap`	1	0	no renaming of weight names; Base was saved by this code
`--load_only_backbone`	0	0	load the heads too, not only the backbone
`--backbone`	ViT-S	`dinov2_vitl14`	DINOv2 ViT-L with 14-pixel patches
`--img_size`	336	672	input side in pixels (48×48 tokens)
`--person_center`	head	head	the joint each detection and its predicted position refer to
`--freeze_backbone`	0	1	freeze the backbone ...
`--unfreeze_last_n_blocks`	0	4	... except its last 4 of 24 blocks and final norm
`--bone_fix`	off	`results/experiments/exp6_bone_fix/anny_s2_shape_v5_00099_n200_s0/bone_fix.json` (`none`)	constant output rotation on neck02, head, metacarpal1.L and finger4-1.R, saved in every checkpoint and re-applied on load (Section 4.5)
Targets and losses			
`--gt_center_joint`	root	`person_center` (`root`)	the joint the depth, translation, detection-cell and offset targets describe: the head, as the network anchors (Control: the pelvis)
`--alpha_dist`, `--alpha_transl`	1, 1	5, 5 (1, 1)	weights of the depth and translation terms (Table 4.2)

Flag	Default	Value (Control)	Meaning
`--alpha_shape`, `--alpha_rotmat`, `--alpha_j2d`, `--alpha_v2d`, `--alpha_global_rot`	1, 0.1, 1, 1, 0	20, 0.1, 0.1, 0.1, 0	loss weights (Table 4.2)
`--boost_neck_weight`, `--boost_hand_weight`	1, 1	4, 4	extra weight on neck, head and hand bones, renormalised to mean 1 (Table 4.2)
`--rotmat_loss`	l1	l1	rotation term as L1 over the nine matrix entries
`--global_rot_bones`	head, neck03, wrists	default	bones of the global-rotation term; unused at weight 0
`--start_2d_epoch`	10	0	2D terms from the first epoch (the epoch counter restarts in every run)
`--mask_helper_joints_train`	0	1	hold the 88 helper bones at identity and drop their rotation targets
`--use_anny_shape`	0	1	take the phenotype targets from the dataset
`--mask_3dpw_hands`	0	1	drop the 38 finger, thumb and metacarpal bones of 3DPW people from the rotation terms
`--mask_3dpw_shape`	0	not passed (0)	shape mask for 3DPW people; on only in Final-masked and Final-partial-mask
`--debug_anchor`	0	2	print predicted and target anchors for the first 2 iterations (a check, no effect on training)
Data			
`--train_data`	BEDLAM	AnnyOne	synthetic training set
`--train_n`, `--val_anny_n`	0, 0	0, 100	all AnnyOne images except the last 100, which form the holdout
`--exclude_indices_file`	none	the 55 manifest images	never sample the diagnostic-manifest images; sampling is then with replacement
`--mix_3dpw_labels`	none	`threedpw_anny_v23/labels`	corrected 3DPW-Anny labels mixed into training
`--mix_3dpw_ratio`, `--mix_3dpw_split`	0.3, train	0.3, train	30 % of the images of a batch from 3DPW train, on average
`--val_3dpw_split`, `--val_3dpw_n`	validation, 0	validation, 300	evaluate on 300 evenly spaced 3DPW validation images at every evaluation
Optimisation and bookkeeping			
`--batch_size`	2	4	images per step
`--n_iters_per_epoch`, `--max_iter`	100, 10,000	1,000, 10,000	10 epochs of 1,000 steps
`--learning_rate`	$5 \cdot 10^{-6}$	10^{-5}	Adam learning rate
`--lr_decay_every`, `--lr_decay_gamma`	30, 0.5	30, 0.5	halve every 30 epochs, so constant over the 10 epochs

Flag	Default	Value (Control)	Meaning
`--amp`, `--tf32`	1, 1	0, 1	no mixed precision (it produced NaNs with the Anny layer); TF32 matrix products
`--num_workers`	0	12	data-loader processes
`--eval_freq`, `--nb_max_ckpt`, `--log_freq`	1, 10, 100	1, 2, 50	evaluate every epoch, keep the last 2 checkpoints, log every 50 steps
`--save_dir`, `--name`	logs, trainval	run folder, train	output folder `results/experiments/exp7_final_model/<run>/train`
Job settings of the script (environment variables)			
`EVAL_TEST`, `N_VIS`	1, 12	1, 12	after training: holdout trace with 12 panels, then the 3DPW test on the v23 labels
`TIME`, `PARTITIONS`	from the epochs	5 h; A100, L40S, RTX A6000	Slurm time limit (60 + 60 + 18 min per epoch) and GPU types

Table C.2.: *Training flags of the Student. Flags with the same value as in Final are grouped in the last block; the targets, masks and loss weights equal Final's so that the Student learns the corrected depth. 30 epochs of 1,000 iterations at batch 8, that is 240,000 images.*

Flag	Default	Value	Meaning
Student network and its start			
`--backbone`	ViT-S	`dinov2_vitb14`	DINOv2 ViT-B with 14-pixel patches
`--pretrained`	none	`models/multiHMR/multiHMR_672_B.pt`	NAVER's mesh-recovery ViT-B as the backbone start
`--pretrained_remap`, `--load_only_backbone`	1, 0	1, 1	rename NAVER's weight names; keep only the backbone
`--min_backbone_match`	90	90	stop if less than 90 % of the backbone loads
`--freeze_backbone`	0	0	train the whole backbone
`--head_dim`	0	1024	build the heads at the teacher's width, fed by a 768 to 1024 linear projector
Teacher and distillation			
`--distill_teacher_ckpt`	none	Final checkpoint	frozen teacher; turns the run into distillation
`--teacher_bf16`	0	1	teacher backbone in bfloat16 (no gradients)
`--init_heads_from_teacher`	0	1	copy every head tensor of matching name and shape from the teacher
`--freeze_heads_epochs`	0	5	keep the copied heads frozen for 5 epochs while backbone and projector adapt

Flag	Default	Value	Meaning
`--lambda_kd`, `--kd_temperature`, `--kd_softmax_dim`	1.0, 4.0, channel	1.0, 4.0, channel	feature distillation: KL divergence between softened per-token channel distributions (Table 4.2)
`--lambda_kd_out`	0	1.0	output distillation: L1 to the teacher's rotations, phenotypes and depth (Table 4.2)
`--bone_fix`	off	the teacher's correction	the four-bone correction of Final during training; after training the Student's own correction is estimated on 200 training images (seed 0) and used for every evaluation
Optimisation and bookkeeping			
`--batch_size`	2	8	images per step
`--n_iters_per_epoch`, `--max_iter`	100, 10,000	1,000, 30,000	30 epochs of 1,000 steps
`--learning_rate`	$5 \cdot 10^{-6}$	$3 \cdot 10^{-5}$	Adam learning rate
`--lr_decay_every`, `--lr_decay_gamma`	30, 0.5	12, 0.5	halve after epochs 12 and 24
`--eval_freq`, `--nb_max_ckpt`, `--log_freq`	1, 10, 100	5, 3, 50	evaluate every 5 epochs, keep the last 3 checkpoints
`--brightness`, `--contrast`, `--saturation`, `--hue`	0	0.2, 0.2, 0.2, 0.05	parsed but read nowhere in the code: no effect, no augmentation was used
`TIME`	14 h	20 h	Slurm time limit
As in Final (Table C.1)			
`-train_data AnnyOne -train_n 0 -val_anny_n 100 -person_center head -img_size 672 -mask_helper_joints_train 1 -use_anny_shape 1 -alpha_shape 20 -alpha_j2d 0.1 -alpha_v2d 0.1 -boost_neck_weight 4 -boost_hand_weight 4 -rotmat_loss l1 -alpha_rotmat 0.1 -alpha_global_rot 0 -start_2d_epoch 0 -gt_center_joint person_center -alpha_dist 5 -alpha_transl 5 -debug_anchor 2 -exclude_indices_file` (the 55 manifest images) `--mix_3dpw_labels` (v23) `-mix_3dpw_ratio 0.3 -mix_3dpw_split train -val_3dpw_split validation -val_3dpw_n 300 -mask_3dpw_hands 1 -num_workers 12 -amp 0 -tf32 1`; no shape mask.			

D. Extra Figures

This appendix holds the figures that the chapters refer to but do not show: the second depth example, the two-frame neck pair, two more bone-correction pairs on the holdout, the other four 3DPW test frames of the demo, the four real photos, all twelve demo images with predicted fingers and the nine rest-pose line-ups that Chapter 6 does not show. Every panel is judged by eye; where a body is placed from the detector box, depth errors are hidden.



Figure D.1.: *Holdout image 579860 rendered at each model's own predicted depth: ground truth, Base, and the depth pilot (head-joint targets, depth weight ×5, 10 epochs, AnnyOne only). Before the fix the ground-truth head shows above a shrunken body placed too far away. Synthetic image; one case, not a typical one. Companion of Figure 6.1.*

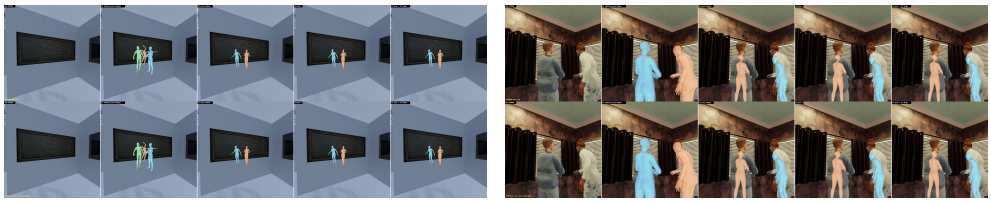
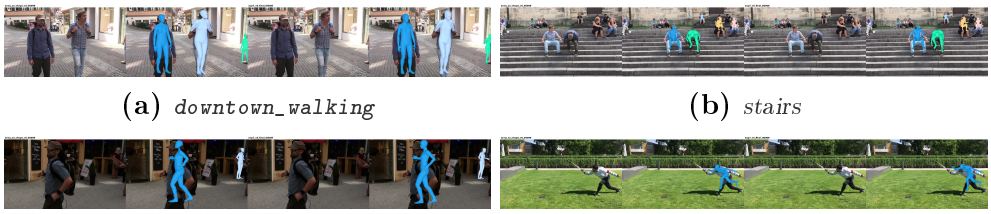


Figure D.2.: *Holdout images 579864 (left) and 579866 (right): Base as trained next to the same checkpoint with the bone correction applied at the output, no training (Section 5.4). Synthetic images, bodies at the model's own predicted depth, so the depth bug still draws them too small; necks judged by eye. Companions of Figure 5.1.*



Figure D.3.: *Real 3DPW test frames* `downtown_bar_00` *701 (top) and* `office_phoneCall_00` *440 (bottom): Base next to the neck pilot (the bone correction and a 2-epoch fine-tune); necks ringed. Bodies placed from the detector box, fingers at rest pose; judged by eye, no metric. The two columns differ by the correction and by the fine-tune.*



(a) `downtown_walking` **(b)** *stairs*

(c) `downtown_runForBus` **(d)** *fencing*

Figure D.4.: *The other four 3DPW test frames of the demo, each shown as photo and Base (left half), then photo and Final (right half; the labels in the image name the two checkpoints). Bodies placed from the detector box, fingers at rest pose; judged by eye, no metric. The first four frames are in Figure 6.7.*



Figure D.5.: *The four real photos of the demo (garden, tennis, stage, party), never used in training, each shown as photo and Base, then photo and Final; bodies placed from the detector box, fingers at rest pose; judged by eye, no metric. The tennis photo also appears in Figure 6.9.*



Figure D.6.: *All twelve demo images with predicted fingers (not set to rest pose), each as photo and Base, then photo and Final (labels in the image). Top to bottom: the garden, tennis and stage photos, the 3DPW test frames fencing, stairs,* `downtown_walking` *and* `downtown_runForBus`*; then the party photo, guitar,* `downtown_bar`*,* `office_phoneCall` *and the weekly market. Demo setting, bodies placed from the detector box; Base still has the flipped left index-finger base (metacarpal1.L) and right ring-finger base, Final has the correction. In most of these images the hands are a few pixels wide and cannot be judged; Figure 6.4 zooms on the three where they can. Judged by eye; no metric.*

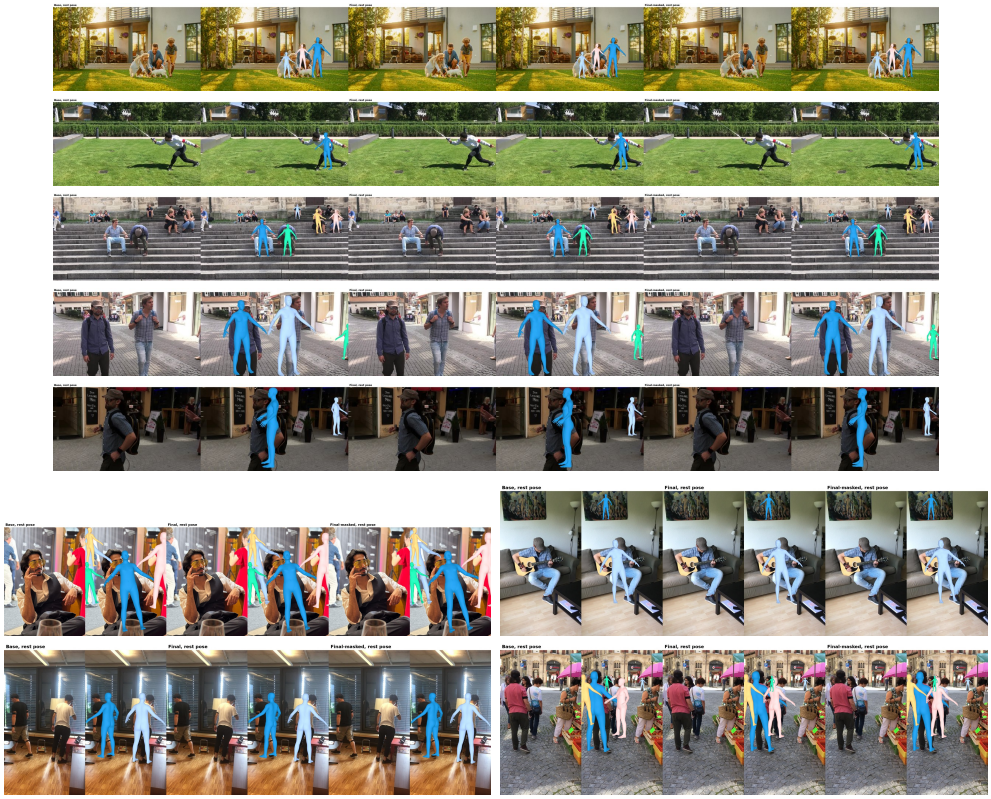


Figure D.7.: *The other nine demo images in rest pose, each as photo and Base, photo and Final, photo and Final-masked (labels in the image). Top to bottom: the garden photo, the 3DPW test frames fencing, stairs, `downtown_walking` and `downtown_runForBus`; then the party photo, guitar, `office_phoneCall` and the weekly market. Every detected person is in rest pose, so only body shape differs. Demo setting, bodies placed from the detector box; at this size only gross differences of build can be seen. Judged by eye; no metric. The other three images are in Figure 6.10.*

E. Pilot Results in Full

Before the final runs, six short pilots tested one change each against a control arm. Table E.1 gives, for each, what it changed, the success rule written into its script before it ran, the result and the control. All pilots fine-tune Base on AnnyOne only, at learning rate 10^{-5} and batch 4, for 2 epochs (8,000 images) unless noted, and are read on the 100-image AnnyOne holdout in setting V1. Each is one run without a seed, so every result must be read against the run-to-run variation of Section 5.7, summarised at the end of this appendix.

Table E.1.: *The six pilot experiments. V1, 100-image AnnyOne holdout, matched people; fine-tunes of Base, one run each, without a seed. The rule is the success rule written into each script before the run. Head direction is the global head orientation error.*

Pilot	What changed	Success rule	Result	Control
Depth pilots (2 and 10 epochs)	training targets at the head joint instead of the pelvis; depth and translation weights ×1, ×5, ×20	slope toward 1 and intercept toward 0 m, with PVE ≤ 71.5 mm and recall ≥ 89.2 %	rule not met (PVE or recall missed in every run, controls too); root error 1.21 → 0.54 m, size 0.63 → 0.84, slope 0.33 → 0.38 (×5, 10 epochs)	same schedule, targets at the pelvis, weight ×1 (one 2-epoch, one 10-epoch)
Sampler pilot	age-balanced sampling: children (age < 0.3) per batch 2.1 → 9.7–11.1 %	10 manifest children: PVE < 47.8 mm and |age bias| < 0.078; holdout PVE ≤ 71.5, recall ≥ 89.2	inconclusive: children 48.5 vs. 53.1 mm, age bias 0.078 vs. 0.085; baby age floor 0.100 → 0.102	same run with the sampler off (age ×3 and height ×2 shape weights in both)
Head-loss pilot	L1 loss on the camera-frame orientation of head, neck03 and both wrists, weight 5	head direction < 12.0 deg; PVE ≤ 71.5 mm, recall ≥ 89.2 %	head direction 11.3 deg, within the controls' 11.7–13.6; safety limit missed (71.8 mm, 89.0 %)	two identical runs at weight 0
Finger pilot	region-normalised rotation loss (body, hands, head-neck averaged separately); a test-time person crop	fingers below the rest-pose floor 15.1 deg without a PVE cost	no effect: fingers 17.5 vs. 17.4 deg; crop 17.2 deg at body PVE 124.8 mm	same run with the loss off (one of the two identical runs above)

Pilot	What changed	Success rule	Result	Control
Geodesic pilot	geodesic angle instead of L1 on the rotation matrices	flagged bones down by ≥ 3 deg more than the control, or neck and head under 15 deg	no movement: neck02 175.6 $\rightarrow$ 175.0, head 177.9 $\rightarrow$ 177.1 deg	same run with the L1 loss
Bone-correction pilot	constant output rotation on neck02, head, metacarpal1.L, finger4-1.R; no training, then a 2-epoch fine-tune	bones < 15 deg, girth ≥ 0.9; no training: head direction $\leq$ Base + 3 deg; fine-tuned: guards against the control	no training: fails (head direction 19.1 vs. limit 15.0 deg); fine-tuned: passes (neck02 4.7 deg, girth 0.997, fingers 8.3 deg)	the geodesic pilot's L1 control

Depth pilots. The pilots built the depth, translation, detection-cell and offset targets at the head joint, which the network anchors, instead of at the pelvis (Section 4.4); three of them also raised the depth and translation weights to 5 or 20. By their own rule they failed: every run, the two controls included, missed the PVE or recall clause, and the slope clause passed even for the 2-epoch control, because it asked only for some move toward 1. Outside the rule the change was large and consistent. The root error fell from 1.21 to 0.53–0.73 m in all five fix runs, against 1.11–1.44 m in the controls; the best 10-epoch run ($\times 5$) reached 0.54 m and size 0.84, but the slope moved only from 0.33 to 0.30–0.44. Every fix run was 2.6–4.9 mm worse in PVE and 1.6–2.4 recall points lower than its control; each gap lies within run-to-run variation, but all five point the same way, and the $\times 5$ and $\times 20$ runs have no control at the same weight. Table 5.3 lists every run. The $\times 5$ variant became part of Final.

Sampler pilot. A weighted sampler drew images with a child (age below 0.3) for a quarter of the draws, which raised the share of children among the supervised people from 2.1 to 9.7–11.1 %. On the 10 children of the diagnostic manifest, who are training images, the pilot reached 48.5 mm against 53.1 mm for its control and 47.8 mm for Base. A 4.6 mm gap on 10 in-sample people cannot be read, and the youngest training people kept their predicted age floor (0.100 $\rightarrow$ 0.102). Both arms missed the adult guard (holdout PVE 75.1 and 74.9 against 70.5 mm), which is the restart cost of any fine-tune rather than an effect of the sampler. The pilot is inconclusive after 8,000 images; it does not show that sampling cannot work, and it has no mechanism for real photos.

Head-loss pilot. An extra L1 term compared the camera-frame orientation of head, neck03 and both wrists with the ground truth, so that the many ways a four-bone neck can reach one head pose would all be judged by where the head points. The head direction reached 11.3 deg and so met the rule's 12.0 deg, but two identical control runs gave 11.7 and 13.6 deg: the pilot differs from them

by less than they differ from each other. Its safety limit was missed narrowly (71.8 mm against 71.5, 89.0 % against 89.2), as it was by every control run. The local neck bones stayed where they were (group mean 62.5 deg), which the half-turned neck02 found later explains (Section 5.4).

Finger pilot. Two ideas were tested. A test-time crop around each person (factor 1.3, no training) gave the hands more tokens; the finger error moved from 17.3 to 17.2 deg while the body PVE rose to 124.8 mm, because the network had never seen such a camera. The region-normalised rotation loss averaged the rotation error within body, hands and head-neck before averaging across them; this cut the hands' share of the rotation loss from about 70 to 33 %, a rebalance rather than an upweighting. The fingers ended at 17.5 against 17.4 deg for the control: no effect. Neither idea reached the rest-pose floor of 15.1 deg. Half of the finger error turned out to be two flipped bones, which the bone-correction pilot removed.

Geodesic pilot. The rotation term measured the geodesic angle between predicted and true bone rotation instead of the L1 distance between their matrix entries. The flip did not move: neck02 went from 175.6 to 175.0 deg and the head bone from 177.9 to 177.1 deg, while the L1 control stayed at 175.7 and 177.9 deg. The likely reason is that neck02 and the head would have to turn back together, and any partial move raises the loss; this was not tested. Its L1 control (PVE 76.9 mm) also served as the control of the next pilot.

Bone-correction pilot. One constant rotation per flipped bone, estimated on 200 training images (178.8 to 179.8 deg), was applied right after the rotation decoding (Section 4.5). Without training it turned the neck bones back (neck02 175.6 $\rightarrow$ 4.5 deg, girth 0.46 $\rightarrow$ 1.004) but failed its rule, because the head direction rose from 12.0 to 19.1 deg against a limit of 15.0: one constant cannot undo a compensation that differs from person to person. After a 2-epoch fine-tune with the correction in place it passed every check against its control: corrected bones under 15 deg, girth 0.997, recall 89.4 against a limit of 89.1, head direction 14.4 against a limit of 16.8 deg. Its PVE of 68.8 mm against 76.9 mm for the control shows no sign of a PVE cost; it does not show a gain. On Fresh-heads-L the same method corrected three flipped metacarpals without training and passed (fingers 32.0 $\rightarrow$ 17.8 deg). The correction became part of Final.

Pilot noise. Table 5.7 in Chapter 5 gives the yardstick for all of the above. Two runs of one configuration differed by 7.6 mm PVE, 1.9 deg head direction and 0.8 recall points, and restarting a fine-tune of Base cost about 2.4 mm. One pilot therefore resolves about 10 mm PVE, 3 deg and 2 recall points. The root error, the bone angles, the girth and the finger error moved far beyond that; the PVE and recall differences of every pilot did not. That is why each pilot is read against its own control, and why the two changes that entered Final were chosen for what the diagnostics showed, not for a PVE gain.

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